SiN photonic circuits for a neutral-atom processor

Volume II: the module-by-module record (compiled from `docs/`)

Simulation study for proposal ANRF/ARG/2026/009937

November 7, 2026

Preface

This volume compiles, without rewriting, the fifteen documents in `docs/` of the repository, in topic order. Each was written as its module was completed, and several carry follow-up notes at their start or in later sections. Where a follow-up note or a later section revises an earlier statement, the later one applies; Volume I (`report/report.pdf`) gives the current state in one place. Values are labelled in each document's verification notes as USER (proposal), SOURCE (read in session), SIM (simulation in the repository), DERIVED (calculation shown), CHOICE (stated assumption) or RECALLED (to be verified), and simulated values should be reviewed before they are quoted.

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Chapter 1

Phase-A simulations of the SiN PIC elements: first results

Proposal ANRF/ARG/2026/009937 (draft v97), Objective 3 and the Phase-A error budget. Code: `picsim/`, scripts `scripts/m1` to `m7`, results in `results/*.json` (inputs with provenance, outputs). Every simulated value below is a SIM output, valid only within the regime simulated, and must be reviewed before it enters the proposal.

1.1 Summary

The passive SiN elements meet the proposal's targets with margin on a 400 nm LPCVD film. Five results bear directly on the proposal text.

(i) **OPA update rate.** A tweezer moved by a staircase of OPA updates is excited when the update rate aliases the move spectrum onto the trap frequency. At the 1.2 mK transport depth (radial trap frequency 84 kHz for w0 = 1.04 um), a 100 kHz update adds about 40 motional quanta to a 100 us, 4 um move and about 2,900 quanta to a 50 us move, against 0.005 and 0.2 for a smooth trajectory. At 200 kHz and above the staircase adds nothing measurable. Hence the >= 100 kHz target of Table 1 holds for 0.3 mK tweezers (42 kHz) but should read >= 200 kHz for moves at 1.2 mK.

(ii) **OPA field and tweezer count.** The number of resolvable positions along the steering axis equals the channel number N, so the range at the atoms is N lambda/(4 NA F), with F the pupil fill. At NA 0.5 and full fill, 64 channels steer over +/-33 um (17 sites at 4 um pitch), not +/-100 um, and cannot place more than about 15 tweezers on one row. More than 20 simultaneous tweezers on one row need 128 channels; 128 channels at 0.75 fill cover +/-89 um at 1.33 times the power per tweezer. A follow-up study (`docs/twenty_tweezers.md`) shows that 20 individually switchable tweezers at 1064.4 and 894.6 nm are best delivered by a 128-emitter OPA fed through a 20-port star coupler, with one switch per port and a common phase ramp for parallel moves; for independent motion of each tweezer, as with AOD tones, a sliding-window focal-plane OPA is required (`docs/independent_steering.md`).

(iii) **Switch architecture.** A single MZI with the simulated 2x2 MMI spread reaches 50 dB in only 49% of devices at 3 mrad phase accuracy. A cascaded pair, as in [ZHAO], reaches >= 50 dB in every trial (5th percentile 66 to 88 dB). The binding constraints are then polarisation and free-space crosstalk, not the interferometer.

(iv) **Addressing crosstalk is optical, not on-chip.** With a uniformly filled NA 0.5 pupil, the 894.6 nm spot puts 1.0e-3 of its peak intensity on the 4 um neighbour, which gives a neighbour error of 2.5e-3, above the 1e-3 budget. A Gaussian pupil fill truncated at 0.7 of the pupil radius brings this to 1.2e-4 (error 2.9e-4) at a spot FWHM of 1.13 um instead of 0.92 um.

(v) **Position noise must be specified as a spectral density.** The error-budget statement "position noise <= 10 nm rms keeps transport survival >= 99%" does not fix the heating rate. If 10 nm rms is white over 1 MHz, a 1.2 mK tweezer heats at about 7,700 quanta/s (about 0.8 quanta per 100 us move, against a depth of about 300 quanta). Heating below 10 quanta/s needs S_x(84 kHz) <= 1.3e-25 m^2/Hz.

The end-to-end loss of the end-fire OPA from fibre to atoms, for 21 tweezers at N = 128, is 5.9 dB (7.1 dB for the best grating emitters), against the ~10 dB assumed in the proposal; the per-tweezer on-chip power falls from 66 mW to about 26 mW. This estimate excludes sidewall scattering and assumes a meta-objective transmission of 0.7.

1.2 Scope, methods and limits

We model the elements named in Sec. 4 (Phase C) of the proposal. The waveguide platform is stoichiometric LPCVD Si3N4, 400 nm thick (the CoE-CPPICS limit), with 200 and 300 nm as foundry alternatives, in SiO2. Material dispersion is from Luke et al. [LUKE] and Malitson [MAL], and the thermo-optic coefficients (Si3N4 2.45e-5 /K, SiO2 1e-5 /K) from [TRC]. The working wavelengths are 894.6 nm (addressing) and 1064.4 nm (transport).

The methods are: vector FEM mode solving (femwell 0.1.12, second-order Nedelec elements) for straight, bent, coupled and leaky guides; effective-index reduction with guided-mode eigenmode expansion for MMIs; scalar overlap with Gaussian fibre modes for edge couplers; 2D FDTD (MEEP 1.35, 40 px/um) for grating antennas, with the Si substrate included; 2x2 transfer matrices with Monte Carlo for switches; 2D thermal FEM for heaters; 1D scalar Fourier optics for the OPA (emitter near field, Fourier-plane slit, aplanatic objective); and an exact harmonic-oscillator result for transport heating. The mode, slab, MZI, OPA and transport models are checked against analytic limits or published examples (`tests/`, 16 tests); the FDTD closes its energy balance to better than 0.5%, and the heater result changes by 1.4% when the domain is doubled.

The principal limits are as follows. The MMI and grating models are 2D. Edge-coupler numbers are mode overlaps without Fresnel or taper conversion loss. The OPA model is scalar and 1D along the steering axis, with an aberration-free objective. Sidewall scattering loss and fabrication roughness are not modelled. The AlN actuator is a single mechanical mode with assumed f0 and Q.

1.3 Platform (Module 1)

The design width is the widest width (0.1 um grid) for which TE1 is cut off or within 0.015 of the cladding index and the TE0/TM0 splitting is at least 0.015. The second condition matters: on 400 nm film at 894.6 nm, a 0.4 um square guide is polarisation-degenerate (TE0 = TM0 = 1.693), which would couple polarisations in bends. Hence we use 0.5 um there.

Film	lambda (nm)	width (um)	n_eff	n_g	dn_eff/dT (1/K)	R for < 0.01 dB/90 deg, TE / TM (um)
400 nm	1064.4	0.6	1.708	2.14	2.36e-5	8 / 10
400 nm	894.6	0.5	1.745	2.17	2.41e-5	6 / 6
200 nm	1064.4	0.9	1.613	1.98	2.02e-5	20 / 60
200 nm	894.6	0.7	1.651	2.05	2.16e-5	20 / 30

Substrate leakage on 400 nm film is below 1e-5 dB/cm for a buried oxide of 2 um or more at both wavelengths; on 200 nm film at 1064.4 nm the oxide must be at least 3 um (TE 7e-8 dB/cm; at 2 um, TM leaks 0.33 dB/cm). The coupling length of two 1064.4 nm guides on 400 nm film is 1.8 mm at 1.6 um pitch and 29 mm at 2.1 um pitch. At 1.4 W the Kerr phase is 0.05 rad/cm and the peak in-guide intensity 420 MW/cm^2.

On 400 nm film the TM bend loss is only about ten times the TE loss at the same radius, hence bends cannot strip TM light. On 200 nm film at 1064.4 nm a 20 um bend radiates 1.5 dB per 90 deg of TM against 5e-4 dB of TE, which makes a compact TM filter possible; at 894.6 nm the same film gives only 0.03 dB per 90 deg of TM.

1.4 Edge couplers and splitters (Module 2)

Inverse tapers on 400 nm film reach a mode overlap of -0.37 dB with a 2.5 um lensed fibre at 1064.4 nm (150 nm tip, +/-0.6 um lateral tolerance at -1 dB) and -0.48 dB with HI1060 (100 nm tip, +/-1.55 um). At 894.6 nm the best values are -0.46 dB (lensed) and -1.1 dB (PM780). Several optima lie at the narrowest tip swept (100 nm), so narrower tips may do better if lithography allows. These overlaps exclude Fresnel and taper conversion loss and are therefore lower bounds on the 1 to 2 dB per facet of the proposal. At 1.4 W through one 400 nm facet the intensity is 45 MW/cm^2; three input fibres keep it below 20 MW/cm^2. We did not find a CW facet-damage threshold for SiN at 1 um in this session.

A 4 um wide 1x2 MMI has 0.044 dB excess loss at both wavelengths (lengths 14.8 um at 1064.4 nm and 17.7 um at 894.6 nm), so the 6-stage, 64-way tree costs 0.27 dB, against the ~1 dB assumed. A symmetric 1x2 MMI is balanced by symmetry; imbalance comes only from asymmetric errors, 0.13 dB for a 20 nm lateral offset of the input guide. With 1-sigma errors of 10 nm (width), 5 nm (thickness) and 5 nm (offset), the 2x2

split ratio varies by 0.0024 (894.6 nm) and 0.0034 (1064.4 nm). Our first-pass 894.6 nm 2x2 design sits at 0.490 rather than 0.500; this offset is removable by redesign and is carried separately below.

1.5 Switches and router (Modules 4 and 5)

Architecture (894.6 nm, split spread 0.0024)	phase error 3 mrad: median / 5th pct (dB)	P(>= 50 dB)	phase error 10 mrad: median / 5th pct (dB)	P(>= 50 dB)
single MZI	49.8 / 41.4	0.49	45.3 / 38.8	0.21
single MZI, trimmed input splitter	55.1 / 48.8	0.89	44.6 / 38.3	0.18
two cascaded MZIs	100 / 88	1.00	92 / 81	1.00

Arm amplitude imbalance (0.5% rms) is included. With the 0.010 design offset added, a single MZI falls to P = 0.15 while the cascade keeps P = 1.00 (5th percentile 66 dB). An incoherent stray-light floor of -70 dB caps the cascade at 70 dB. A single balanced MZI with a phase error delta has ER = -20 log10(delta/2), so 50 dB needs delta <= 6.3 mrad; for the AlN half-wave voltage at 894.6 nm, which we bracket between 84 V (lambda-scaled from 74.7 V at 795 nm [ZHAO]) and 132 V (interpolated to 200 V at 1013 nm [ZHAO]), the driver noise must stay below 0.17 to 0.27 V rms.

Polarisation sets a separate floor. TM light is not nulled when the TE path is. For an input polarisation extinction of 20 dB, 50 dB at the output needs 24 to 30 dB of TM suppression after the chip; for 25 dB input, 19 to 25 dB. On 400 nm film bends cannot provide this (Sec. 3), hence a polariser in the free-space delivery path or a dedicated on-chip TE-pass element is required.

The settling of the switch is set by mechanical ringing, not by the 26 ns rise [ZHAO]. For an assumed single mode at f0 = 20 MHz and Q = 10, an unshaped step needs 0.98 us to reach -50 dB after switch-off. A two-impulse shaper (ZV) gives 24 ns but 0.43 us with a 2% error in f0; a three-impulse shaper (ZVD) gives 49 ns with a 2% error and 0.19 us with 5%, at a 28 ns optical rise. Hence <= 100 ns needs ZVD shaping with f0 known to about 2%. These f0 and Q values are assumptions to be replaced by measured ones.

The 16-output router (four stages of thermo-optically tuned MZI splitters) leaks at most 1.4e-3 into the worst unaddressed output (95th percentile) for a 30 mrad trim error, so with the cascaded output switch the idle-site extinction exceeds 90 dB. Its insertion loss is 5.0 dB, of which 3 dB is the assumed free-space delivery, so 16 sites at 20 uW need about 1 mW at the input.

Thermo-optic shifters with a 2 um wide heater 2 um above the core need 84 mW for pi at 1064.4 nm (68 mW at 894.6 nm), with a 34 us rise and 16% thermal crosstalk to a guide 10 um away; isolation trenches reduce these to 54 mW and 2%, and an additional undercut to 3 mW, with a 1.5 ms rise. They suit the slow router trims but not OPA updates at >= 100 kHz. For the OPA, the PZT shifters of [MONT] (V_pi 2.8 V at 532 nm, about 5.6 V at 1064.4 nm by wavelength scaling, 0.4 MHz bandwidth) settle in about 2 us, within a 5 us update period.

1.6 Optical phased array (Module 3)

Range and resolution. The atom-plane field is a scaled copy of the slit-filtered OPA far field.

N	fill	usable half-range (um)	sites on one row (4 um)	spot FWHM along x (um)	power for equal depth
32	1.0	16	9	0.95	1.00
64	1.0	33	17	0.95	1.00
64	0.5	68	35	1.89	1.99
128	1.0	67	33	0.94	1.00
128	0.75	89	45	1.26	1.33
256	1.0	134	67	0.94	1.00

"Usable" means the steered peak keeps at least half its on-axis value. Uniform channel amplitudes give 4.7% first sidelobes along x; a Gaussian channel apodisation (1/e^2 at the aperture edge) lowers them to 0.9% at a FWHM of 1.09 um.

Emitter pitch. A Fourier-plane slit passes only the central diffraction zone, so the efficiency into it falls with pitch: 0.65 at 1.5 um, 0.52 at 2 um, 0.34 at 3 um (for the 0.6 um emitter guide; a wider emitter mode

at the facet would raise these). The minimum pitch is set by coupling between neighbouring emitters over their parallel length, for < -20 dB power transfer:

Emitter	parallel length (um)	minimum pitch (um)	upward fraction	emission angle
end-fire facet	50 (assumed)	1.46	1 (in plane)	0 deg, wavelength independent
grating, 200 nm etch, 2 um oxide	20	1.30	0.76	8.9 deg
grating, 100 nm etch, 3 um oxide	90	1.56	0.72	9.7 deg
grating, 50 nm etch, 3 um oxide	384	1.82	0.68	10.0 deg

The grating directionality oscillates with oxide thickness through the substrate reflection (0.42 to 0.76 over 2 to 4 um), and the emission angle moves by -0.10 deg/nm. The 200 nm etch reflects 6% back into the guide. Hence the end-fire array is the better first choice: denser pitch for short parallel sections, no directionality loss, and no wavelength dependence of the pointing. The grating array becomes attractive if a bottom reflector or two-layer grating is available.

Simultaneous tweezers. With weighted Gerchberg-Saxton alone, light at untargeted lattice sites reaches 11 to 21% of the tweezer intensity. A refinement that also penalises stray light and off-centre spots holds it at 3 to 7%:

N	K	max stray at untargeted sites	max (Imax - Imin)/(Imax + Imin)	power in tweezers / emitted
64	8	0.056	0.062	0.44
64	12	0.053	0.061	0.46
128	16	0.054	0.106	0.45
128	21	0.057	0.106	0.45

(Four random target sets per case, one row along the steering axis.) A 5.7% stray at a static site corresponds to about 70 uK under a 1.2 mK transport tweezer.

Phase accuracy. For independent channel phase errors sigma, the position jitter at the atoms is sigma sqrt(12/N) lambda/(4 pi NA) (Monte Carlo agrees to 2%). Hence 10 nm needs sigma <= 0.14 rad at N = 64 (0.10 rad at 32, 0.19 rad at 128); 8-bit phase resolution (7 mrad) is sufficient, and a PZT hysteresis error must stay below about 4% of the applied phase at N = 64.

Loss budget (end-fire, N = 128, K = 21, 2 um pitch). Edge coupler -0.37 dB, MMI tree -0.27 dB, propagation 1 cm at 0.3 dB/cm (assumed) -0.30 dB, emitters with slit, pupil and synthesis -3.43 dB, meta-objective transmission 0.7 (assumed) -1.55 dB: total -5.9 dB. The grating case adds -1.2 dB of directionality. At 1.5 um pitch the slit term improves by about 1 dB.

1.7 Error-budget links and power handling (Modules 6 and 7)

The proposal's footnote 4 is reproduced: an idle-qubit error of (pi^2/4) 10^(-ER/10), 2.5e-5 at 50 dB when one Raman beam is addressed. The trap depth of 6.6 mW in a Gaussian of w0 = 1.04 um with 1160 a.u. is 1.01 mK, not 1.2 mK; 1.2 mK needs 7.8 mW, or w0 = 0.96 um. The OPA spot along the steering axis is 22% brighter per watt than that Gaussian, which brings 6.6 mW to about 1.24 mK.

Intensity noise must satisfy S_I(2 nu) <= 1.4e-11 /Hz at 1.2 mK for a parametric heating time of 1 s. At 1.4 W into the input guide, with 0.3 dB/cm fully absorbed, the guide warms by 3.8 K and accrues 5.3 rad/cm of common-mode phase before the split; after the 64-way split each channel accrues at most 0.08 rad/cm, which the per-channel calibration absorbs.

1.8 Suggested changes to the proposal

(i) Table 1, OPA update rate: ">= 200 kHz" for transport at 1.2 mK. Alternatively, keep >= 100 kHz and lower the transport depth to <= 0.4 mK during moves (trap frequency 48 kHz) with move times of >= 100 us, for which a 4 um move adds 0.025 quanta (0.008 for a smooth trajectory). (ii) Table 1, OPA

tweezers: state ">= 20 individually switchable tweezers with 128 emitters and a 20-port star coupler" (see `docs/twenty_tweezers.md`), or ">= 15 with 64 channels on one row", and state the steering field (+/-33 um at N = 64, +/-68 um at N = 128, F = 1) separately from the +/-100 um meta-objective field. (iii) Sec. 4, "~10 dB of chip and OPA loss": the model gives about 6 dB (end-fire) to 7 dB (grating); keep 10 dB as the conservative value but note the modelled one. (iv) Sec. 4 error budget: replace "position noise <= 10 nm rms" by a spectral density at the trap frequency, S_x(nu_trap) <= 1.3e-25 m^2/Hz for < 10 quanta/s at 1.2 mK, and add the 894.6 nm pupil apodisation needed for neighbour crosstalk <= 1e-4. (v) Sec. 4(ii): 6.6 mW gives 1.0 mK for w0 = 1.04 um; quote 7.8 mW or the OPA spot. (vi) Phase C: specify cascaded MZIs (or trimmed single MZIs with <= 3 mrad phase accuracy), an output polariser of >= 30 dB on the 894.6 nm path, and shaped (ZVD) drive for <= 100 ns settling.

1.9 Next steps

(i) 3D FDTD or EME of the inverse taper, MMI and end-fire facet to replace the 2D and overlap estimates; (ii) end-fire emitter with an expanded facet mode to raise slit efficiency at 1.5 um pitch; (iii) mechanical FEM of the AlN actuator to replace the assumed f0 and Q; (iv) a 2D model of the OPA spot through the meta-objective design of Phase B, including its aberrations, since a Strehl of 0.8 adds a halo not included in the crosstalk estimate; (v) GDS layout of the 64-channel OPA and 16-channel router on one 10 x 10 mm die.

1.10 Verification notes

References read in this session. [LUKE] K. Luke et al., Opt. Lett. 40, 4823 (2015), doi:10.1364/OL.40.004823 (Sellmeier via refractiveindex.info). [MAL] I. H. Malitson, JOSA 55, 1205 (1965), doi:10.1364/JOSA.55.001205 (via refractiveindex.info). [TRC] arXiv:2606.05673, introduction: dn/dT of Si3N4 and SiO2 at 193 THz. [ZHAO] M. Zhao et al., arXiv:2508.09920: 71.4 dB, 26 ns, V_pi 74.7 V (795 nm) and 200 V (1013 nm), cascaded MZIs, -68 dB on-chip and -50.8 dB free-space crosstalk. [MONT] N. Montifiore et al., arXiv:2601.15695: PZT coil MZM, V_pi 2.8 V at 532 nm, 0.4 MHz, 21.5 dB. [OHARA] K. M. O'Hara et al., arXiv:physics/0003049: Gamma = pi^2 nu^2 S_I(2 nu), Q_dot = 4 pi^4 M nu^4 S_x(nu). AlN index: Pastrnak and Roskovcova, Phys. Stat. Sol. 14, K5 (1966), via refractiveindex.info.

Recalled, not verified (check before use). Weighted Gerchberg-Saxton: R. Di Leonardo et al., Opt. Express 15, 1913 (2007) (publisher page blocked); fibre MFDs (HI1060 6.2 um, PM780 5.3 um); Si index 3.55; Kerr index 2.5e-19 m^2/W; thermal conductivities and heat capacities; Cs mass.

Provenance ledger (quoted values).

value	location	type	pointer
40 and 2,900 quanta at 100 kHz; 200 kHz threshold	Sec. 1(i), 8(i)	SIM	results/m3_update_rate.json (review before submission)
0.025 and 0.008 quanta at 0.4 mK	Sec. 8(i)	SIM	picsim.transport.n_bar_staircase, run in session (review)
84 kHz, 42 kHz trap frequencies	Sec. 1(i)	DERIVED	picsim.transport.radial_freq_Hz, w0 = 1.04 um (USER)
+/-33, 67, 89 um ranges; 17, 33, 45 sites	Sec. 1(ii), 6	SIM	results/m3_field_tradeoff.json (review)
49%, 89%, P = 1.00, 66-88 dB	Sec. 1(iii), 5	SIM	results/m4_extinction.json (review)
1.0e-3, 1.2e-4 neighbour intensity; 2.5e-3, 2.9e-4 error	Sec. 1(iv)	SIM	results/m7_budget.json (review)
7,700 quanta/s; 1.3e-25 m^2/Hz	Sec. 1(v), 8(iv)	DERIVED	[OHARA] formula, results/m7_budget.json
-5.9 dB, -7.1 dB, 26 mW	Sec. 1, 6	SIM	results/m3_loss_budget.json (review)
design widths, n_eff, n_g, dn/dT, bend radii	Sec. 3	SIM	results/m1_*.json (review)
-0.37 to -1.1 dB overlap, 45 MW/cm^2	Sec. 4	SIM	results/m2_couplers.json (review)
0.044 dB per MMI, 0.27 dB tree, 0.0024/0.0034 spread	Sec. 4	SIM	results/m2_mmi.json (review)
84-132 V, 0.17-0.27 V rms	Sec. 5	DERIVED	from [ZHAO] values, results/m4_dynamics.json
0.98 us, 24 ns, 49 ns, 0.19 us	Sec. 5	SIM	results/m4_dynamics.json (f0, Q assumed; review)
1.4e-3 leak, > 90 dB, 5.0 dB, 1 mW	Sec. 5	SIM	results/m4_router.json (review)

value	location	type	pointer
84, 54, 3 mW; 34 us, 1.5 ms; 16%, 2%	Sec. 5	SIM	results/m5_heaters.json (review)
min pitch 1.30-1.82 um; slit efficiency 0.65/0.52/0.34	Sec. 6	SIM	results/m3_pitch_efficiency.json (review)
directionality 0.42-0.76; -0.10 deg/nm; 6% reflection	Sec. 6	SIM	results/m3_grating.json, m3_grating_fdtd_raw.json (review)
stray 3-7%; uniformity to 0.106	Sec. 6	SIM	results/m3_multibeam.json (review)
sigma 0.10/0.14/0.19 rad; 4% hysteresis	Sec. 6	SIM/DERIVED	results/m3_noise.json (review)
1.01 mK, 7.8 mW, 0.96 um, 1.22	Sec. 7	DERIVED/SIM	results/m7_budget.json
3.8 K, 5.3 rad/cm, 0.08 rad/cm	Sec. 7	SIM	results/m6_power.json (review)
6.6 mW, 1.2 mK, 1.04 um, 1160 a.u., 10 dB, 1.4 W, >= 50 dB, <= 100 ns, 10 nm, NA 0.5, 4 um	throughout	USER	proposal draft v97
meta-objective transmission 0.7; 0.3 dB/cm; 50 um end-fire section; f0 = 20 MHz, Q = 10; fab sigmas	Sec. 5, 6	CHOICE	stated assumptions, not claims

Open placeholders. [VALUE?] CW facet-damage threshold of SiN/SiO2 at 1 um; [VALUE?] PZT hysteresis magnitude; [VALUE?] AlN actuator f0 and Q at 894.6 nm; [VALUE?] measured V_pi at 894.6 nm; [REF?] verified citation for weighted Gerchberg-Saxton.

Chapter 2

Parameter ledger

Generated by `scripts/m7_budget.py` from the input records of every `results/*.json`. Types: USER (proposal text), SOURCE (read in session, pointer given), DERIVED, SIM, CHOICE (design or sweep choice), RECALLED (not verified; check before use). Simulation outputs are SIM and must be reviewed before submission.

value	unit	type	pointer	used in
4.0	um	CHOICE	buried thermal oxide; checked against substrate leakage in scripts/m1_platform.py	m5_heaters
2.5	um	CHOICE	lensed/tapered fibre spot, typical catalogue range 2-3 um	m2_couplers
[0.2, 0.3, 0.4]	um	CHOICE	thickness sweep agreed with PI; 200 nm covers foundry runs	m1_width_sweep
5.60228947368421	V	DERIVED	2.8 V x 1064.4/532 assuming the same coil length and stress-optic coefficient; real value depends on mode overlap	m5_actuators
886.0600542866746	um	DERIVED	R = N p_a p_port / lambda_s (one port per resolvable spot)	m9_E_window
15.0	mm	DERIVED	focal length taken as the 15 mm working distance (order of magnitude)	m9_routes
84000.0	Hz	DERIVED	radial, 1.2 mK, w0 = 1.04 um (m3_update_rate)	m9_routes
5000.0	m/s	RECALLED	AlN surface-acoustic-wave velocity, order of magnitude; verify for the AlN/SiN/SiO2 stack	m9_routes
6.2	um	RECALLED	Corning HI1060 MFD at 1060 nm approx. 6.2 um; verify datasheet	m2_couplers
2.5e-19	m^2/W	RECALLED	Kerr index of LPCVD Si3N4, approx. 2.4-2.5e-19 m^2/W (Ikeda 2008); verify	m1_design_modes
20.0	W/(m K)	RECALLED	LPCVD SiN film, literature spread ~10-30 W/(m K); verify	m5_heaters
5.3	um	RECALLED	PM780-HP MFD at 850 nm approx. 5.3 um; verify datasheet	m2_couplers
20.0	W/(m K)	RECALLED	TiN/NiCr heater film order of magnitude; verify	m5_heaters
3.55	-	RECALLED	crystalline Si near 1.06 um, approx. 3.55 + 0.0001i; verify (Green 2008)	m1_box_leakage
1628000	J/(m^3 K)	RECALLED	density 2200 kg/m^3 x c_p 740 J/(kg K)	m5_heaters
1631000	J/(m^3 K)	RECALLED	density 2330 kg/m^3 x c_p 700 J/(kg K)	m5_heaters
1.38	W/(m K)	RECALLED	fused/thermal silica thermal conductivity; verify	m5_heaters
148.0	W/(m K)	RECALLED	silicon thermal conductivity at 300 K; verify	m5_heaters
0.0023748698871815	-	SIM	SIM m2_mmi (895 nm, 4 um 2x2)	m4_extinction
0.010466565944314093	-	SIM	SIM m2_mmi (895 nm, 4 um 2x2)	m4_extinction
2.14	-	SIM	m1_design_modes 1064_0.4	m9_routes
2.36e-05	1/K	SIM	m1_design_modes 1064_0.4	m9_routes

value	unit	type	pointer	used in
(model)	-	SOURCE	Luke et al., Opt. Lett. 40, 4823 (2015), doi:10.1364/OL.40.004823; Sellmeier via refractiveindex.info Si3N4/Luke (340 nm LPCVD film), read 2026-10-31	m1_width_sweep
(model)	-	SOURCE	Malitson, JOSA 55, 1205 (1965), doi:10.1364/JOSA.55.001205; via refractiveindex.info SiO2/Malitson, read 2026-10-31	m1_width_sweep
400000.0	Hz	SOURCE	Montifiore et al., arXiv:2601.15695 (abstract, read 2026-10-31): coil MZM 3-dB bandwidth 0.4 MHz	m3_update_rate, m5_actuators
2.8	V	SOURCE	Montifiore et al., arXiv:2601.15695 (abstract, read 2026-10-31): coil MZM V_pi 2.8 V at 532 nm	m5_actuators
2.6e-08	s	SOURCE	Zhao et al., arXiv:2508.09920 (abstract and Sec. 2, read 2026-10-31): 10-90% rise 26 +/- 7 ns	m4_dynamics, m5_actuators
200.0	V	SOURCE	Zhao et al., arXiv:2508.09920 (abstract and Sec. 2, read 2026-10-31): V_pi 200 V at 1013 nm	m4_dynamics, m5_actuators
74.7	V	SOURCE	Zhao et al., arXiv:2508.09920 (abstract and Sec. 2, read 2026-10-31): single-ended V_pi 74.7 +/- 3.7 V at 795 nm	m4_dynamics, m5_actuators
2.45e-05	1/K	SOURCE	arXiv:2606.05673, introduction (values at 193 THz; ~7% dispersion to visible), read 2026-10-31	m1_design_modes
1e-05	1/K	SOURCE	arXiv:2606.05673, introduction (values at 193 THz; ~7% dispersion to visible), read 2026-10-31	m1_design_modes
(model)	-	SOURCE	heating rates: O'Hara et al., arXiv:physics/0003049 (PRL 82, 4204 (1999)), Gamma = pi^2 nu^2 S_I(2 nu), Q_dot = 4 pi^4 M nu^4 S_x(nu); read 2026-11-01	m7_budget
1160.0	a.u.	USER	ANRF/ARG/2026/009937 draft v97, Sec. 3 citing Le Kien et al. EPJD 67, 92 (2013)	m3_update_rate, m7_budget, m8_power, m9_E_power
4.0	um	USER	ANRF/ARG/2026/009937 draft v97, Sec. 3: 'about five sites at 4 um pitch'	m3_field_tradeoff, m3_multibeam, m7_budget, m8_final, m8_star_design
1.04	um	USER	ANRF/ARG/2026/009937 draft v97, Sec. 4 error budget: 1.04 um waist	m3_spot, m3_update_rate, m7_budget, m8_power, m9_E_power
0.01	um	USER	ANRF/ARG/2026/009937 draft v97, Sec. 4 error budget: <= 10 nm rms	m3_noise, m7_budget
0.0012	K	USER	ANRF/ARG/2026/009937 draft v97, Sec. 4(ii): transport tweezers ~1.2 mK	m3_update_rate, m7_budget, m8_power, m9_E_power
1.4	W	USER	ANRF/ARG/2026/009937 draft v97, Sec. 4(ii): ~1.4 W on chip for 21 tweezers	m1_design_modes, m6_power
10.0	dB	USER	ANRF/ARG/2026/009937 draft v97, Sec. 4(ii): ~10 dB of chip and OPA loss	m3_loss_budget
0.0066	W	USER	ANRF/ARG/2026/009937 draft v97, Sec. 4(ii): ~6.6 mW per tweezer at the atoms	m3_loss_budget, m7_budget
16	-	USER	ANRF/ARG/2026/009937 draft v97, Sec. 4: 16 channels on the 895 nm router	m4_router
1.0	dB	USER	ANRF/ARG/2026/009937 draft v97, Sec. 4: 5-6 MMI stages, ~1 dB	m2_mmi
[1.0, 2.0]	dB	USER	ANRF/ARG/2026/009937 draft v97, Sec. 4: ~1-2 dB per facet	m2_couplers

value	unit	type	pointer	used in
[1e-06, 2e-05]	W	USER	ANRF/ARG/2026/009937 draft v97, Sec. 4: ~1-20 uW per site at the atoms	m4_router, m8_power
0.4	um	USER	ANRF/ARG/2026/009937 draft v97, Sec. 7.4: CoE-CPPICS stoichiometric LPCVD SiN up to 400 nm	m2_mmi
50.0	dB	USER	ANRF/ARG/2026/009937 draft v97, Table 1: >= 50 dB at 895 nm	m4_extinction, m4_polarisation
100.0	um	USER	ANRF/ARG/2026/009937 draft v97, Table 1: Strehl >= 0.8 over >= +/-100 um	m3_field_tradeoff
0.5	-	USER	ANRF/ARG/2026/009937 draft v97, Table 1: meta-objective NA >= 0.5	m3_field_tradeoff, m3_multibeam, m3_noise, m3_spot, m7_budget, m8_star_design, m9_E_window
1.064435	um	USER	ANRF/ARG/2026/009937 draft v97, Table 2: side-B tweezers at 1064.4350 nm (45S1/2 double magic)	m1_width_sweep, m3_field_tradeoff
0.8946	um	USER	ANRF/ARG/2026/009937 draft v97: 895 nm Raman addressing (Cs D1 line, 894.6 nm)	m1_width_sweep, m7_budget

2.1 Result files

- `results/m10_bpm_small.json`:
- `results/m10_bpm_vs_fdtd.json`: fidelity = |<T_fdtd|T_bpm>|^2 / (|T_fdtd|^2 |T_bpm|^2): 1 means identical arm amplitude and phase distributions up to a constant.
- `results/m10_eim_calibration.json`: A: fitted to indices of 0.6/1.0/1.7 um guides and the 1.7 um / 0.3 um-gap coupling length (dense ports). B: additionally the 0.6 um / 1.0 um-gap coupling length (compromise). The spread between A and
- `results/m10_fpr_full_A.json`: T saved in m10_T_full_.npz. Arm phases calibrated with the two central ports; the residual is the rms deviation of each port's arm phases from a straight line (aberration).
- `results/m10_fpr_opt_A.json`: T saved in m10_T_opt_A.npz. Arm phases calibrated with the two central ports; the residual is the rms deviation of each port's arm phases from a straight line (aberration).
- `results/m10_meep_port0_res20.json`:
- `results/m10_meep_port2_res20.json`:
- `results/m10_meep_port3_res20.json`:
- `results/m10_route_E_LS_full_A.json`:
- `results/m10_route_E_LS_opt_A.json`:
- `results/m10_route_E_full_A.json`: P_eff: harmonic mean over |x| <= 60 um of the peak intensity per unit port power; compare with m11 rows of the same geometry.
- `results/m10_route_E_opt_A.json`: P_eff: harmonic mean over |x| <= 60 um of the peak intensity per unit port power; compare with m11 rows of the same geometry.
- `results/m11_stage1.json`:
- `results/m11_stage2.json`:
- `results/m1_bend_loss.json`: Values below ~1e-6 dB/90 are at the numerical floor. Mode-mismatch loss at straight-bend junctions is not included.
- `results/m1_box_leakage.json`: Values below ~1e-5 dB/cm are at the numerical floor.
- `results/m1_coupling.json`: Power transfer over a parallel length L is sin^2(pi L / (2 L_c)); crosstalk below -30 dB needs L < 0.02 L_c.
- `results/m1_design_modes.json`: Kerr phase uses gamma_core x n2 x P/Aeff (upper bound: Aeff from transverse |E|^2). dneff/dT for uniform heating of core and cladding.
- `results/m1_width_sweep.json`: Design width: widest 0.1 um grid width with TE1 cut off or within TE1_margin of the cladding index, and TE0-TM0 splitting >= birefringence_min (a square 400 x 400 nm guide at 895 nm is polarisation-de
- `results/m2_couplers.json`: Excludes Fresnel loss (index-matched or AR-coated facet assumed) and taper conversion loss.
- `results/m2_mmi.json`: 2D effective-index model; excess loss from unguided content only (no sidewall

scattering, no back-reflection). Absolute loss should be confirmed by 3D EME/FDTD.

- `results/m3_field_tradeoff.json`: 1D scalar Fourier optics; aplanatic objective (pupil radius F NA). The number of resolvable positions equals N; the range at the atoms is N lambda/(4 NA fill).
- `results/m3_grating.json`: 2D model of a uniform grating. Directionality oscillates with buried-oxide thickness through the Si-substrate reflection; a bottom reflector or a two-layer grating is needed for > 0.8.
- `results/m3_loss_budget.json`: Per-tweezer share of the emitted power for K = 21 at N = 128, 2 um pitch. The 6.6 mW per tweezer target uses the proposal's w0 = 1.04 um; the OPA spot differs (see m3_spot).
- `results/m3_multibeam.json`: One row of the lattice along the steering axis. K is limited by the number of sites in range.
- `results/m3_noise.json`: Static (frozen) channel errors. For a time-varying error the same mapping gives the position noise spectrum, which sets heating (see m7_budget).
- `results/m3_pitch_efficiency.json`:
- `results/m3_spot.json`: Uniform channel amplitudes give a sinc^2-like profile along the steering axis with ~4-5% first sidelobes; edge apodisation trades spot width for sidelobes.
- `results/m3_update_rate.json`: Harmonic approximation, radial axis, atom initially in the ground state; the staircase aliases the move spectrum to |f_update - f_trap|.
- `results/m4_dynamics.json`: Vpi at 894.6 nm is bracketed, not known: lambda scaling of the 795 nm device and linear interpolation to the 1013 nm device (a different design) differ.
- `results/m4_extinction.json`: Trimmed: the input splitter is an MZI trimmed to match the output coupler, limited by the same phase error. Floor: incoherent stray light relative to input.
- `results/m4_polarisation.json`: TM light is not nulled when the TE path is at its null. T_TM = 1 is the worst case for one MZI; 0.25 is the mean of two cascaded MZIs with random TM phase. Required suppression S = ER_target - PER + 1
- `results/m4_router.json`: Router leakage is coherent-free (powers add); the output switch extinction multiplies.
- `results/m5_actuators.json`: Thermo-optic bandwidth ~0.35/rise. 64-channel power assumes on average P_pi per channel (phases uniformly spread over 0..2 pi).
- `results/m5_heaters.json`: 2D cross-section (long heater). Si substrate 100 um with the bottom at ambient; doubling the domain changes dT by rel_change. Metal absorption of the optical mode is not computed.
- `results/m6_power.json`: Temperature rise is the steady state of a uniformly absorbing straight guide; the phase is common-mode before the split and differential only through channel-to-channel loss spread.
- `results/m7_budget.json`: PSF: scalar Debye, aberration-free; a Strehl of 0.8 adds a halo not included here.
- `results/m8_apod.json`: Stronger apodisation narrows the port far field (smaller FPR radius and port pitch), widens the spot and suppresses the tails that make lit ports interfere.
- `results/m8_final.json`: Adjacent 1.4 um port tapers: coupling length 1.33 mm at 1064.4 nm (0.49 um gap); at 894.6 nm 1.4 um tapers with a 0.45 um gap give 3.25 mm, and the final 1.0 um tapers have a wider 0.64 um gap (femwel
- `results/m8_power.json`: Peak intensity from the simulated port spot along x and a Gaussian (w0 = 1.04 um) along y. The tunable router sends power only to lit ports, so per-tweezer power is independent of the number lit.
- `results/m8_resynth.json`:
- `results/m8_star_design.json`: delivered_frac: fraction of port input power reaching the atom plane through FPR, arms, emitters, slit and pupil (meta-objective transmission not included).
- `results/m8_star_multi.json`:
- `results/m8_star_steer.json`:
- `results/m8_summary.json`:
- `results/m8_taper_scan.json`: Basis for the FINAL choice: 1064.4 nm exp(-6), 1.4 um; 894.6 nm exp(-6), 1.0 um.
- `results/m9_E_independence.json`:
- `results/m9_E_power.json`: Depth equalised over the field by the tunable tree; mean factor from m9_E_independence.
- `results/m9_E_window.json`: Uncalibrated sliding Gaussian window; ripple and nonlinearity are deterministic in the window position and are removed by a lookup table (see m9_E_independence).
- `results/m9_papr.json`: Complex synthesis on the arms must scale every arm to the largest |c_n|; the efficiency is mean/max. With frequency tags (route F) the maximum is over arms and time.
- `results/m9_routes.json`:

Chapter 3

Twenty steerable, individually switchable tweezers at 1064.4 and 894.6 nm

Design study for Objective 3 (proposal ANRF/ARG/2026/009937, draft v97). Code: `picsim/opa/star.py`, `scripts/m8_twenty.py`; results `results/m8_*.json`. All simulated values are SIM outputs, valid in the regime simulated, and must be reviewed before they enter the proposal.

> Follow-up: this design moves all lit tweezers together. For independent motion of each tweezer, as with AOD tones, see `docs/independent_steering.md` (sliding-window focal-plane OPA). The switched-port design below remains the recommendation for the 894.6 nm addressing beams.

3.1 Result

Twenty tweezers that are steerable and individually switchable can be delivered on the 400 nm SiN platform by a multi-port star-coupler OPA with 128 emitters, one chip per wavelength. Each tweezer has its own input port and its own MZI switch; the ports illuminate a common 128-arm phased array through a free-propagation region (FPR); a phase ramp on the arms moves all lit tweezers together. On simulated designs, with the 20 ports placed on a row of 4 um sites:

	1064.4 nm (transport tweezers)	894.6 nm (addressing beams)
spot FWHM along the row	1.52 um	1.27 um
intensity at the 4 um neighbour / peak	1.0e-6	3.5e-6 (neighbour error 8.5e-6)
peak spread, all 20 lit, random port phases	<= 0.73%	<= 0.89%
stray at unlit sites (10 of 20 lit)	<= 1.2e-5	<= 2.1e-5
static position error of a spot	<= 6 nm	<= 5 nm
switch per port	one trimmed thermo-optic MZI	two cascaded AlN MZIs
extinction needed	>= 31 dB (1 uK residual at 1.2 mK)	>= 50 dB (idle error 2.5e-5)
fibre power	33-43 mW per tweezer at 1.2 mK; 0.74 W for 20	<= 0.14 mW per site; 2.4 mW for 20

The price is that steering is collective: all lit tweezers move by the same displacement. Which sites carry tweezers is set by which ports are lit. This is the primitive an AOD provides for parallel rearrangement (select a subset of a row, move it), but at one optical frequency, so two tweezers brought together do not beat. Fully independent steering of 20 tweezers through one aperture requires phase-only synthesis on a shared OPA, and that architecture is a poor on/off engine (Sec. 4).

3.2 Architecture

One chip per wavelength, both on the shared 400 nm LPCVD SiN process:

(i) edge-coupled input (two fibres at 1064.4 nm, to keep each facet below 20 MW/cm^2); (ii) a 1x32 tunable-splitter tree (five stages of thermo-optic MZIs with 3 um 2x2 MMIs), which sends power only to lit ports, so the power per tweezer does not depend on how many are lit; (iii) 20 port switches, thermo-optic at 1064.4 nm (tweezer on/off ramps of >= 100 us), two cascaded AlN MZIs at 894.6 nm (gate pulses, ZVD-shaped drive, <= 100 ns); (iv) 20 input tapers, 1.4 um (1064.4 nm) or 1.0 um (894.6 nm) wide, spaced 1.89 um and 1.64 um at the FPR; (v) an FPR of radius 223 um (1064.4 nm) or 197 um (894.6 nm); (vi) 128 arms, 1.7 um wide at the FPR edge (0.3 um gaps), each with a phase shifter, routed to (vii) 128 end-fire emitters at 1.5 um pitch; (viii) the relay of Sec. 6 of `docs/findings.md`: cylindrical collimation, Fourier-plane slit, anamorphic telescope, meta-objective (NA 0.5).

The two chips are combined before the meta-objective by a dichroic. Their port geometries map to the same 4 um sites; the residual registration is set by the common phase ramp of each chip.

Each port illuminates all 128 arms with the same envelope, the far field of its taper mode, which acts as a Gaussian-like apodisation of the array. Port j imposes a linear phase k_s d_j p_a / R per arm (k_s, the slab wavenumber; d_j, port offset; p_a, arm pitch), which steers its beam to sin(theta_j) = n_s d_j p_a / (R p_out). The port pitch that places beams 4 um apart at the atoms follows from the relay scale x = sin(theta) N p_out / (2 NA): a 4 um step is sin(theta) = 0.0208 for N = 128 and p_out = 1.5 um.

3.3 Design choices and trade-offs

Apodisation. The arm-power envelope sets the spot tails, and the tails set two things: the light that a lit tweezer leaves on its neighbours, and the interference between lit ports, whose relative phases drift with the router temperature. With the envelope truncated at 1/e^2 at the outer arms (as for a standard star coupler), the neighbour intensity is 1.2e-3 and, with all 20 lit and random port phases, peak intensities vary by up to 19%. Narrowing the envelope removes both at the cost of a wider spot:

envelope at outer arm (nominal)	FWHM at 1064.4 nm	neighbour / peak	peak spread, 20 lit
e^-2	1.10 um	1.2e-3	19%
e^-4	1.24 um	4.2e-4	11%
e^-6	1.38 um	9.6e-5	5.8%
e^-8	1.52 um	1.3e-5	2.0%

(Input taper 0.6 um.) A narrower envelope needs a smaller FPR radius, which brings the ports closer together (1.0 um at e^-8). Widening the input taper restores the radius: with 1.4 um tapers the port pitch is 1.89 um, and the taper mode, being closer to Gaussian than the 0.6 um guide mode, also lowers the tails further. Two 1.4 um tapers 0.49 um apart have a coupling length of 1.33 mm at 1064.4 nm, so the ~20 um over which they converge on the FPR transfers well below -20 dB. The final designs were chosen from a scan of taper width and envelope (`results/m8_taper_scan.json`): 1.4 um tapers at 1064.4 nm, 1.0 um tapers at 894.6 nm, both at a nominal e^-6 envelope. Because the wide-taper modes are not Gaussian, the realised envelope at the outer arm is 2.7e-5 (1064.4 nm) and 7.2e-6 (894.6 nm), and a nominally narrower envelope (e^-8) puts the outer arms on the side lobes of the port far field (realised 1e-3 to 2.5e-3, worse tails). The designs are therefore specified by FPR radius and taper width, not by the nominal edge level.

We keep the cleaner design at 1064.4 nm although a wider envelope (e^-5) gives 9% more depth per watt, because its 3% peak spread and 4e-5 neighbour light would load the static atoms and make tweezer depths depend on drifting port phases.

Emitter pitch. 1.5 um, the smallest pitch at which a 50 um straight end-fire section keeps emitter-to-emitter transfer below -20 dB (`m3_pitch_efficiency`). It raises the slit efficiency to 0.65 from 0.52 at 2 um.

Channel count. With N = 96 the steering half-range is 51 um (1064.4 nm) or 43 um (894.6 nm), so a row of 20 sites (+/-38 um) leaves only 5 to 13 um for collective moves; N = 128 gives +/-68 um and +/-57 um.

3.4 Why not phase synthesis on one OPA

On a shared 128-channel OPA, 20 tweezers can be computed with independent positions (`m3_multibeam`), but a tweezer is switched off only by re-synthesis. Removing one of 20 (warm-started refinement with that site among the dark points) leaves 1.9 to 3.3% of the tweezer intensity at its site (-15 to -17 dB), changes the

19 kept tweezers by up to 12%, and along a linear phase path between the two patterns the kept tweezers deviate by up to 19%. Stray light at untargeted sites stays at about 2%. Hence it cannot gate 894.6 nm light at 50 dB, and it perturbs the other atoms whenever one tweezer is switched. It remains the route to fully independent steering at 1064.4 nm if that is ever required.

3.5 Steering

All lit tweezers move by the displacement set by a linear phase ramp across the 128 arms. With all 20 lit (peak change quoted at 1064.4 nm; at 894.6 nm it is within 0.6% of these values):

collective shift	position error (static, max)	peak change (min / max)
4 um	25 nm (1064.4), 5 nm (894.6)	-2.5% / +2.3%
8 um	25 nm, 5 nm	-5.1% / +4.3%
16 um	25 nm, 6 nm	-10.7% / +7.8%
24 um	26 nm (1064.4 only)	-16.7% / +10.3%

The position error is a static offset of the outer spots, caused by the element-factor slope across them; it changes by less than 2 nm over a 24 um sweep and is calibrated out. The peak change follows the emitter element factor and can be pre-compensated through the port powers. At 894.6 nm the outer spots leave the +/-57 um field beyond a 19 um shift; subsets of the row can move farther.

Because the collective move is one linear ramp, its fast part needs a single waveform: the 128 arm shifters take fixed gains n = 0 ... 127 from one analog ramp, with slow per-arm offsets for calibration. A continuous analog ramp has no staircase, which removes the update-rate heating of `m3_update_rate` (40 quanta per 100 us move at 100 kHz and 1.2 mK). The shifters must be fast and low-power: PZT on SiN (0.4 MHz, V_pi about 5.6 V at 1064.4 nm by wavelength scaling of [MONT]) rather than thermo-optic (34 us rise).

3.6 Switching

At 1064.4 nm a tweezer is ramped on or off adiabatically over >= 100 us, so a thermo-optic MZI (34 us rise) suffices; >= 31 dB keeps the residual below 1 uK at 1.2 mK, which a trimmed single MZI exceeds at phase errors up to 10 mrad (5th percentile 38 dB, `m4_extinction`; computed at 894.6 nm, to be repeated at 1064.4 nm). At 894.6 nm the switch gates Raman pulses. The proposal's 50 dB gives an idle-qubit error of 2.5e-5; the switch leak equals the 3.5e-6 optical tail of a lit neighbour at 54.6 dB. Two cascaded AlN MZIs give a 5th percentile of 66 to 88 dB, so the switch is not the limit; ZVD-shaped drive reaches -50 dB within 49 ns for a 2% error in the actuator resonance (`m4_dynamics`, with an assumed resonance).

3.7 Power

At 1064.4 nm, per tweezer, from the fibre: facet -0.37 dB, router -0.28 dB, switch -0.06 dB, propagation (1 cm, 0.3 dB/cm assumed) -0.30 dB, FPR, arms, emitters, slit and pupil 0.40 to 0.52 of the port power depending on the port, meta-objective 0.7 (assumed): -6.3 to -7.4 dB to the atom plane. For a 1.2 mK depth (1160 a.u., Gaussian 1.04 um along the cross axis) this needs 33 to 43 mW at the fibre per tweezer, 0.74 W for 20, against 1.4 W in the proposal. At 894.6 nm, 20 sites at the intensity of 20 uW in a 1.04 um Gaussian need 2.4 mW in total.

3.8 Implementation load

item	1064.4 nm chip	894.6 nm chip
router MZIs (thermo-optic)	31	31
port switches	20 thermo-optic	40 AlN (two per port, high voltage)
arm phase shifters	128 PZT (one fast ramp + slow offsets)	128 (slow; registration only)
FPR radius	223 um	197 um

item	1064.4 nm chip	894.6 nm chip
electrical channels	about 180	about 200

About 380 channels across the two chips exceed the 32-64-channel OPA and 16-channel router of the proposal, and the 32-channel high-voltage driver of Sec. 8 covers only the 894.6 nm switches if one AlN MZI per port is replaced by a trimmed thermo-optic stage (a trimmed single MZI reaches 50 dB in 89% of devices at 3 mrad). Thermo-optic power, at half of P_pi on average, is about 42 mW per heater without trenches (`m5_heaters`), i.e. about 2.1 W for 51 heaters on the 1064.4 nm chip; trenches with undercut bring this to about 0.1 W.

3.9 What this does not provide, and limits of the model

It does not move different tweezers by different amounts in one step; a sorting sequence is built from parallel moves, as with AODs. It steers along one row; the second axis needs a second chip, a row-select element, or an OPA-AOD hybrid (`docs/findings.md`, Sec. 6). The FPR model is 2D and paraxial (flat arcs, no Rowland aberration, isolated arm modes), and the arm phases from fabrication are assumed calibrated. The relay is scalar and aberration-free; a meta-objective with a Strehl of 0.8 adds a halo that is not included in the crosstalk figures.

3.10 Suggested wording for the proposal

Hypothesis H3b: "An OPA fed through a multi-port star coupler forms up to 20 tweezers, each switched on and off by its own integrated MZI, and moves all lit tweezers together by setting a single phase ramp across its 128 emitters. All tweezers share one optical frequency, hence ... ." Table 1, OPA row: "simultaneous, individually switchable tweezers >= 20 on one row; collective move >= +/-16 um with all 20 lit; neighbour intensity <= 1e-4 at 894.6 nm".

3.11 Verification notes

Sources read in session: as listed in `docs/findings.md` ([MONT] arXiv:2601.15695 for the PZT figures). No new literature values were introduced in this study; all new numbers are SIM or DERIVED.

value	location	type	pointer
FWHM, neighbour, spread, stray, position error, delivery	Sec. 1, 3, 5	SIM	results/m8_final.json (review before submission)
apodisation table	Sec. 3	SIM	results/m8_apod.json (review)
taper-width scan, realised envelope 2.7e-5 / 7.2e-6	Sec. 3	SIM	results/m8_taper_scan.json, m8_final.json (review)
coupling length 1.33 mm (0.49 um gap, 1.4 um tapers)	Sec. 3	SIM	femwell, run in session; noted in m8_final.json
re-synthesis residual 1.9-3.3%, 12%, 19%	Sec. 4	SIM	results/m8_resynth.json (review)
collective-shift table	Sec. 5	SIM	results/m8_final.json, steer (review)
33-43 mW, 0.74 W, 2.4 mW	Sec. 1, 7	SIM/DERIVED	results/m8_final.json, power (review)
31 dB, 50 dB, 54.6 dB	Sec. 6	DERIVED	results/m8_final.json, switching
emitter pitch 1.5 um; arm pitch 2.0 um; arm width 1.7 um; propagation 0.3 dB/cm; meta-objective 0.7	Sec. 2, 7	CHOICE	stated assumptions
1.2 mK, 1160 a.u., 1.04 um, 4 um sites, NA 0.5, 20 uW per site	throughout	USER	proposal draft v97

Open placeholders: [VALUE?] PZT shifter footprint and hysteresis for 128 arms; [VALUE?] AlN resonance and Q at 894.6 nm; [VALUE?] measured arm phase errors after fabrication (calibration range).

Chapter 4

Independently steerable tweezers: an integrated equivalent of the AOD

Design study for Objective 3 (proposal ANRF/ARG/2026/009937, draft v97). Code: `scripts/m9_independent.py`, `picsim/opa/star.py`; results `results/m9_*.json`. Simulated values are SIM outputs, valid in the regime simulated, and must be reviewed before they enter the proposal.

> Follow-up: the star coupler has since been designed by beam propagation and optimised (`docs/fpr_design.md`). The optimised design (192 arms, Rowland-type port arc, 1.2 um emitter facets, least-squares port synthesis) needs about 43 mW per 1.2 mK tweezer (0.86 W for 20) and keeps the independence figures below. The numbers in this note are for the first, paraxial design.

4.1 Result

A sliding-window focal-plane OPA moves 20 tweezers independently along a row, switches each on and off on its own, and does so at one optical frequency. A dense row of 128 input ports, one per resolvable spot, feeds a star coupler, 128 arms and 128 emitters; port k forms a spot at x_k = k lambda/(2 NA) = 1.064 um at the atoms. A tweezer is a Gaussian-weighted group of about five ports, and it moves by sliding the window. In this (position) basis different tweezers use different ports, so their fields add linearly and do not interfere except through their tails. With 20 tweezers at random positions over +/-60 um (window width sigma = 0.9 ports, calibrated):

quantity	value
moved tweezer, 0.1 to 4 um moves: position error / peak change	<= 0.9 nm / <= 0.10%
the other 19 during the move: position / peak change	<= 1.5 nm / <= 0.19%
tweezer switched off: residual at its site / change of the others	8.9e-6 / <= 0.08%
static peak spread of the 20 (depth equalised by the tree)	1.4%
two tweezers in phase quadrature at 2.5 um: pull / peak change	3.5 nm / 0.16%
spot FWHM along the row	1.66 um
fibre power per tweezer at 1.2 mK: centre / field mean / field edge	48 / 64 / 102 mW
20 tweezers spread over the field	1.3 W

The cost is in actuators and their accuracy: 127 fast tunable splitters and 128 port phase shifters per chip, and 10 nm position noise needs per-port amplitude and phase errors of about 1% and 10 mrad rms (Sec. 4).

4.2 What makes AOD tones independent

Two physical ingredients, and the integrated device must supply an equivalent of at least one (standard AOD physics; [REF?] for a citable source):

(i) frequency tagging: each RF tone shifts its diffracted light by its own frequency, so tweezers from different tones add incoherently on average, and each tone's amplitude sets one tweezer's depth; (ii) a time-delay medium: the acoustic transit time across the aperture converts each tone's frequency into its own angle; the number of resolvable spots is the RF bandwidth times the transit time.

The independence is per axis. A single AOD moves its spots independently along one line; crossed AODs give a grid of row and column tones, not independent 2D positions [17]. Hence the fair target here is a 1D AOD along a row. Our earlier shared-arm OPA (`docs/twenty_tweezers.md`) has neither ingredient: every tweezer uses every arm, so phase-only synthesis couples them all.

4.3 Routes considered

route	how independence arises	efficiency for 20	show-stopper or cost
E sliding-window focal-plane OPA	tweezers on separate ports (position basis), linear superposition	0.40 at centre, 0.19 at field edge (as the star coupler of m8)	127 fast splitters + 128 phases; ~1% per-port accuracy
A' complex synthesis on the arms	linear superposition of 20 phase ramps (amplitude and phase per arm)	0.75 (peak-to-mean 1.34, phases optimised)	128 amplitude and phase modulators; moving one tweezer changes the optimum, so power headroom is needed
F AOD emulation on the arms (one RF tone per tweezer)	frequency tags + per-tone phase ramp	0.44 (peak-to-mean 2.28 over arms and time)	256 RF-driven modulators; modulator nonlinearity makes intermodulation ghosts, as in an AOD
D wavelength-addressed dispersive OPA (one laser per tweezer)	optical frequency tags + delay lines	high (passive chip)	a diffractive meta-objective shifts focus by 14 um/nm, so only +/-71 pm (+/-19 GHz) is allowed; even 5 GHz per site then needs 6.7 m of delay lines to cover the field
G integrated surface-acoustic-wave deflector in the slab	a literal AOD	unknown (photoelastic response of SiN not established)	128 spots need 640 MHz of RF bandwidth on a 1 mm aperture, 128 MHz on 5 mm (v_SAW ~5 km/s, recalled)
T time-multiplexed OPA (one tweezer at a time)	time-averaged potentials	1 per slot	17 MHz pattern rate, 60 ns slots, 20x peak intensity

Route E is the only one that is single-frequency, uses the components already in the proposal (SiN star coupler, MZI trees, PZT or AlN shifters), and needs no new physics. A' is a useful fallback with fewer optical elements but needs amplitude modulation on every arm and power headroom. D is excluded by the chromatic focal shift unless the meta-objective is made achromatic over several nanometres around 1064 nm.

4.4 Route E: design and simulated performance

Geometry. 128 input ports, 1.7 um wide at 2.0 um pitch on the input arc; FPR radius R = N p_a p_port / lambda_s = 886 um, which makes one port per resolvable direction; 128 arms (1.7 um wide, 2.0 um pitch at the FPR); 128 end-fire emitters at 1.5 um pitch; the slit-filtered relay and NA 0.5 meta-objective of `docs/findings.md`. Adjacent ports at 0.3 um gaps have a coupling length of 625 um, so the ~20 um over which they converge on the FPR transfers below -20 dB (40 um would still be within -20 dB).

Window width. The Gaussian window in port space sets the spot apodisation (uncalibrated values):

sigma (ports)	FWHM	position nonlinearity	peak ripple	neighbour / peak at 4 um
0.8	1.51 um	18 nm	3.4%	1.1e-4
0.9	1.66 um	8.6 nm	1.5%	2.0e-5
1.0	1.82 um	3.7 nm	0.55%	2e-6
1.2	2.16 um	0.5 nm	0.07%	7.6e-5

The nonlinearity and ripple are deterministic functions of the window position and are removed by a lookup table, together with the fall-off of delivered power from 0.40 at the centre to 0.19 at +/-64 um. We use sigma = 0.9 as a compromise between spot size (power per depth) and the residual that calibration must remove.

Independence. Moving one of 20 tweezers by up to 4 um changes the other 19 by at most 1.5 nm and 0.19%; removing its window leaves 8.9e-6 at its site. Two tweezers approach without interference if their

relative phase is set to pi/2, since the cross term of two real fields then vanishes: at 2.5 um the pull is 3.5 nm and the peak change 0.16%, against 146 nm and 9.5% in phase. This is a control an AOD lacks, where approaching tones beat at their frequency difference.

Actuator accuracy. Each tweezer is set by about five ports, so per-port errors average little:

per-port amplitude / phase error (rms)	position noise	peak noise
0.3% / 3 mrad	3.1 nm	0.5%
1% / 10 mrad	9.4 nm	1.9%
3% / 30 mrad	28 nm	5.6%

This is about 15 times tighter than the arm-side OPA (0.14 rad for 10 nm at N = 64, `m3_noise`). PZT hysteresis must therefore be corrected, for example by closed-loop calibration against on-chip taps.

Speed. A moving tweezer is updated through the tree. The staircase result of `m3_update_rate` applies:

> = 200 kHz updates at 1.2 mK. PZT shifters (0.4 MHz [MONT]) meet this; thermo-optic shifters (34 us rise) do not. The node settings follow in closed form from the desired port amplitudes (each node splits in the ratio of the powers of its two subtrees), so an FPGA can compute them in real time.

Power. Facet -0.37 dB, 7-stage tree -0.39 dB, propagation -0.3 dB (assumed), meta-objective 0.7 (assumed), and the simulated FPR, arms, emitters, slit and pupil. A 1.2 mK tweezer needs 48 mW at the fibre at the field centre and 102 mW at the edge; equalising depth over the field costs a mean factor of 1.34, so 20 tweezers spread over +/-60 um need about 1.3 W. Restricting operation to +/-40 um lowers the mean factor.

On/off. A tweezer is ramped on or off by scaling its window through the tree (adiabatic, >= 100 us), with the residual set by the tree extinction; the simulated residual with the window removed is 8.9e-6.

4.5 Where this leaves the 894.6 nm addressing beams

Raman gates act on atoms held in static traps, and the addressed Raman leg must keep a fixed frequency relative to the global leg, so frequency tagging is not available at 894.6 nm. The switched-port star coupler of `docs/twenty_tweezers.md` (20 ports, two cascaded AlN MZIs each, neighbour intensity 3.5e-6) remains the recommended addressing engine. Route E at 894.6 nm would give steerable addressing, but its off-state would be set by the extinction of the tree nodes rather than by a cascaded switch per beam.

4.6 Limits and next steps

The FPR model is 2D and paraxial with isolated port and arm modes; dense ports at 0.3 um gaps couple, so the port array and FPR need EME or 3D beam-propagation design, as for AWG star couplers. The relay is scalar and aberration-free. Steering is along one row; the second axis needs a second engine or an OPA-AOD hybrid. Next steps: (i) EME/BPM design of the dense-port FPR; (ii) a closed-loop calibration scheme for 255 actuators; (iii) a rearrangement-sequence simulation (atom survival per move, including the actuator noise spectrum and the >= 200 kHz update).

4.7 Suggested wording for the proposal

Hypothesis H3b: "A focal-plane OPA, in which each tweezer is a sliding window over a dense array of input ports, moves 20 tweezers independently along a row and switches each on and off, as the tones of an AOD do, but at one optical frequency, so that tweezers can approach within 2.5 um without beating or intermodulation." Table 1, OPA row: "independently steerable, individually switchable tweezers >= 20 on one row; update rate

> = 200 kHz; position noise <= 10 nm rms".

4.8 Verification notes

No new literature values were used except the recalled surface-acoustic-wave velocity (route G, order of magnitude only). [17] is the proposal's reference for crossed AODs; [MONT] is arXiv:2601.15695 (read in session). The AOD physics in Sec. 2 is stated without a citation: [REF?].

value	location	type	pointer
independence, switch-off, approach, spread	Sec. 1, 4	SIM	results/m9_E_independence.json (review before submission)
window table, field fall-off 0.40 to 0.19	Sec. 4	SIM	results/m9_E_window.json (review)
noise table	Sec. 4	SIM	results/m9_E_independence.json, amplitude_phase_noise (review)
48 / 64 / 102 mW, 1.3 W	Sec. 1, 4	SIM/DERIVED	results/m9_E_power.json (review)
R = 886 um, 1.064 um per port	Sec. 4	DERIVED	m9_E_window.json; tests/test_star.py
coupling length 625 um (1.7 um ports, 0.3 um gap)	Sec. 4	SIM	femwell, run in session (not stored)
A' 0.75, F 0.44	Sec. 3	SIM	results/m9_papr.json (review)
D: 14 um/nm, +/-71 pm, 6.7 m; G bandwidths; T rates	Sec. 3	DERIVED	results/m9_routes.json
v_SAW ~5 km/s	Sec. 3	RECALLED	verify for an AlN/SiN/SiO2 stack
port pitch 2.0 um, port width 1.7 um, sigma 0.9, propagation 0.3 dB/cm, meta-objective 0.7	Sec. 4	CHOICE	stated assumptions

Open placeholders: [REF?] AOD multi-tone independence and intermodulation; [VALUE?] PZT hysteresis and long-term drift after closed-loop calibration; [VALUE?] dense-port FPR excess loss from EME/BPM.

Chapter 5

Focal-plane OPA: beam-propagation design and geometry optimisation

> Follow-up: the power figures below omit the cross axis and the emitter facet loss (about 21% into air). With both included and a collimator glued to the facets, a 1.2 mK tweezer needs about 57 mW instead of 43 mW (`docs/stacked_relay.md`, `docs/collimator.md`).

Follow-up to `docs/independent_steering.md` (route E, sliding-window focal-plane OPA). Code: `picsim/bpm.py`, `picsim/opa/fpr_layout.py`, `fpr_bpm.py`, `star_rs.py`, `focal_plane.py`; scripts `m10_fpr.py`, `m10_fpr_meep.py`, `m11_optimise.py`; results `results/m10_*`, `results/m11_*`. Simulated values are valid in the regime simulated and should be reviewed before they are quoted. Design choices below are made on physical and fabrication grounds, not to match the original proposal's numbers.

5.1 Result

A dense-port star coupler designed by wide-angle beam propagation, with a Rowland-type port arc, 1.5x oversampled arms and wide emitter facets, gives a focal-plane OPA in which 20 tweezers move and switch independently along a row, at 1.5 times the power efficiency of the first design. Verified by beam propagation of the full 128-port, 192-arm structure (calibrated effective-index model, least-squares port synthesis):

quantity	first design (128 x 128, confocal)	optimised design D1
field-averaged peak per unit port power (relative)	1.00	1.50
edge (60 um) / centre peak	0.52	0.70
laser power per 1.2 mK tweezer, field mean (fibre)	~65 mW	~43 mW
20 tweezers over +/-60 um	~1.3 W	~0.86 W
residual arm-phase aberration, worst port	up to 1.3 rad pk-pk	<= 0.08 rad rms
others' change when one of 20 moves by up to 4 um	<= 3.5 nm, <= 0.12%	<= 3.5 nm, <= 0.11%
residual at a switched-off site	1.3e-6	1.6e-6
static peak spread of 20 tweezers	0.56%	0.52%
spot FWHM along the row	1.71 um	1.71 um

Laser power uses the loss chain of `m9_E_power` (facet -0.37 dB, 7-stage tunable tree -0.39 dB, 1 cm at 0.3 dB/cm, meta-objective transmission 0.7 assumed) and a Gaussian 1.04 um profile across the row.

Design D1. 128 input ports and 192 arms, both 1.7 um wide at 2.0 um pitch at the slab (0.3 um gaps); slab radius R = 1329 um; ports on an arc of radius 0.555 R through the slab vertex, pointing at the arm-array centre; arms on an arc of radius R; 30 um tapers to 0.6 um guides; 192 end-fire emitters at 1.5 um pitch with 1.2 um facets (facet taper <= 15 um); static phase trims on all arms and ports; port amplitudes from least-squares synthesis on the calibrated response.

5.2 Method and its validation

Vertical dimension. The 400 nm film is uniform and fully etched, so every guide and the slab share one vertical mode, and the lateral effective-index model carries the star-coupler physics (standard AWG practice). The plain effective-index model overestimates lateral confinement: its coupling lengths are 31 to 36% short of the 3D vector-FEM values. We calibrate the lateral core and cladding indices (1.8423, 1.3551) to vector-FEM effective indices of 0.6, 1.0 and 1.7 um guides (residuals <= 4.3e-4) and to the coupling length of the dense ports (1.7 um, 0.3 um gap: 622 vs 625 um). Held-out checks: 1.4 um guides at 0.49 um gap, 1373 vs 1330 um (+3%); 0.6 um guides at 1.0 um gap, 2287 vs 1777 um (+29%; narrow, widely spaced guides couple negligibly in the design, so this residual has little effect). A second calibration that fits both coupling lengths degrades the dense-port value by 10% and is kept as a bound.

Propagation. 2D wide-angle (Pade 1,1) Crank-Nicolson BPM with the TM-type lateral operator that matches the chip TE mode; all ports propagated together. Unit tests: power conserved to 1e-3 over 300 um; phase index error < 3e-4; half a coupling length transfers half the power. Grid: dx = 0.04 um, dz = 0.125 um (capture within 0.5% and arm-field fidelity >= 0.992 against dz = 0.0625 um).

Against 2D FDTD. On a reduced 8-port, 8-arm coupler with the same angular range, MEEP (resolution 20/um) and the BPM agree with arm-field fidelity 0.997 (centre port), 0.985 (middle) and 0.93 (edge port), and capture within 2 to 5%. At FDTD resolution 30/um the edge-port fidelity rises to 0.981 (rms phase difference 0.13 rad, capture 0.369 vs 0.371), and the FDTD result itself changes by fidelity 0.98 between the two resolutions; hence the residual difference is dominated by FDTD discretisation rather than by the BPM.

Exact-path (Huygens) model. For fast optimisation, a 2D Rayleigh-Sommerfeld model with exact path lengths (`star_rs.py`) reproduces the full 128 x 128 BPM to within 1% in capture and with fidelity 0.996 to 0.9996 after arm-phase calibration (0.96 at the two extreme ports, where taper coupling, absent from this model, matters). On the optimised design the BPM and the exact-path model give the same field-averaged efficiency to 0.6%.

5.3 What the beam propagation found

(i) **Aberration of the confocal coupler.** With ports on an arc of radius R, each off-axis port's arm phases deviate from a straight line by up to 1.3 rad pk-pk at the edge port (0.43 rad half-way out), confirmed by exact path lengths. Arm trims cannot remove it because it depends on the port. Placing the ports on an arc of radius 0.555 R (close to the Rowland circle) halves it over the field of use; with 192 arms the worst residual is 0.08 rad rms. The Rowland-type arc gives each port a different piston phase, which the static port trims of route E remove (without them, adjacent ports in a window do not add in phase).

(ii) **Coupling between dense ports.** In the converging 1.7 um tapers (0.3 um gaps) neighbouring ports exchange light. With fixed Gaussian windows this broadens the spot tails, and moving one of 20 tweezers then changes the others by up to 7 nm and 1.1%. All of this is linear and static: least-squares synthesis on the calibrated port-to-atom response removes it (<= 3.5 nm, <= 0.12%; switch-off residual 1.3e-6) at 1 to 2% cost in efficiency. Synthesis also equalises depth over the field without a lookup. It is one matrix product per update (precomputed 128 x n matrix).

5.4 Geometry optimisation

Figure of merit: the field-averaged laser power per tweezer for equal depth, i.e. the harmonic mean of the peak intensity per unit port power over |x| <= 60 um. Constraints: neighbour intensity at 4 um <= 1e-4; gaps

= 0.3 um; 128 ports (fast actuators) fixed.

The roll-off across the field has one physical origin on both sides of the chip: scan loss. The arms capture tilted light from off-axis ports, and the emitters radiate tilted beams, each with an element pattern that falls across its angular range. Two levers act on it:

arms (128 ports)	facet 0.6 um: gain / edge ratio	facet 0.9 um	facet 1.2 um
128	1.01 / 0.53	1.10 / 0.51	1.23 / 0.46
160	1.12 / 0.67	1.24 / 0.65	1.41 / 0.61
192	1.18 / 0.76	1.31 / 0.74	1.51 / 0.71
256	1.24 / 0.86	1.39 / 0.85	1.61 / 0.83

(Gain relative to the confocal 128 x 128 design with 0.6 um facets; exact-path model, Rowland-type arc.)

Oversampling (more arms and emitters than ports) confines the ports to the central part of the arms' angular range and raises the edge capture from 0.44 to 0.54 (192 arms). Arms and emitters carry only static trims in route E, so this costs area and slow heaters, not fast actuators. 192 arms take most of the benefit; 256 add 7% for 64 more trims and a 1.8 mm slab.

Wider emitter facets narrow the element pattern and raise the relay transmission at the centre from 0.64 (0.6 um) to 0.84 (1.2 um at 1.5 um pitch). At 0.5 um gaps the transmission saturates near 0.74 to 0.76 for any pitch from 1.5 to 2.2 um, so the 0.3 um gap of the 1.2 um facet is what buys the last 10%. Its coupling length is 227 um, which limits the wide facet section to about 15 um.

Slab widths. The widest arms and ports (1.7 um at 2.0 um pitch) win: field-averaged efficiency 0.261 against 0.18 to 0.24 for 1.2 to 1.5 um, because fill factor outweighs the slightly better angular acceptance of narrow arms. The 0.3 um gaps are safe at the slab: the coupling length is 622 um and the tapers converge over about 20 um.

Window width. Narrowing the Gaussian window from 0.9 to 0.85 ports gains 5% but raises the neighbour intensity to 4.8e-5; least-squares synthesis with a 1.68 um target is the adopted compromise.

5.5 Limits and open points

The model is 2D with a calibrated effective index; the residual 3D effects are polarisation coupling at sidewalls and slab-edge scattering. Fabrication variation of the 0.3 um gaps and of the taper geometry has not been simulated; static errors are absorbed by the trims and the measured response matrix, but the 1% actuator accuracy for 10 nm position noise (`m10_route_E_LS_*`, noise) remains. The relay and objective are scalar and aberration-free. Steering is along one row.

Next steps: (i) tolerance analysis of gap and width errors in the dense tapers; (ii) 3D FDTD of a few port-taper-slab cells for polarisation and scattering; (iii) GDS layout of D1 with its 255 fast actuators (127 tree splitters, 128 port phase shifters, which also carry the port trims) and 192 static arm trims; (iv) the calibration procedure that measures the port-to-atom response on the device.

5.6 Verification notes

All numbers are SIM outputs of this repository except the loss-chain assumptions marked above (CHOICE).

value	location	type	pointer
calibrated indices, coupling lengths	Sec. 2	SIM	results/m10_eim_calibration.json
BPM convergence, BPM vs FDTD	Sec. 2	SIM	results/m10_bpm_small.json, m10_bpm_vs_fdtd.json
confocal 128 x 128 capture and aberration	Sec. 3	SIM	results/m10_fpr_full_A.json
optimised D1 capture and aberration	Sec. 1	SIM	results/m10_fpr_opt_A.json
window and least-squares evaluations	Sec. 1, 3	SIM	results/m10_route_E_{full,opt}*A.json, m10_route_E_LS*{full,opt}_A.json
optimisation tables	Sec. 4	SIM	results/m11_stage1.json, m11_stage2.json
power per tweezer	Sec. 1	DERIVED	m9_E_power.json scaled by the field-averaged peak per unit power

Chapter 6

How far the focal-plane OPA scales

Follow-up to `docs/fpr_design.md`. Code: `scripts/m12_scaling.py`; results `results/m12_*.json`. Packing is simulated on the beam-propagation matrix of design D1; aberration with exact path lengths; the rest of the scaling table is derived from those results. Values are SIM or DERIVED and valid in the regime stated.

6.1 Result

One focal-plane engine moves about 40 independent tweezers along a row at its present size (128 ports), and about 60 when the row is matched to a +/-100 um objective field (192 ports). The packing limit is a spacing of 3 um, i.e. 2.8 resolution cells or 1.75 spot widths. Below that the spots merge. Power per tweezer does not grow with the number of tweezers. A second dimension needs one engine per row: stacking chips with fast-axis micro-lenses, as in laser-diode stacks, gives an estimated 8 to 16 rows. Hence several hundred independent tweezers (16 rows x 40 to 60) are a realistic target, against a median of 13 (maximum 38) moved in parallel with crossed AODs in the 1,024-atom assembly cited in the proposal.

The binding limits, in order: (i) the diffraction-limited field of the objective, which sets the ports per row; (ii) the number, speed and accuracy of fast actuators (2 per port, >= 200 kHz, ~1% and 10 mrad); (iii) laser power through each chip; (iv) slab size. Star-coupler aberration is not a limit below about 512 ports.

6.2 Tweezers per row: packing

A +/-60 um row of design D1 filled at spacing s, neighbours in phase quadrature, least-squares synthesis:

spacing	tweezers	intensity midway between neighbours	others' change when one moves 0.25 um	residual after switch-off
6 um	21	5e-4	0.11%, 1.6 nm	< 1e-5
4 um	31	3.8%	0.10%, 2.6 nm	1e-5
3 um	41	22%	0.11%, 3.3 nm	9e-4
2.5 um	49	43%	0.79%, 10 nm	9e-3
2 um	61	76% (spots merge, FWHM 3 um)	5.2%, 74 nm	7e-2

The peak per tweezer per unit port power stays at 0.256 from 6 to 2.5 um spacing, so power per tweezer is independent of how many are lit. At 3 um the barrier between neighbours is 78% of the trap depth, which keeps the atoms separate; at 2.5 um independence degrades to the 1% level. We take 3 um as the practical limit. It is set by the spot width (1.71 um FWHM), hence by NA: at fixed NA, a narrower spot needs a less apodised window, which raises the tails.

6.3 Ports per row: scaling of one engine

With 1.5x oversampled arms, 2.0 um slab pitch and 1.5 um emitter pitch:

ports	row field	tweezers at 4 um / 3 um	slab radius	slab area	fast actuators	static trims	data rate (200 kHz, 16 bit)	worst aberration (rms)	full row at 4 um, 1.2 mK / 0.5 mK
128	136 um	30 / 40	1.3 mm	0.5 mm^2	255	192	0.8 Gbit/s	0.03 rad	1.3 W / 0.54 W
192	204 um	45 / 60	2.0 mm	1.1 mm^2	383	288	1.2 Gbit/s	0.05 rad	1.9 W / 0.81 W
256	272 um	60 / 80	2.7 mm	2.0 mm^2	511	384	1.6 Gbit/s	0.07 rad	2.6 W / 1.1 W
512	545 um	120 / 160	5.3 mm	8.2 mm^2	1023	768	3.3 Gbit/s	0.13 rad	5.2 W / 2.2 W
1024	1090 um	240 / 320	10.6 mm	33 mm^2	2047	1536	6.6 Gbit/s	0.27 rad	10.4 W / 4.3 W

(Row field used: 88% of the port field. Power: 43 mW per 1.2 mK tweezer, the D1 value, assumed to hold at larger N.)

Aberration. With the arm envelope produced by the synthesis, the worst-port residual grows linearly with size, about 0.033 rad x (ports/128), on the Rowland circle (port arc radius R/2). A quartic correction of the port curve gains nothing, so the remainder comes from the circular arm arc; an aspheric arm curve is the next step beyond ~512 ports. These geometric values are about half of the BPM figure for D1 (0.079 rad), so they are lower bounds. Even doubled, the Strehl ratio stays above 0.93 up to 512 ports.

6.4 What sets the limit, and what pushing it requires

(i) **Objective field.** Each port adds one resolution cell (1.06 um at NA 0.5) to the row, so the row can only be as long as the objective's diffraction-limited field. The +/-100 um target corresponds to about 192 ports and 60 tweezers per row. Going further needs a larger corrected field at NA 0.5, which is the hardest optical requirement in the system.

(ii) **Fast actuators.** Two per port: 383 for a 192-port row, each updated at >= 200 kHz (the transport-heating limit at 1.2 mK) and accurate to about 1% in amplitude and 10 mrad in phase for 10 nm position noise. Beyond a few hundred channels per chip this needs drivers flip-chip bonded or co-integrated with the photonics, and closed-loop calibration (on-chip power taps or camera feedback) against PZT hysteresis and drift. This is the main engineering limit.

(iii) **Power.** About 43 mW per tweezer at 1.2 mK, 18 mW at 0.5 mK. A full 192-port row needs 1.9 W (0.8 W at 0.5 mK), which should enter through 2 to 3 fibres feeding sub-trees, to keep each facet below about 20 MW/cm^2 (`m6_power`). Lowering the transport depth is the cheapest lever: it scales power linearly and relaxes the update rate (`m3_update_rate`).

(iv) **Chip size.** The slab radius grows linearly with ports (2 mm at 192, 5 mm at 512). This is within one die but beyond a single electron-beam field, so stitching errors must be absorbed by the static trims, or the slab written by DUV lithography.

(v) **Second dimension.** One engine addresses one row. Stacking chips at 0.5 mm pitch, each with its own cylindrical collimator (f = 175 um, NA 0.9), and imaging the stack with 125x vertical demagnification gives rows 4 um apart at NA 0.5. This relay is now simulated (`docs/stacked_relay.md`). Rows are independent provided every chip runs at its own laser frequency; with one shared frequency, crosstalk of about 1e-4 between rows interferes and changes neighbouring depths by up to 2.6%. The collimators must be aligned to about 0.1 um in height. The cross axis and the emitter facet loss (`docs/collimator.md`) raise the power per tweezer from 43 mW to about 57 mW at 1.2 mK (factor 1.34) with a collimator glued to the facets; the power column of the table above does not include this factor. The practical limit is mechanical, set by stacking and alignment, and plausibly 8 to 16 rows. With 192 ports per chip, 16 rows give about 720 (4 um spacing) to 960 (3 um) independent tweezers over about 200 x 64 um, needing about 41 to 55 W at 1.2 mK, or 17 to 23 W at 0.5 mK, spread over 16 chips (one input fibre each).

Follow-up (`docs/scaling_beyond.md`): on a 4 um site lattice the useful number per row is the row length / 4 um (30 for D1, 45 for 192 ports). Transport at 0.2 to 0.5 mK with 200 to 400 us moves cuts the power 2.4 to 6 times, and a row-select tilt lets one 16-chip stack serve all 50 rows of the +/-100 um field (2,250 sites).

6.5 Next steps

(i) Done: stacked-chip relay, `docs/stacked_relay.md` (next: real collimator and relay lens design). (ii) Design an aspheric arm curve for > 512 ports. (iii) Specify the driver and calibration architecture for 383 fast channels per chip. (iv) Tolerance study of the dense port tapers (0.3 um gaps).

6.6 Verification notes

value	location	type	pointer
packing table	Sec. 2	SIM	results/m12_packing.json (BPM matrix of D1; review before quoting)
aberration vs size	Sec. 3	SIM	results/m12_aberration.json (exact path lengths)
scaling table	Sec. 3	DERIVED	results/m12_table.json
43 mW per tweezer	Sec. 3, 4	DERIVED	m10_route_E_LS_opt_A.json with the m9_E_power loss chain
crossed-AOD median 13, max 38	Sec. 1	USER	proposal, Table 1, citing its ref. [17]
stacked-chip relay (0.5 mm pitch, 125x, 8 nm per um, factor 1.10, frequency per chip)	Sec. 4	SIM/DERIVED	results/m13_*.json, docs/stacked_relay.md
8 to 16 rows	Sec. 4	CHOICE	engineering judgement (laser-diode-stack analogy), not a result

Chapter 7

Scaling beyond one row of about 40 tweezers

> Follow-up (`docs/megascale.md`): scaling to 1e4, 1e5 and 1e6 sites. A 32 x 4 engine block assembles 1e6 sites in 1.5 s and holds them at the loss-limited floor rho ~ 1.5-1.8 (eps_img + T_img/tau); static trapping power, the objective's field and the 3 um lane distance are the walls.
>
> Follow-up (`docs/transport3d.md`): the 3D simulation with handoff refines lever 1. The moving trap must be at least three times the static hold depth, the hold about 20 kT, and moves must run along lanes between the rows. The minimum moving depth is therefore 0.3 mK for 5 uK atoms (0.6 mK for 10 uK), not 0.2 mK. That gives 10 to 21 W for 720 tweezers (17 to 34 W for 1 to 2 uK heating per move).

Follow-up to `docs/scaling.md`, `docs/stacked_relay.md`, `docs/collimator.md` and `docs/row_coherence.md`. Code: `scripts/m17_scale.py`; results `results/m17_moves.json` (+ .png), `results/m17_table.json`. Transport results are SIM from a 1D classical model; the scaling table is DERIVED. They should be reviewed before they are quoted.

7.1 Result

The 41 tweezers per row of design D1 are at 3 um spacing. On the 4 um site lattice of the proposal, only 30 are useful over the same +/-60 um. In general the useful number per row is the row length divided by the site pitch, so it cannot be raised by packing tweezers more densely. Hence scaling means longer rows and more rows.

The per-tweezer costs do not fall with scale: 8.7 fast actuators and about 57 mW at 1.2 mK per simultaneous tweezer. Two levers nevertheless remove most of the penalty.

(i) **Shallower transport.** Moves of three sites (12 um) in 200 us survive at 0.2 mK with the 200 kHz update rate (loss < 1e-3, energy gain < one trap quantum). Moves of ten sites in 200 us survive at 0.5 mK. This cuts the laser power 2.4 to 6 times: 720 simultaneous tweezers need 6.9 to 17 W instead of 41 W.

(ii) **Time-sharing of rows (row select).** A mirror at a plane conjugate to the pupil in the y path tilts all rows of a stack rigidly in y. A 16-chip stack (720 tweezers) then serves all 50 rows of a +/-100 um field, 2,250 sites, in four passes. The same tilt moves every row of the stack together in y, so the stack can carry atoms in 2D: independent along x within each row, common along y.

configuration	simultaneous tweezers	sites served	passes	chips	fast actuators	power at 1.2 mK / 0.2 mK
one D1 chip (128 ports)	30	30	1	1	255	1.7 / 0.3 W
one 192-port chip (+/-90 um row)	45	45	1	1	383	2.6 / 0.4 W
stack of 16	720	720	1	16	6,128	41 / 6.9 W
stack of 16 + row select	**720**	**2,250**	**4**	**16**	**6,128**	**41 / 6.9 W**
two stacks (orthogonal polarisation) + row select	1,440	2,250	2	32	12,256	83 / 14 W

7.2 What limits the number of tweezers

Per row. One input port per resolution cell (1.06 um at NA 0.5), so a row of L um needs L/1.06 ports and holds L/4 + 1 tweezers on 4 um sites. That is 3.8 ports and 7.5 to 8.7 fast actuators per tweezer, independent of L. The objective's corrected field sets L: the proposal's +/-100 um gives 192 ports and 45 tweezers per row. Beyond that, the slab grows (5 mm at 512 ports) and the aberration grows (0.13 rad), but neither is a wall below about 512 ports (`docs/scaling.md`). The wall is the objective.

Rows. One chip per row, each with a collimator aligned to 0.1 um (`docs/collimator.md`). Sixteen chips make an 8 mm stack and 64 um of rows, against a 200 um field.

Power. Linear in the number of simultaneous tweezers and in the depth.

7.3 Lever 1: transport depth

The depth of 1.2 mK in the proposal is what fast long moves need. A 40 um move in 100 us survives only at 1.2 mK even with a smooth trajectory. With a 200 kHz staircase it does not survive at any simulated depth, because each update step (up to 3.75 um) exceeds the 1.43 um waist. Shorter or slower moves need much less depth. Minimum depth for loss <= 1e-3 and energy gain <= one trap quantum per move, 200 kHz staircase, atoms at 10 uK:

move	100 us	200 us	400 us
12 um (3 sites)	0.8 mK	**0.2 mK**	0.1 mK
40 um (10 sites)	none <= 1.2 mK	0.5 mK	0.2 mK

(1D classical Monte Carlo, 4,000 atoms, Gaussian potential along the move with w = 1.43 um, minimum-jerk trajectory.) The excitation is not monotonic in depth: at some depths the trap frequency aliases with the update steps, as in module 3. A schedule would therefore use the shallowest depth that clears all the resonances, around 0.2 to 0.3 mK, for moves of a few sites lasting 200 to 400 us. Rearrangement mostly needs short moves, so a cycle of four passes at about 0.6 ms each (pick-up and release ramps assumed 100 us each, 200 us move, 200 us for the y tilt to settle) takes 2 to 3 ms. Depth can be raised for the occasional long move.

7.4 Lever 2: row select

The pupil of the y relay is conjugate to every chip's facet, and the atom plane to the collimators' focal plane S (`docs/stacked_relay.md`). A tilt at any plane conjugate to the pupil therefore translates all rows rigidly, without aberration in the paraxial limit. Shifting the stack's 16 rows by 64 um is a 3.7 mrad tilt at the objective pupil. At the y telescope's internal Fourier plane, where the beam is 21.6 times smaller, it is 80 mrad optical, i.e. a 40 mrad mechanical rotation. Placing rows to 10 nm needs about 6.5 urad of mechanical repeatability. That is galvanometer-scanner class (RECALLED; verify against a specific scanner). The mirror is narrow in y (about 1 mm) and as wide as the x beam (about 20 mm). Translating the whole stack by 8 mm with 1 um accuracy is the slower alternative (1 um becomes 8 nm at the atoms).

Atoms wait in the static 1075 nm array between passes. The cost is time (four passes for 50 rows) and one precision scanner. Hardware per site drops by a factor 3.1.

7.5 Other options considered

A second stack, polarisation-combined. A polarising beam splitter before the objective combines a second 16-chip stack, orthogonally polarised and offset by 64 um, so the stacks never overlap. This doubles the simultaneous capacity (32 rows, two passes). It costs a second set of hardware, so the penalty per tweezer is unchanged. It needs a polarisation-insensitive objective (as in `docs/row_coherence.md`, Sec. 4).

A larger corrected field. This is the only way to longer rows. The +/-100 um field is a meta-objective target in the proposal. Refractive plan objectives of NA 0.5 have flat fields of several hundred micrometres (RECALLED; verify). With a +/-250 um field, 480 ports would give 110 tweezers per row, at the slab and aberration costs of `docs/scaling.md` (R ~ 5 mm, about 0.12 rad).

Planar tiling of surface-emitting chips (speculative, not simulated). Chips side by side on a common carrier, each with long grating emitters, imaged in y onto the atoms with demagnification D = tile pitch / 4 um. For a 6 mm tile pitch, D = 1500. The object-side NA in y is then 3.3e-4, which a 2 mm weak

grating radiates directly, so no collimator is needed. Chip placement errors shrink by D (10 um becomes 7 nm). The 0.1 um decentre tolerance is replaced by an emission-angle tolerance of about 30 urad. That corresponds to delta n_eff ~ 3e-5, trimmable with a heater (about 1.5 K) or by tilting the chip. The open questions are grating efficiency (up and down emission) and fitting the engine (slab of about 2 mm) and a 2 mm grating within the tile. This is worth a separate study, because it removes the stack and the collimators altogether.

What does not work. Packing tweezers below the site pitch adds nothing on a 4 um lattice. Several emitter rows per chip cannot be separated at the atoms: rows a few um apart on the facet land within nanometres of each other, and a per-row collimator cannot be fitted. Spectral multiplexing of rows through the dispersive star coupler is excluded by the objective's chromatic focal shift (16 um/nm).

7.6 Recommendation and next steps

The low-penalty path is: 192-port chips (45 per row) filling the +/-100 um field; transport at 0.2 to 0.5 mK with moves of 200 to 400 us; a 16-chip stack with row select serving 2,250 sites. This gives 720 simultaneous independent tweezers on 6.9 to 17 W of laser power and 6,128 fast actuators. A second, polarisation-combined stack doubles the parallelism if the rearrangement time matters more than hardware.

Next steps: (i) a 3D transport simulation (radial and axial motion, handoff to and from the static array) at 0.2 to 0.5 mK; (ii) a rearrangement scheduler that uses independent x moves per row and the common y tilt, to count passes and time for realistic loading; (iii) an optical design of the row-select scanner plane; (iv) a first study of the planar surface-emitting tiling.

7.7 Verification notes

value	location	type	pointer
minimum depths, loss, excitation	Sec. 1, 3	SIM	results/m17_moves.json (1D classical; review)
configuration table	Sec. 1	DERIVED	results/m17_table.json
57 mW per tweezer at 1.2 mK	Sec. 1, 2	DERIVED	results/m14_plan.json
ports per row, slab, aberration	Sec. 2, 5	SIM/DERIVED	results/m12_*.json
tilt angles and magnification 21.6	Sec. 4	DERIVED	conjugations of docs/stacked_relay.md; f_a = 100 mm, f_FAC = 175 um, beta = 0.7 um
atom temperature 10 uK, ramps 100 us, criteria	Sec. 3	CHOICE	stated assumptions
galvanometer repeatability, objective fields	Sec. 4, 5	RECALLED	verify
planar tiling numbers	Sec. 5	DERIVED	order of magnitude, not simulated

Chapter 8

Stacked-chip relay: a 2D array of independently steered rows

> Follow-up (`docs/megascale.md`, Sec. 12): this module and `docs/collimator.md` treat the y axis for light with no x component (s_x = 0). A tweezer at x_a is made of light with s_x = x_a / 288 um, and the cylindrical collimator then acts with effective indices sqrt(n^2 - s_x^2). The adopted collimator defocuses the facet by -99 s_x^2 um. The y figures below therefore hold near the row centre: the peak falls to 0.97 at x_a = 30 um and 0.48 at 60 um. With the best common focus offset a row is usable over |x_a| <= 45 um.

Follow-up to `docs/scaling.md`, Sec. 4(v). Code: `picsim/opa/stack.py`, `scripts/m13_stack.py`; results `results/m13_*.json` (+ .png); tests `tests/test_stack.py`. Each chip is design D1 (`docs/fpr_design.md`): 128 input ports, 192 arms and emitters, one row of independently steered tweezers along x. The model is scalar wave optics. The x axis uses the beam-propagation response of D1. The y axis uses the femwell facet mode of the D1 emitters. Both axes pass through the circular NA 0.5 pupil. Values are SIM or DERIVED and valid in the regime stated. They should be reviewed before they are quoted.

8.1 Result

Sixteen D1 chips stacked at 0.5 mm, each with its own fast-axis collimator, can be imaged onto the atoms as 16 rows 4 um apart. The rows then behave as independent as the tweezers within a row, on one condition: every chip runs at its own laser frequency. Design point, 1064.4 nm, NA 0.5:

quantity	value
demagnification of the stack in y, D = chip pitch / row pitch	125
collimator (FAC) per chip: focal length / numerical aperture	175 um / 0.9
y spot FWHM at the atoms (x: 1.71 um)	1.31 um
cross-axis transmission (FAC capture x pupil) / loss to the circular pupil	0.973 / 3%
laser power per 1.2 mK tweezer, field mean (was 43 mW with an assumed Gaussian cross axis)	47 mW (57 mW with the facet loss, `docs/collimator.md`)
crosstalk of a full row (41 tweezers) at the neighbouring row, 4 um / 3 um	1.8e-4 / 4.3e-4
neighbouring row, chips at different frequencies: depth / position change when a tweezer moves	2.4e-5 / 0.04 nm
the same with all chips at one frequency (coherent)	2.8e-3 / 4.5 nm (static: 2.6%, 11 nm)
FAC alignment for <= 3% loss: decentre / focus / roll	+/-0.1 um / +/-0.5 um / 1 mrad
FAC wavefront error for <= 6% loss	<= 0.05 waves rms
row position at the atoms per um of chip height error	8 nm
facet-to-pupil length of the first-order layout	0.38 m

Hence the stacked relay is optically sound. Its costs are (i) one frequency shifter per chip, (ii) collimator alignment to about 0.1 um in height, which is set by the small vertical facet mode, and (iii) a diffraction-

limited collimator at NA 0.9. With 192-port chips, 16 rows give 720 (4 um spacing) to 960 (3 um) independent tweezers. Including the emitter facet loss found later (`docs/collimator.md`), they need 41 to 55 W at 1.2 mK, or 17 to 23 W at 0.5 mK, spread over 16 chips.

8.2 Optical layout

In x, the relay of `picsim/opa/array.py` images the emitter plane onto the objective pupil, so the atoms see the far field of the chip. In y, each chip's facet sits in the front focal plane of its own cylindrical collimator. The collimators' back focal planes form a plane S, which a common y lens group of effective focal length F_y Fourier transforms onto the pupil. Hence S is imaged onto the atoms with magnification f_obj/F_y = 1/D, and chip m lands at y_m = m d_c/D. Here, d_c is the chip pitch and f_obj = 17.3 mm the objective focal length (pupil radius WD tan(asin NA) = f_obj NA at WD = 15 mm). At the same time, the pupil is an image of the facet in both axes: in x with magnification 60, and in y with 12,400. Because each chip has its own collimator centred on its facet, all 16 facet images overlap in the pupil, and their row offsets come only from the tilt y_m/F_y. A single common collimator would not work: it would place the 16 facet images side by side.

The free parameter of the y design is beta, the pupil radius referred back to the facet, beta = NA f_FAC/D. It plays the role of the pupil fill (Sec. 4). The row pitch then fixes D, and beta fixes f_FAC.

First-order layout (thin cylindrical lenses; ABCD matrices of both axes checked in `m13_design`):

row pitch	D	F_y	f_FAC	collimated beam half-height	length facet to pupil
3 um	167	2.89 m	233 um	210 um	439 mm
4 um	125	2.17 m	175 um	158 um	382 mm
5 um	100	1.73 m	140 um	126 um	347 mm
6 um	83	1.44 m	117 um	105 um	324 mm

At 4 um, the long effective focal length is realised as a 25x demagnifying y telescope (f = 100 mm and 4 mm) followed by an 86.6 mm Fourier lens. The x relay is a 4f pair (3.12 mm and 187.7 mm), with the Fourier slit 6.2 mm from the facets. Both pupils coincide 382 mm from the facets. The Lagrange invariant matches on both sides (stack half-height 4 mm x collimated NA 0.0022 = atom half-field 30 um x 0.5 x fill). The collimated beams (158 um half-height) fit within the 250 um half-pitch, also at 3 um row pitch (210 um).

8.3 The facet sets the y optics

The D1 emitter facet (1.2 x 0.4 um SiN in oxide) has a vertical mode of 1/e^2 equivalent radius 0.39 um. It radiates into air with a far-field 1/e^2 half-angle of sin(theta) = 0.86 (`m13_facet`). A collimator of NA 0.8 captures 96% of the light, and one of NA 0.9 captures 98.8%. Thinning the facet section does not help much:

SiN at the facet	far-field NA (1/e^2)	equivalent radius	capture at NA 0.8	coupling length of adjacent emitters (0.3 um gap)
400 nm	0.86	0.39 um	0.961	228 um
200 nm	0.79	0.43 um	0.971	106 um
150 nm	0.69	0.49 um	0.980	72 um
100 nm	0.51	0.67 um	0.991	47 um

A thinner facet expands the vertical mode, but it also expands the lateral one. At 1.5 um emitter pitch the coupling length drops to 47 um, which leaves no room for an adiabatic vertical taper. Hence the dense emitter array and a large vertical facet mode are in conflict. The collimator alignment tolerances below scale as the vertical mode radius (decentre) and its square (focus). We therefore keep the 400 nm facet and accept the tolerances.

8.4 Pupil fill and the crosstalk floor

The y profile at the atoms is the facet's far field, clipped by the collimator aperture and scaled by beta/NA, then low-pass filtered by the pupil. Crosstalk between rows has two sources: (i) the pupil clips the sharply peaked image of the facet (small beta); (ii) the collimator aperture cuts the wide far field with a hard edge, and the pupil makes this edge ring (large beta).

FAC NA 0.9, single tweezer, 1D (y only):

beta	cross-axis transmission	y FWHM	power vs Gaussian w0 = 1.04 um	crosstalk at 3 / 4 / 6 / 8 um
0.5 um	0.946	1.09 um	0.92	3.5e-3 / 1.8e-3 / 7.2e-4 / 3.9e-4
0.6 um	0.968	1.18 um	0.97	1.2e-3 / 6.2e-4 / 2.5e-4 / 1.3e-4
0.7 um	**0.973**	**1.31 um**	**1.07**	**2.8e-4 / 1.1e-4 / 3.2e-5 / 1.5e-5**
0.8 um	0.973	1.49 um	1.22	1.1e-3 / 2.6e-4 / 4.9e-5 / 1.9e-5
1.0 um	0.977	2.08 um	1.65	2.1e-3 / 6.4e-4 / 1.9e-4 / 9.3e-5

The two sources balance near beta = 0.7 um, which we adopt. Raising the collimator NA to 0.95 lowers the crosstalk at 4 um to 7.4e-5. A graded (raised-cosine) collimator edge removes source (ii), but it costs 7 to 8% of the light and reaches only 9.4e-5 at 4 um (beta = 0.8 um). None of these options gets below about 5e-5 at 4 um. That is adequate for mutually incoherent rows (Sec. 5) but not for coherent ones. At beta = 0.7 um the y spot (1.31 um FWHM) is smaller than the Gaussian assumed so far (w0 = 1.04 um, 1.22 um FWHM). Together with the 2.7% cross-axis loss and the 3% loss to the circular pupil, the power per tweezer rises by a factor 1.10, the same at the field centre and edge.

8.5 Crosstalk between rows

For the 2D test, chip m forms a full row: 41 tweezers at 3 um over +/-60 um, alternating phases 0 and pi/2. The neighbouring chip forms 11 tweezers offset by 1.5 um in x. We evaluate them at row separations of 3 to 16 um. "Coherent" means both chips use the same laser frequency, with the worst of 8 relative phases. "Incoherent" means the chips use different frequencies.

row separation	full-row crosstalk (max over x)	incoherent: static depth / position	incoherent, row-m tweezer moves 0.25 um	coherent: static depth / position	coherent, moving
3 um	4.3e-4	2.0e-4 / 0.13 nm	6.2e-5 / 0.04 nm	2.9% / 18 nm	6.5e-3 / 5.6 nm
4 um	1.8e-4	1.7e-4 / 0.002 nm	2.4e-5 / 0.04 nm	2.6% / 11 nm	2.8e-3 / 4.5 nm
6 um	6.3e-5	4e-6 / 0.02 nm	3e-6 / 0.01 nm	0.41% / 8.9 nm	1.7e-3 / 2.1 nm
8 um	4.3e-5	< 1e-6	< 1e-6	0.08% / 1.8 nm	1.1e-3 / 1.7 nm
12 um	1.5e-5	< 1e-6	< 1e-6	0.01% / 5.9 nm	8e-4 / 0.8 nm
16 um	1.3e-5	2e-6	< 1e-6	0.27% / 1.6 nm	4e-4 / 0.6 nm

With incoherent rows, a moving tweezer changes the next row by at most 6e-5 in depth and 0.04 nm in position, even at 3 um row pitch. This is two orders of magnitude below the in-row figures of D1 (0.11%, 3.5 nm). With coherent rows, the crosstalk field interferes with the neighbour's field, so the depth changes by about 2 sqrt(epsilon). This interference persists at all distances simulated, because a full row has a tail plateau of about 1e-5 out to 16 um. Its strength follows the relative optical phase of the two chips, which drifts with the fibre paths, so the 2.6% and 11 nm at 4 um become slow noise. Hence alternating two frequencies is not enough. Every chip needs its own frequency, with all pairwise differences of at least 1 MHz, far above twice the 84 kHz trap frequency. That is 16 offsets (for example one acousto-optic shifter per chip feed) for 16 chips. The beat notes then modulate the depth at MHz frequencies with amplitude about 2 sqrt(epsilon) (3% at 4 um). This lies far from the parametric resonance at 2 nu_trap, and the atoms see the time average. The tweezers within one row stay coherent, as route E requires.

Follow-up (`docs/row_coherence.md`): the chips need not run at different frequencies. Any randomisation of their relative phase much faster than the trap period suffices. One phase modulator on the common seed, driven with Gaussian noise, plus different fibre delays per chip does this as well as 16 frequency shifters. Alternating polarisation is a passive partial alternative.

8.6 Alignment tolerances

Per-chip errors, single tweezer at x = 0 (the field edge, x = 60 um, gives the same values within 0.1%). The table gives the peak at the nominal atom plane relative to no error, and the worst crosstalk at 4 um. None of these errors moves the row in y by more than 0.01 nm.

error	value	peak	crosstalk at 4 um	note
none		1	1.1e-4	y FWHM 1.33 um (2D)
FAC decentre (facet vs collimator axis, y)	0.05 / 0.1 / 0.2 / 0.3 um	0.992 / 0.969 / 0.881 / 0.750	2e-4 / 4e-4 / 1.2e-3 / 2.8e-3	moves the facet image across the pupil (delta/beta)
FAC focus (along z)	0.25 / 0.5 / 1 / 2 um	0.993 / 0.973 / 0.893 / 0.641	3e-4 / 9e-4 / 2.8e-3 / 4.3e-3	y focus moves by about 1 um at 1 um error
FAC roll about z	0.5 / 1 / 2 mrad	0.998 / 0.992 / 0.968	1e-4 / 1e-4 / 2e-4	decentre varies along the 288 um emitter row
smile (bow, peak-to-valley)	0.1 / 0.2 / 0.4 um	0.998 / 0.992 / 0.967	1e-4 / 2e-4 / 4e-4	
FAC wavefront, spherical-like	0.05 / 0.1 waves rms	0.942 / 0.783	3e-4 / 1.2e-3	best focus 0.95 / 0.82
FAC wavefront, coma-like	0.05 / 0.1 waves rms	0.937 / 0.767	5e-4 / 2.0e-3	
chip piston along z (x axis, D1 weights unchanged)	10 / 100 / 300 um	>= 0.999 / >= 0.992 / >= 0.908		x position <= 1.5 / 0.9 / 6.8 nm, static

The collimator decentre is the tightest tolerance. A decentre delta shifts the facet image in the pupil by delta/beta, so the tolerance scales with the vertical facet mode (Sec. 3). It costs power and crosstalk but does not move the row. The wavefront requirement (about 0.03 waves rms for <= 3%, at NA 0.9) follows from the same imaging: the collimator is the aperture of the facet-to-pupil image, and light it scatters out of that image misses the pupil. Defocus-like aberration is harmless after refocusing. A larger beta would relax the wavefront tolerance, but it raises the crosstalk and the power (Sec. 4).

Row position at the atoms (DERIVED from the conjugations): a height error of chip plus collimator moves its row by 1/D = 8 nm per um. A collimator-array placement error has the same effect. A chip tilt about x of 1 mrad moves the row by f_FAC x 1 mrad / D = 1.4 nm. A chip piston of dz defocuses its row in y by dz/D^2 (6 nm per 100 um). A stack built to +/-2 um therefore places rows to +/-16 nm. A static offset of this size must be matched by the static 1075 nm array, or measured and absorbed into its design. Thermal expansion of a Si stack (8 mm) moves the outer rows by about 0.2 nm/K at the atoms (DERIVED, CTE 2.6e-6/K, RECALLED).

8.7 Limits and next steps

The model is scalar and uses ideal (sine-condition) thin cylindrical lenses with the error types listed. The objective's apodisation factor and polarisation are omitted, as in the x model. The facet radiates into air with Kirchhoff boundary conditions; Fresnel reflection and the angle dependence of transmission are not included. We have not designed real lens prescriptions: the 0.38 m layout is first order. Gluing the collimator to the facets with index matching would lower the ray angles inside the lens for the same NA 0.9; it is not simulated. Fibre and electrical access to chips stacked at 0.5 mm pitch is not addressed. Facets must stay flush to within about 100 um (Sec. 6), so stair-step stacking for wire bonds is excluded, and access must come from the chip sides or back.

Next steps (i) to (iii) below are addressed in `docs/collimator.md` (ray-traced collimators, assembly and drift, power and frequency plan) together with the first x lens of the relay. Remaining: a full lens design of the anamorphic relay over its 8 mm field.

Original list: (i) ray-trace a real collimator (acylinder or two-photon-printed lens) to NA 0.9 and check the 0.03-wave target; (ii) an assembly study of the 0.1 um collimator alignment, including adhesive shrinkage; (iii) the frequency plan and power distribution for 16 chips (16 shifters, about 2 to 3 W each); (iv) a real lens design of the anamorphic relay, including its field (16 chips over 8 mm).

8.8 Verification notes

No literature values were retrieved in this module. Recalled values are labelled.

value	location	type	pointer
facet NA, capture, coupling lengths	Sec. 3	SIM	results/m13_facet.json (review before quoting)
fill table, graded edges	Sec. 4	SIM	results/m13_fill.json (review)
2D crosstalk, coherent and incoherent	Sec. 1, 5	SIM	results/m13_crosstalk.json (review)
tolerances	Sec. 1, 6	SIM	results/m13_tolerance.json (review)
power factor 1.10, 47 mW, circle loss 3%	Sec. 1, 4	SIM/DERIVED	results/m13_power.json; 43 mW from docs/fpr_design.md
layout table, ABCD checks	Sec. 2	DERIVED	results/m13_design.json
row position per um, tilt, piston	Sec. 6	DERIVED	conjugations of Sec. 2
chip pitch 0.5 mm, 16 chips, beta 0.7 um, FAC NA 0.9, telescope 25x	Sec. 1, 2	CHOICE	stated design choices
f_obj 17.3 mm	Sec. 2	DERIVED	WD 15 mm (USER) and NA 0.5
Si CTE 2.6e-6/K	Sec. 6	RECALLED	verify
>= 1 MHz frequency spacing	Sec. 5	DERIVED	2 x 84 kHz trap frequency (m3_update_rate)

Open placeholders: [REF?] achievable wavefront error of NA 0.9 micro-collimators (commercial acylinders, two-photon-printed lenses); [REF?] alignment accuracy and long-term stability of actively aligned, bonded collimators; [VALUE?] power handling of fibre-coupled or free-space frequency shifters at 2 to 3 W, 1064 nm.

Chapter 9

Fast-axis collimator, facet loss, relay lens and the power plan

> Follow-up (`docs/megascale.md`, Sec. 12): this module and `docs/collimator.md` treat the y axis for light with no x component (s_x = 0). A tweezer at x_a is made of light with s_x = x_a / 288 um, and the cylindrical collimator then acts with effective indices sqrt(n^2 - s_x^2). The adopted collimator defocuses the facet by -99 s_x^2 um. The y figures below therefore hold near the row centre: the peak falls to 0.97 at x_a = 30 um and 0.48 at 60 um. With the best common focus offset a row is usable over |x_a| <= 45 um.

Follow-up to `docs/stacked_relay.md`, which assumed an ideal collimator (focal length 175 um, NA 0.9, Abbe sine condition). Code: `picsim/raytrace.py`, `picsim/opa/collimator.py`, `scripts/m14_collimator.py`, `scripts/m14_facet_meep.py` (2D FDTD); results `results/m14_*.json`. Values are SIM or DERIVED, valid in the regime stated. They should be reviewed before they are quoted.

9.1 Result

A collimator that meets the requirement exists in a classical form. It is an aplanatic (Weierstrass) cylinder on a fused-silica block glued to the facets, followed 10 um later by a two-asphere aplanat of index 1.8. Ray traced, it collimates to NA 0.9 with the sine condition exact and no residual wavefront error. Rays meet every surface at 25 deg or less, and the whole lens is 0.40 mm long and 0.32 mm tall. In the wave model it performs like the ideal lens in glass. Two findings change the system numbers:

(i) **The emitter facet loses about a quarter of the light into air.** A 2D FDTD of the facet cross-section gives 79% transmission into air (12.8% reflected back into the chip) and 96% into glass (2.3% reflected). Within the collimator aperture ($|s| < 0.9$) the figures are 77% (air) and 85% (glass). This loss was not in any earlier power figure. Gluing the collimator to the facet recovers about 10% and cuts the back-reflection sixfold.

(ii) **Power per 1.2 mK tweezer rises to 57 mW** with the adopted collimator, from 47 mW (`docs/stacked_relay.md`, facet loss omitted). With an ideal lens in air it would be 61 mW.

quantity	adopted collimator (Weierstrass block + aplanat)
focal length / NA / sine-condition error over the aperture	175 um / 0.9 / < 4e-5
length (facet to last vertex) / clear height	395 um / +/-158 um
largest incidence angle on any surface	25 deg
peak per guided power vs ideal lens in air	1.10 (2D check: 1.107)
row crosstalk at 4 um	8e-5
decentre / focus for <= 3% loss	+/-0.1 um / +/-1 um
surface form error for <= 2% loss	<= 20 nm rms
laser power per 1.2 mK tweezer	57 mW

The decentre tolerance (+/-0.1 um) does not depend on the lens design. It is set by the small vertical mode of the facet. The first x lens of the relay, by contrast, needs no special accuracy, because calibration absorbs its static aberration (Sec. 6). One HI1060-matched input fibre per chip carries the 2.6 to 3.4 W a row needs, at 8 to 11 MW/cm^2 (Sec. 8).

9.2 Why NA 0.9 at 175 um is hard

A collimator must bend rays from sin(theta) = 0.9 (64 deg in air) to the axis within a lens no taller than the 0.5 mm chip pitch. Single refracting surfaces fail:

(i) a plano-hyperbolic lens with its curved side to the facet collimates exactly, but only up to sin(theta) = sqrt(1 - 1/n^2): 0.72 for fused silica and 0.83 for n = 1.8; (ii) a plano-asphere with its flat side to the facet reaches 0.9. In fused silica its exit surface slopes at 81 deg and rays meet it at 43 deg, at the edge of total internal reflection. With n = 1.8 the slope falls to 58 deg, but the flat surface then sees 64 deg incidence (51% bare transmission), which an AR coating cannot easily recover; (iii) one elliptical surface on a block glued to the facet (source inside glass, eccentricity 1/n) collimates exactly at any NA up to sqrt(n^2 - 1) = 1.05. However, it violates the sine condition: the ray at s = 0.9 exits at 68% of the height an aplanatic lens would give.

The Weierstrass construction resolves this. A circular surface (radius R) on a glass block, with the facet R/n from its centre, is aplanatic: every ray appears to come without aberration from a virtual point, and the NA drops by n^2 (0.9 -> 0.43 for fused silica). A second lens of focal length f n^2 = 368 um then collimates NA 0.43. We design it as a two-asphere aplanat in 2D (Wassermann-Wolf construction: exact collimation and exact sine condition, `collimator.aplanat`). A scan of R, the air gap, the aplanat thickness and its index (96 combinations, 29 reaching NA 0.9; `m14_scan`) selected R = 80 um, a 10 um gap, a 250 um thick aplanat and n = 1.8. That combination gives the lowest largest incidence angle (25.4, 18.2 and 19.2 deg on the three surfaces) and slopes of 64, 7 and 36 deg.

design	sine condition y_S/(f s) at s = 0.3 / 0.6 / 0.9	largest incidence	surfaces	Fresnel transmission at the edge, bare
Weierstrass + aplanat (glued)	1.000 / 1.000 / 1.000	25 deg	3 (one spherical)	0.74
ellipse (glued or printed)	0.954 / 0.834 / 0.675	43 deg	1	0.43
plano-asphere, n = 1.8 (air)	0.988 / 0.960 / 0.971	64 deg	2	0.51

9.3 Facet transmission

The x model of the chip treated the emitter facet as lossless, and the cross-axis model of module 13 normalised to the power radiated into air. A Kirchhoff estimate puts the loss into air at 27%. It comes from 8% Fresnel reflection plus the 20% of the angular spectrum of the 0.39 um vertical mode that lies beyond the light cone. We checked this by 2D FDTD (MEEP, resolution 40/um) of the vertical cross-section. The SiN core index was set to 1.965 so that the slab mode index equals that of the 3D mode (1.7995). The guide ends in air, or in fused silica for a glued lens.

into	reflected	transmitted	within the collimator aperture, FDTD	same, Kirchhoff
air	12.8%	79.1%	77.2%	72.3%
fused silica	2.3%	95.9%	85.2%	82.9%

The two methods agree to 3 to 7%. We use the FDTD values to scale the Kirchhoff spectra in the power estimate. The remainder (8% in air, 2% in glass) leaves through the cladding near the facet. It is a 2D result with the lateral confinement folded into the core index; a 3D FDTD of the facet would refine it.

9.4 The designs in the wave model

The wave model takes the facet spectrum in air or glass, each lens's traced map (exit height and wavefront vs s, Fresnel transmission), the pupil fill beta and the NA 0.5 pupil. Peak values are per unit guided power, relative to the ideal lens in air at beta = 0.7 um, with AR-coated lens surfaces:

design	beta	peak	y FWHM	crosstalk at 3 / 4 um
ideal, air	0.7 um	1.00	1.31 um	2.8e-4 / 1.1e-4
Weierstrass + aplanat	0.7 um	**1.10**	1.35 um	4.0e-4 / 8.0e-5

design	beta	peak	y FWHM	crosstalk at 3 / 4 um
ellipse, printed	0.9 um	1.03	1.35 um	6.6e-4 / 1.8e-4
ellipse, printed	0.7 um	1.22	1.13 um	2.1e-3 / 1.0e-3
plano-asphere n = 1.8	0.7 um	1.03	1.27 um	4.0e-4 / 1.7e-4

The aplanatic design reproduces the ideal lens in glass exactly. The ellipse narrows the y spot because its mapping compresses the outer far field, but the same distortion raises the crosstalk. Opening the pupil fill to beta = 0.9 um restores 1.8e-4 at 4 um at a cost of 15% in peak. Without AR coatings, all designs lose 10 to 30% (n = 1.8 surfaces reflect 8% each at normal incidence), so coatings are needed in every case.

9.5 Tolerances with the real lenses

error	ideal (air)	Weierstrass + aplanat	ellipse	plano-asphere n 1.8
decentre 0.1 / 0.2 um	0.973 / 0.897	0.973 / 0.896	0.962 / 0.855	0.971 / 0.887
focus 0.5 / 1 um	0.989 / 0.937	0.995 / 0.977	0.974 / 0.905	0.978 / 0.906
form error 20 / 50 nm rms (corr. 20 um), mean		0.984 / 0.916	0.998 / 0.988	0.989 / 0.916
form error 50 nm, 10th percentile		0.880	0.981	0.865

(Peak relative to no error, 1D y line, beta = 0.7 um for all designs.)

The tolerance model was checked against exact traces of a displaced source. For decentres of 0.2 and 0.5 um and a 0.5 um focus error, the exactly traced path change differs from the first-order model by at most 0.003 waves rms (0.03 waves peak-to-valley for the ellipse at 0.5 um, about 2e-4 waves for the aplanatic design).

Three points follow. (i) The decentre tolerance is a property of the facet, not of the lens: +/-0.1 um for every design. (ii) Focus is looser for the glued designs because the facet radiates into glass (the defocus phase scales as n_m cos theta): about +/-1 um for 3%. (iii) The ellipse has one surface with a small index step (n - 1 = 0.45), so it tolerates 50 nm rms form error; the n = 1.8 surfaces need about 20 nm.

9.6 The first x lens of the relay

The x relay's first lens (f = 3.12 mm, NA 0.355 through the slit) sees the whole emitter row, +/-144 um, so it works off axis. We traced two fused-silica designs from 7 emitter heights. Each off-axis wavefront was applied to each emitter's angular spectrum in the D1 model (BPM matrix), and tweezers at 0, +/-30 and +/-60 um were evaluated before and after least-squares recalibration:

lens	wavefront at 144 um field (tilt removed)	edge tweezer, uncalibrated: peak / shift	after recalibration: peak per power / shift	moving tweezer, others' change
aplanat (sine condition)	0.15 waves rms	0.960 / 164 nm	1.001 / <= 1.5 nm	4e-4 / < 1 nm
plano-asphere (no sine condition)	0.60 waves rms	0.672 / 590 nm	0.964 / <= 0.5 nm	4e-4 / 0.7 nm

The static aberration of the x relay is a fixed linear transformation of the chip's response. The least-squares synthesis, which already works on the measured response matrix, removes it. Hence the x relay needs ordinary lens quality: an aplanat costs nothing, and even a lens without the sine condition costs only 3.6%. The same argument does not apply to the collimator, because the y axis has no actuators.

9.7 Assembly and drift

(DERIVED, with RECALLED coefficients marked.)

Thermal. For a collimator glued directly onto its own facet, vertical registration is fixed at the facet. Differential expansion then acts only over the lens's own height (+/-80 um at the glued block): with Delta

alpha ~ 2e-6/K (Si 2.6e-6/K vs fused silica 0.55e-6/K, RECALLED), that is about 0.2 nm/K, negligible. A monolithic collimator array for the whole stack, referenced at its base, would see the full 8 mm stack height: 16 nm/K at the top chip, so 0.1 um after 6 K. Hence one collimator per chip, or a stack held to +/-1 K.

Adhesive. The glue line of the immersed design lies between the facet and the block, along z. Shrinkage during cure therefore changes the focus, which tolerates about +/-1 um, and only weakly the decentre. The vertical position must be held during cure to +/-0.1 um. Typical shrinkage and creep of UV adhesives need checking [REF?].

Alignment signal. Decentre moves the facet image across the pupil by delta x 12,400 (`docs/stacked_relay.md`, Sec. 2): 0.1 um becomes 1.2 mm on the 17 mm pupil. Imaging the pupil, or its conjugate in the y telescope, while each chip is lit gives a direct and very sensitive alignment signal. A camera with 10 um pixels resolves 1 nm of decentre in principle. Active alignment against this signal, before and during cure, is the recommended process. A collimator printed by two-photon lithography directly on the facet would be registered lithographically; its placement accuracy needs a source [REF?].

9.8 Power and frequency plan for 16 chips

Laser power per 1.2 mK tweezer, field mean, per unit guided power (FDTD-scaled facet, AR-coated lenses):

collimator	power per tweezer
ideal lens in air	61 mW
Weierstrass + aplanat, glued (adopted)	57 mW
ellipse, printed, beta 0.9	61 mW
plano-asphere n = 1.8 in air	59 mW

tweezers per row	depth	power per chip	16 chips	input fibres per chip: lensed tip / HI1060-matched tip
45 (4 um)	1.2 mK	2.6 W	41 W	5 / 1 (8 MW/cm^2)
60 (3 um)	1.2 mK	3.4 W	55 W	6 / 1 (11 MW/cm^2)
45	0.5 mK	1.1 W	17 W	2 / 1
60	0.5 mK	1.4 W	23 W	3 / 1

One input per chip. The lensed-fibre inverse taper of module 2 has a 3.1 um^2 mode at the chip facet, so a 20 MW/cm^2 cap (CHOICE; no damage threshold was found) allows 0.62 W per fibre and 5 to 6 fibres per chip. Several inputs would split the 128 ports into sub-trees whose relative optical phases drift with their fibres, and a tweezer window straddling a sub-tree boundary would then fluctuate. The HI1060-matched taper of module 2 (tip 0.1 um, -0.48 dB, 0.11 dB worse than the lensed tip) has a 30.6 um^2 mode and carries 6.1 W at the same cap. Hence one fibre per chip is possible and preferred. Its 0.1 um tip width is at the assumed minimum feature size.

Frequency offsets. Superseded by `docs/row_coherence.md`: per-chip shifters are not needed. One phase modulator on the seed, driven with Gaussian noise (3 rad rms, 1 to 2 GHz), plus fibre delays in 0.3 m steps gives the same row independence. (The original plan was one acousto-optic shifter per chip feed at 80 + 1.13 m MHz, m = 0 to 15, pairwise differences 1.13 to 17 MHz, far above 2 nu_trap = 168 kHz, each handling 2.6 to 3.4 W [VALUE?].)

Laser. 41 to 55 W of single-frequency 1064.4 nm light at 1.2 mK, or 17 to 23 W at 0.5 mK. Lowering the transport depth remains the cheapest lever: power scales linearly with depth, and the lower trap frequency also relaxes the required update rate (`m3_update_rate`).

9.9 Limits and next steps

The ray trace is 2D (cylindrical lenses), with ideal AR coatings in the wave model. Polarisation effects at the steep Weierstrass surface (64 deg slope, 25 deg incidence) are limited to s-polarised Fresnel transmission. The facet FDTD is 2D with an effective core index. The n = 1.8 glass is a class, not a selected catalogue glass. Tolerances are evaluated one at a time.

Next steps: (i) a tolerance Monte Carlo of the adopted collimator with all errors combined (decentre, focus, form, thickness, index); (ii) a 3D FDTD of the emitter facet, glued and in air; (iii) a fabrication route

for the Weierstrass block and aplanat at 0.5 mm pitch (moulded glass array, wafer-level optics or two-photon printing); (iv) alignment trials using the pupil-walk signal.

9.10 Verification notes

value	location	type	pointer
designs, limits, geometry	Sec. 1, 2	SIM	results/m14_designs.json (review)
facet R, T, capture	Sec. 1, 3	SIM	results/m14_facet_meep_{air,glass}.json (review)
wave-model table	Sec. 1, 4	SIM	results/m14_wave.json (review)
tolerances, exact checks	Sec. 1, 5	SIM	results/m14_tolerance.json (review)
form Monte Carlo	Sec. 5	SIM	results/m14_form.json (24 samples per point; review)
x lens	Sec. 6	SIM	results/m14_xlens.json (review)
power plan, fibres, offsets	Sec. 1, 8	DERIVED	results/m14_plan.json
parameter scan of the Weierstrass design	Sec. 2	SIM	results/m14_scan.json
2D check of the adopted design (1.107)	Sec. 1	SIM	run in session, not stored
n(SiO2) 1.4496	throughout	SOURCE	Malitson (picsim/materials.py)
n = 1.8, R 80 um, gap 10 um, T 250 um, AR coatings	Sec. 2, 4	CHOICE	design choices
CTE Si 2.6e-6/K, fused silica 0.55e-6/K	Sec. 7	RECALLED	verify
20 MW/cm^2 facet cap	Sec. 8	CHOICE	m6 (no damage threshold found)

Open placeholders: [REF?] UV-adhesive shrinkage and creep at the 0.1 um level; [REF?] placement accuracy of two-photon-printed micro-optics on chip facets; [VALUE?] CW power handling of acousto-optic shifters at 3 W, 1064 nm; [VALUE?] available single-frequency 1064 nm power at 40 to 55 W.

Chapter 10

Independent rows without acousto-optic frequency offsets

Follow-up to `docs/stacked_relay.md` (Sec. 5) and `docs/collimator.md` (Sec. 8), which proposed one acousto-optic frequency shifter per chip. Code: `scripts/m15_coherence.py` (vector focusing in `picsim/opa/stack.py`, `Focus2D.field_vector`); results `results/m15_*.json`. Values are SIM or DERIVED, valid in the regime stated, and should be reviewed before they are quoted.

10.1 Result

Per-chip frequency offsets are not needed. The same decoupling is obtained with one electro-optic phase modulator on the seed laser, driven with Gaussian noise of about 3 rad rms and 1 to 2 GHz bandwidth, ahead of the 1:16 split. Each chip's feed is then given a different fibre length, in steps of at least three correlation lengths (about 0.3 m at 2 GHz, at most 5 m in total). Within one chip all paths agree to better than a millimetre, so the light stays coherent and route E is unaffected. Between chips the relative phase becomes a random process with correlation time below 1 ns, so the atoms see the time average.

A fully passive alternative, orthogonal linear polarisation on alternate chips, meets the dynamic criteria. It leaves a slow depth wander of 0.4% rms and needs a polarisation-insensitive objective.

Perturbation of a tweezer by all other rows within +/-16 um (both sides; rms over drifting relative phases, pairs in quadrature; 4 um rows, full rows of 41 tweezers):

option	added hardware	power	static depth / position (rms)	when a tweezer moves 0.25 um: depth / position (rms)
all chips coherent, no measure	none	1	2.3% / 9.2 nm	2.3e-3 / 5.1 nm
per-chip AOM offsets	16 AOMs at 2.6 to 3.4 W	1 (+ AOM loss)	1.8e-4 / 0.09 nm	2e-5 / 0.05 nm
common phase noise + fibre delays	**1 EOM on the seed, noise source, <= 5 m fibre**	**1**	**1.6e-4 / 0.07 nm**	**2e-5 / 0.04 nm**
alternating x / y polarisation	half-wave-plate strips (passive)	1	4.2e-3 / 2.5 nm	1.1e-3 / 1.6 nm
y apodiser a = 0.65	graded mirror (passive)	1.84	5.2e-3 / 4.1 nm	1.0e-3 / 1.8 nm
polarisation + apodiser a = 0.8	passive	1.55	2.8e-3 / 1.8 nm	5.7e-4 / 0.7 nm

(Criteria used for the rows within a chip: others' depth change $< 3e\text{-}3$ and position change < 5 nm when a tweezer moves; proposal budget 10 nm rms position noise.) Leaving all chips coherent breaks the position budget: 9.2 nm rms of slow wander plus 5.1 nm when neighbours move.

10.2 What the frequency offset actually does

Light from chip m reaches row m+1 at about 1e-4 of the peak intensity (module 13). If the two chips are mutually coherent, this field interferes with the neighbour's own field, and the depth changes by about 2 sqrt(epsilon) cos(phi), where phi is the relative optical phase of the two feeds. Separate fibres let phi drift

with temperature and acoustics. A frequency offset delta f makes phi advance at 2 pi delta f. When delta f is far above the trap frequencies, the atom sees only the time average, and the interference term vanishes. Any process that randomises phi on time scales much shorter than the trap period does the same. A frequency offset is one such process; a common random phase seen with different delays is another.

10.3 Common phase modulation with fibre delays

If the seed carries a phase phi(t) and chip m receives it after a delay tau_m, the relative phase of chips m and n is phi(t - tau_m) - phi(t - tau_n). Its mean contribution to the interference is the field coherence |<exp(i [phi(t) - phi(t - tau)])>| at tau = tau_m - tau_n.

modulation	residual coherence, delays 2 to 15 bits	with finite modulator bandwidth (10-90% rise 0.3 bit)	at arbitrary (non-integer) delays
binary 0/pi m-sequence, N = 1023	9.8e-4 (= 1/N)	4.1e-2 (carrier 0.12)	as integer
same, depth error +5% / +10%	5.2e-3 / 2.4e-2		
Zadoff-Chu polyphase, N = 1021	< 1e-13	4.7e-3	6.2e-2
Gaussian phase noise, 3 rad rms	4.2e-3 (record-limited; ensemble exp(-9) = 1.2e-4)	inherent	7.2e-3
Gaussian phase noise, 4 rad rms	5.6e-3 (record-limited)	inherent	8.2e-3

Binary modulation is fragile. Every 0-to-pi and pi-to-0 transition passes through +pi/2, so a band-limited modulator leaves a carrier (0.12 at 0.3-bit rise), and the delays must be exact multiples of the bit length. Polyphase codes have perfect autocorrelation but also need exact delays. Gaussian phase noise of 3 rad rms needs neither: the carrier is exp(-sigmaˆ2/2), and any delay longer than about three correlation times decorrelates. The 4e-3 to 8e-3 above is the statistical floor of a 32,768-bit record. What the atom sees is set by the low-frequency density of the interference term, 0.40/B. For the coherent term of module 13 (2 sqrt(epsilon) = 2.1e-2), this gives S_I(2 nu_trap) = 8.8e-14/Hz at B = 2 GHz. The parametric heating rate Gamma = piˆ2 nuˆ2 S_I(2 nu) (O'Hara et al., arXiv:physics/0003049, used in module 3) is then 6e-3 sˆ-1 at nu = 84 kHz, an energy e-folding time above 2 minutes. At B = 0.5 GHz it is 2.4e-2 sˆ-1.

Spectral width. With 3 rad of noise at correlation rate B, the optical spectrum is Gaussian with rms width 0.67 B (1.35 GHz, 5 pm, at B = 2 GHz). Its effect on the tweezers comes through the diffractive objective's axial chromatic shift (16 um/nm, DERIVED for f = 17.3 mm with f proportional to 1/lambda) and the lateral scaling x proportional to lambda:

B	optical rms width	axial rms smear	peak	lateral rms smear at 60 um
0.5 GHz	0.34 GHz	0.02 um	0.99997	0.07 nm
2 GHz	1.35 GHz	0.08 um	0.9998	0.29 nm
10 GHz	6.7 GHz	0.41 um	0.995	1.4 nm

The spectrum is symmetric about the carrier, so the mean light shift at the double-magic wavelength is unchanged to first order. The second-order term needs the curvature of the differential polarisability [VALUE?].

Coherence inside a chip. Path differences within a chip and its relay are at most about 1 mm (about 5 ps). With a 0.5 ns correlation time (B = 2 GHz) and sigma = 3 rad, the coherence loss is sigmaˆ2 (tau/tau_c)ˆ2 / 2 ~ 5e-4. The least-squares synthesis of route E therefore sees an essentially coherent source.

Hardware. A fibre-coupled LiNbO3 phase modulator on the low-power seed and a Gaussian noise source amplified to about 1 V_pi rms (3 rad) over 1 to 2 GHz. Extra fibre is 0, 0.3, 0.6 ... 4.5 m on the 16 chip feeds, before or after the amplifiers. A seed broadened to about 1 GHz also raises the stimulated-Brillouin threshold of the high-power amplifier and delivery fibres, which carry 3 W each [REF?, RECALLED: SBS suppression by phase modulation is standard in high-power single-frequency fibre amplifiers]. If the chosen amplifier already broadens its seed for this reason, the only addition is the fibre delays. Compared with 16 AOMs, this removes 16 RF drivers and the AOM insertion loss (about 15% each [VALUE?]).

10.4 Passive options

Orthogonal polarisation on alternate chips. A half-wave-plate strip in front of every other chip, in the collimated region after the collimators, rotates its light to y. We computed vector focal fields (Richards-Wolf, aplanatic NA 0.5 objective) and summed the intensity over Ex, Ey and Ez (scalar light shift):

separation	polarisations	static: depth / position (worst phase)	moving: depth / position
4 um	orthogonal	8.4e-4 / 1.2 nm	1.6e-4 / 0.3 nm
8 um	parallel	3.3e-3 / 2.1 nm	1.1e-3 / 1.4 nm
12 um	orthogonal	5.0e-5 / 0.3 nm	4.4e-5 / 0.1 nm
16 um	parallel	2.4e-3 / 0.4 nm	3.5e-4 / 0.6 nm

At NA 0.5 the orthogonal neighbours' interference falls by a factor of 26 (the residual comes from the depolarisation of high-angle rays, mainly via Ez). The rows two apart (8 um) share a polarisation and then dominate. The moving-tweezer criteria are met, and the position budget too (2.5 nm rms). The static depth wanders by 0.4% rms with the fibre phases, at Hz to kHz rates, far below twice the trap frequencies.

Three costs follow. (i) The meta-objective must be polarisation-insensitive (propagation-phase meta-atoms with fourfold symmetry rather than geometric phase). (ii) Where the orthogonal fields overlap, the polarisation becomes elliptical in proportion to their relative phase, which produces a vector light shift. Our estimate, from the far-detuned alkali formula for Cs at 1064 nm, is a vector-to-scalar ratio of about 0.09 g_F m_F per unit ellipticity (RECALLED, verify). With an ellipticity of up to about 2e-2 at 4 um, this is about 2e-3 g_F m_F of the trap depth: up to tens of kHz for stretched states, zero to first order for clock states (m_F = 0). (iii) Only two orthogonal states exist, so rows two apart stay coherent.

Passive y apodiser. A Gaussian graded-transmission (or graded-reflectivity) filter at the plane of the y telescope conjugate to the pupil apodises all chips at once. It lowers the 1D crosstalk at 4 um from 8.4e-5 to 3.5e-6 (a = 0.65 pupil radii, beta = 0.7 um). However, it costs a factor 1.84 in power and widens the y spot from 1.35 to 1.73 um. In 2D, the coherent sum of a full row, with the circular pupil, still leaves 4.7e-3 of static depth change at 4 um. It is not competitive.

10.5 Recommendation

Use common phase modulation with fibre delays in place of per-chip AOMs: one modulator, no power penalty, and the same row independence as frequency offsets. Alternating polarisation costs nothing to add, and it gives a second, independent factor of about 26 on the nearest rows if the modulation fails or is switched off.

Next step (i) is simulated in `docs/pm_am.md` (filters narrower than about 100 pm and reflection pairs with R1 R2 > 1.7e-3 must be avoided; a representative chain gives -134.5 dB/Hz). Next steps: (i) measure the delayed self-coherence of the modulated seed after the amplifier (PM-to-AM conversion by any spectral filtering adds intensity noise at GHz; its density at 2 nu_trap should be checked); (ii) choose the noise drive (or a deterministic multi-tone drive with the same autocorrelation); (iii) obtain the second-order differential light shift of the 1064.4350 nm double-magic condition over +/-5 pm.

10.6 Verification notes

value	location	type	pointer
option table	Sec. 1	SIM/DERIVED	results/m15_summary.json from m15_coherent and m15_prbs (review)
2D coherent / incoherent, vector fields	Sec. 1, 4	SIM	results/m15_coherent.json (review)
residual coherence, spectra, chromatic smear	Sec. 3	SIM	results/m15_prbs.json (review)
apodiser scan	Sec. 4	SIM	results/m15_apod.json (review)
heating rates	Sec. 3	DERIVED	O'Hara et al., arXiv:physics/0003049 (used and checked in module 3), density from m15_prbs
16 um/nm axial chromatic shift	Sec. 3	DERIVED	f/lambda for a diffractive objective, f = 17.3 mm

value	location	type	pointer
vector-to-scalar ratio 0.09 g_F m_F	Sec. 4	RECALLED	far-detuned formula (Grimm et al., arXiv:physics/9902072), not read this session; counter-rotating terms neglected
3 rad rms, B 1-2 GHz, 0.3 m delay steps	Sec. 1, 3	CHOICE	design choices

Open placeholders: [REF?] SBS suppression by phase modulation in high-power 1064 nm fibre amplifiers; [VALUE?] AOM insertion loss at 3 W; [VALUE?] second-order differential light shift of the double-magic condition over a 5 pm spectrum; vector polarisability of Cs at 1064 nm from a primary source.

Chapter 11

Phase-to-intensity conversion of the decorrelating phase noise

Follow-up to `docs/row_coherence.md`, which replaces per-chip frequency shifters by common Gaussian phase noise (3 rad rms, correlation rate B = 2 GHz) on the seed. Code: `scripts/m16_pm_am.py`; results `results/m16_pm_am.json` (+ .png). Values are SIM, valid for the element models and parameter ranges stated (CHOICE ranges, not measured components), and should be reviewed before they are quoted.

11.1 Result

Phase noise alone leaves the intensity constant: the simulated control gives -376 dB/Hz, the numerical floor. Intensity noise appears only where an element filters the spectrum. Two elements matter: narrow band-pass filters and multipath (any delayed copy, from two reflections). Gain tilt, fibre dispersion and polarisation crosstalk in PM fibre are harmless. The criterion is a parametric heating rate Gamma = pi^2 nu^2 S(2 nu) of at most 0.1 s^-1 at nu = 84 kHz, i.e. a relative intensity noise (RIN) density at 168 kHz of S <= 1.4e-12/Hz (-118 dB/Hz). A representative chain gives -134.5 dB/Hz (Gamma = 2.5e-3 s^-1), 16 dB inside the criterion. The design rules that follow are: (i) no filter narrower than about 100 pm (20 times the 1.35 GHz rms modulation width) in the light path; (ii) the product of any two reflections in the chain, R1 R2, below 1.7e-3 (amplitude r <= 0.042), i.e. no pair of uncoated glass surfaces in series.

11.2 Elements

RIN density in the 100-300 kHz band, record 131 us at 16 GS/s, worse of two relative phases for copies:

element	case	RIN rms (all frequencies)	RIN at 168 kHz	heating rate
amplifier gain tilt	0.05 / 0.5 / 5 dB/nm	6e-5 / 6e-4 / 6e-3	-237 / -216 / -182 dB/Hz	< 1e-6 s^-1
band-pass, FWHM 1 nm, offset 250 pm	ASE filter	7e-3	-173 dB/Hz	< 1e-6 s^-1
band-pass, FWHM 100 pm, centred / offset 25 pm		1.0e-2 / 7.0e-2	-130 / -133 dB/Hz	7e-3 / 3e-3 s^-1
band-pass, FWHM 30 pm, centred / offset 7.5 pm		9.6e-2 / 0.21	**-110 / -113 dB/Hz**	**0.70 / 0.37 s^-1**
multipath, R1 = R2 = 1e-2, delay >= 0.3 ns	two -20 dB reflections	1.4e-2	-130 dB/Hz	7e-3 s^-1
multipath, R1 = R2 = 1e-2, delay 0.1 / 0.01 ns	on chip, mm scale	7e-3 / 1e-4	-133 / -171 dB/Hz	3e-3 / 1e-6 s^-1
multipath, R1 = R2 = 1e-3 or 1e-4, any delay		<= 1.4e-3	<= -149 dB/Hz	<= 8e-5 s^-1
PM fibre, PER -20 dB, 5 / 20 m (DGD 1.2 ps/m)	then the chip TE guide as polariser	3e-4 / 5e-3	-161 / -137 dB/Hz	6e-6 / 1.4e-3 s^-1
dispersion, 20 ps^2/km, 5 m to 1 km		<= 8e-4	< -280 dB/Hz	0
chain (Sec. 3)		6.9e-3	-134.5 dB/Hz	2.5e-3 s^-1

The structure follows from the physics. Gain tilt and dispersion respond to the instantaneous frequency, the time derivative of the phase, whose spectrum vanishes towards low frequency. So their intensity noise lies at GHz, not at 168 kHz. A delayed copy of amplitude r produces an intensity term 2 r cos(phi(t) - phi(t - tau)). Once tau exceeds the correlation time (about 0.3 ns), this term is a random telegraph-like signal of variance 2 r^2 spread over about B, so S ~ 4 r^2 x 0.40 / B (the 0.40/B is the low-frequency density found in module 15). For tau far below the correlation time the term is quadratic in the phase increment and much smaller. A band-pass narrower than the spectrum is the multipath effect again: it is a delay distributed over its impulse response, about 1/(pi x FWHM) = 40 ps for 30 pm.

Dependence on B (r = 1e-2): at tau = 1 ns, S is -125, -130 and -138 dB/Hz for B = 0.5, 2 and 8 GHz, because the noise spreads as 1/B. At tau = 0.1 ns, B = 0.5 GHz is the quietest (-170 dB/Hz): there the copy is still inside the correlation time. A higher B therefore helps against long-delay reflections and hurts against short ones. B = 2 GHz is a reasonable middle.

11.3 Representative chain

Amplifier tilt 0.5 dB/nm; ASE filter of 1 nm FWHM, offset 250 pm; PM fibre 5 m with -20 dB polarisation crosstalk; two -30 dB connector reflections 5 m apart (tau = 25 ns); an on-chip etalon between a -30 dB input facet and the 2.3% glued emitter facet (module 14), 190 ps round trip; 5 m of dispersion. Result: RIN rms 6.9e-3 (at GHz), -134.5 dB/Hz at 168 kHz, Gamma = 2.5e-3 s^-1 (energy e-folding about 7 min). The on-chip etalon and the connector pair dominate.

11.4 Comparison with no modulation

Without phase modulation, the same reflections do not produce noise at 168 kHz. Instead they give a slow intensity offset 2 r cos(Phi) that follows the drift of the path phase Phi: 2% for r = 1e-2, at Hz to kHz rates. The modulation converts this slow, uncontrolled offset into broadband noise of the same rms. Spread over GHz, it deposits only 1e-13 to 1e-12 /Hz near the trap frequencies. Phase noise therefore helps against parasitic etalons as well as against row crosstalk, provided rule (ii) holds.

11.5 Limits and what to measure

The elements are linear and treated separately. Gain dynamics of the amplifier (which can suppress or shape slow intensity noise) and nonlinear effects (self-phase modulation, stimulated Brillouin scattering) are not modelled. The criterion uses the radial frequency; the axial frequency is lower and its parametric resonance lies where the RIN density is the same, so the conclusion holds. Measurement: delayed self-heterodyne or direct RIN of the modulated, amplified light after the full chain, with the modulation on and off, at 10 kHz to 1 MHz. The prediction to test is a floor below -130 dB/Hz with a 1 nm ASE filter and AR-coated components.

11.6 Verification notes

value	location	type	pointer
all RIN densities, heating rates	Sec. 1-4	SIM	results/m16_pm_am.json (review)
element parameter ranges	Sec. 2, 3	CHOICE	stated ranges, not component data
PM-fibre DGD 1.2 ps/m, beta2 20 ps^2/km	Sec. 2	RECALLED	verify for the chosen fibre
Gamma = pi^2 nu^2 S(2 nu)	Sec. 1	SOURCE	O'Hara et al., arXiv:physics/0003049 (module 3)
0.1 s^-1 criterion	Sec. 1	CHOICE	energy doubling time ~7 s

Chapter 12

3D transport with handoff between the static array and the moving tweezers

> Follow-up (`docs/rearrangement.md`): slower handoff steps (200 us ramps, 400 us lane steps) cut the heating per move to 0.12 to 0.16 uK at 0.1 / 0.3 mK, confirmed both quantum mechanically and by 3D classical simulation. The shallow 0.3 mK point is then also the low-heating point. The static array's depth uniformity is not a constraint.

Follow-up to `docs/scaling_beyond.md`, whose 1D model left out handoff, the static array and the third dimension. Code: `picsim/transport3d.py`, `scripts/m18_transport3d.py`; results `results/m18_*.json`, `results/m18_trap.npz`; tests `tests/test_transport3d.py`. Values are SIM (classical Monte Carlo) and valid in the regime stated. They should be reviewed before they are quoted.

12.1 Result

A moving tweezer of the adopted design picks an atom out of a static 1075 nm trap, carries it 12 or 40 um and drops it into an empty static trap. With the right depths and path, it delivers 3,000 of 3,000 simulated atoms (loss < 1e-3 at 95% confidence) and heats them by 1 to 2 uK per move. Three conditions are needed:

(i) **Move along lanes between the rows, not along the rows.** A tweezer that crosses empty static sites on its row loses 45 to 95% of atoms at most settings. Only a moving trap five times deeper than the static traps, moving slowly (400 us), recovers 99.1 to 99.7%. Along a lane 2 um off the row, reached with the row-select y tilt, the static traps are 1.6e-5 of their depth away and transport is clean. All tweezers of a stack enter and leave the lanes together.

(ii) **Moving depth at least three times the static hold depth.** At a ratio of 1.5, 5% of atoms stay behind; at 2, 3 to 5% are lost; at 3, 99.5 to 99.9% are delivered with about 10 uK of heating per move; at 5, 99.6 to 100% with 1 to 2.6 uK.

(iii) **Hold depth of about 20 kT.** Atoms waiting next to a lane feel the passing tweezer's out-of-focus light (6 to 15% of its peak at 1.5 to 2.5 um, a few um from focus). At a hold depth of 10 kT, 1 to 2% of them are lost from the hot tail; at 20 kT, none of 3,000.

Operating points (12 um move in 400 us, 100 us ramps, 200 kHz updates, lane 2 um):

atoms	hold	moving	delivered	heating per move	waiting atoms lost	power per tweezer	720 tweezers
5 uK	0.1 mK	0.3 mK	3000/3000	9.9 uK	0/3000	14 mW	10 W
5 uK	0.1 mK	0.5 mK	3000/3000	1.8 uK	<= 2/3000	24 mW	17 W
10 uK	0.2 mK	0.6 mK	3000/3000	10.0 uK	0/3000	29 mW	21 W
10 uK	0.2 mK	1.0 mK	3000/3000	0.9 uK	<= 3/3000	48 mW	34 W

(Power scales from 57 mW at 1.2 mK, `docs/collimator.md`.) Hence the required depth is set by the atom temperature, not by the move: about 60 kT for the moving trap, or 100 kT for low heating. Cooling the atoms to 5 uK halves the laser power. The 1D estimate of 0.2 mK in `docs/scaling_beyond.md` is too optimistic once handoff and waiting atoms are included.

12.2 Model

Moving tweezer. The intensity of one tweezer of design D1 (BPM response, least-squares synthesis), through the Weierstrass + aplanat collimator and the circular NA 0.5 pupil, computed by scalar focusing with defocus on a grid of 7 x 6 x 20 um (`m18_field`). FWHM 1.73 um (x, along the row), 1.37 um (y), 9.8 um (z); trap frequencies at 1 mK 54, 67 and 22 kHz. Its shape is held fixed while it moves; positions and depth are sampled and held at 200 kHz.

Static array. Gaussian beams at 1075 nm, waist 0.85 um (CHOICE), Rayleigh range 2.1 um, 4 um pitch, three rows of 15 sites, all at one global hold depth (a fixed meta-optic array cannot switch single sites). The two wavelengths add incoherently.

Atoms. Cs, classical, velocity Verlet with 0.25 us steps, gravity along y, 1,500 to 3,000 atoms per case. Each starts in a thermal distribution (harmonic sampling), truncated to atoms bound in the real Gaussian site.

Sequence. Ramp the moving trap up on the source site (sinˆ2, 100 us); move 2 um in y to the lane (100 us); move in x (minimum jerk); move back to the row (100 us); ramp down on the target (100 us); hold 100 us. An atom counts as delivered if it is bound in the target static trap after the moving trap is off.

12.3 Tolerances

Hold 0.1 mK, moving 0.3 / 0.5 mK, 10 uK, lane, 12 um in 400 us (fraction delivered):

parameter	value	moving 0.3 mK	moving 0.5 mK
none		0.9993	0.9993
axial offset of the 1064 and 1075 nm foci	0.5 / 1 / 2 / 4 um	0.998 / 0.995 / 0.939 / 0.455	0.995 / 0.997 / 0.957 / 0.902
lateral offset, y	0.1 / 0.2 / 0.4 um	1.000 / 0.998 / 0.999	1.000 / 1.000 / 0.998
lateral offset, x	0.1 / 0.2 / 0.4 um	0.998 / 1.000 / 0.998	1.000 / 1.000 / 0.999
ramp time	25 / 200 us	0.994 / 0.998	0.993 / 1.000
move	40 um in 400 / 800 us	0.998 / 0.998	0.998 / 1.000
updates	smooth instead of 200 kHz	0.999	0.999
gravity	off	0.999	0.999

Three points follow. (i) **Axial registration** of the two foci is the one tight alignment, to about +/-1 um. A diffractive objective shifts focus by about 170 um between 1064.4 and 1075 nm (16 um/nm, DERIVED), so the two paths must be focused independently before they are combined, and held to 1 um. (ii) **Lateral registration** is loose: 0.4 um costs nothing. The 16 nm row-placement budget of the stacked relay is therefore not a constraint for transport. (iii) **Speed** is not limiting in this regime: 40 um in 400 us is as safe as 12 um.

12.4 On the row versus along a lane

Along the row, the moving trap passes over empty static sites. The total potential then develops a dip that moves with it and kicks the atom. Losses depend strongly on the move time, with resonances where the transit time over a site matches the trap period:

hold / moving	200 us	400 us
0.1 / 0.3 mK	5 to 7% delivered	54 to 57%
0.1 / 0.5 mK	42 to 48%	99.1 to 99.7% (heating 14 uK)
0.2 / 0.5 mK	11%	99.3 to 99.6% (heating 24 uK)

The lane removes this entirely. It costs one shared y tilt per pass (100 us each way), which the row-select mirror already provides.

12.5 Limits and next steps

The model is classical. Quantum ground-state effects matter when the energy gain is near one trap quantum (about 1 to 3 uK here), so the 1 to 2 uK operating points should be checked quantum-mechanically. Light-assisted processes, photon scattering (a few per second at 0.5 mK, estimated from the far-detuned rate, not simulated) and the vector light shift are not included. The moving trap's shape is assumed invariant along the row; least-squares synthesis keeps it within 0.1% (module 10). Its neighbours in the same row (independent tweezers 4 um away) are not included. The static traps are ideal Gaussian beams.

Next steps: (i) a rearrangement scheduler using lanes and passes, to count moves per atom (heating accumulates at 1 to 10 uK per move); (ii) a quantum check of the lowest-heating points; (iii) the tolerance of the static array's own depth uniformity.

12.6 Verification notes

value	location	type	pointer
moving-trap field, widths, frequencies	Sec. 2	SIM	results/m18_field.json, m18_trap.npz
scan (hold, moving, ramp, move, path)	Sec. 1, 4	SIM	results/m18_scan.json (1,500 atoms; review)
tolerances	Sec. 3	SIM	results/m18_tol.json (review)
temperature rule	Sec. 1	SIM	results/m18_temp.json
waiting atoms, hold margin, operating points	Sec. 1	SIM	results/m18_bystander.json, m18_margin.json (3,000 atoms)
power per tweezer	Sec. 1	DERIVED	57 mW at 1.2 mK (m14_plan) scaled with depth
static waist 0.85 um, ramps, lane 2 um, 5 and 10 uK	Sec. 2	CHOICE	stated assumptions
170 um chromatic focal shift	Sec. 3	DERIVED	16 um/nm x 10.6 nm, diffractive objective

Chapter 13

Rearrangement cycles, quantum check of the handoff, and static-array uniformity

Follow-up to `docs/transport3d.md`. Code: `picsim/rearrange.py`, `scripts/m20_rearrange.py` (scheduler), `scripts/m19_quantum.py` (quantum check), `scripts/m18_transport3d.py` parts `uniformity` and `slowlane`; results `results/m20_rearrange.json`, `m19_quantum.json`, `m18_uniformity.json`, `m18_slowlane.json`. Values are SIM and valid in the regime stated. They should be reviewed before they are quoted.

13.1 Result

A 16-chip stack with row select assembles a defect-free 32 x 32 block from a 50 x 45 array loaded at 50% in 17 ms (mean of 40 loadings; worst 23 ms), using about 8 passes. Every loading succeeded before losses. The average atom in the block moves 1.12 times; none moves more than twice. With slower handoff steps (200 us ramps, 400 us lane steps), each move heats the atom by 0.2 to 0.3 uK at a moving depth of only 0.3 mK. The quantum and 3D classical models agree on this. The finished block is therefore no hotter than about 0.6 uK above its starting temperature. Loss sets the outcome: at 1e-3 loss per move, a cycle leaves 1.15 defects on average; at 1e-4, 0.12. Vacuum loss during the 17 ms adds 0.59 defects for a 30 s lifetime and 0.06 for 300 s. The static array's depth uniformity is not a constraint: 20% rms random depth errors cost nothing.

target (loading)	addressed rows	passes (vertical + horizontal)	cycle time, mean / max	moves per atom, mean / max	defects from moves at 1e-3 / 1e-4	defects from vacuum at 30 / 300 s
30 x 30 (50%)	16	4.3 + 2	13.3 / 17.1 ms	1.11 / 2	1.00 / 0.10	0.40 / 0.04
32 x 32 (50%)	**16**	**5.8 + 2**	**17.2 / 23.0 ms**	**1.12 / 2**	**1.15 / 0.12**	**0.59 / 0.06**
32 x 32 (60%)	16	5.0 + 2	14.8 / 17.1 ms	1.03 / 2	1.06 / 0.11	0.50 / 0.05
36 x 36 (60%)	16	8.3 + 3	25.1 / 35.6 ms	1.07 / 2	1.38 / 0.14	1.08 / 0.11
32 x 32 (50%)	32 (two stacks)	5.8 + 1	15.5 / 21.3 ms	1.12 / 2	1.15 / 0.12	0.53 / 0.05

The probability that a 32 x 32 block is defect-free after one cycle is 0.84 at 1e-4 loss per move and a 300 s lifetime. Reaching 0.99 in one cycle would need about 5e-6 loss per move and a vacuum lifetime above about an hour; a repair round after imaging is the realistic route.

13.2 Scheduler

The hardware sets the rules: (i) a stack addresses 16 consecutive rows at a time; (ii) every tweezer of the stack shares the y motion (into the lane 2 um off the row, optionally along mid-column lanes by j rows, and

back), while each moves independently in x; (iii) tweezers in one row keep their order and so stay at least 4 um apart.

Vertical balancing. Each pass chooses a stack position k and a common row shift j that move the most surplus atoms into target rows lacking atoms. Arriving atoms go straight to empty target columns where possible. Atoms that arrive in a pass are not picked again in the same pass.

Horizontal compaction. One pass per group of 16 target rows. Each row's atoms are matched in order to the target columns, with extras pushed just outside the block.

A pass takes two ramps, two lane steps, the x moves (min-jerk; 400 us per 40 um, scaled as sqrt(d), at least 200 us), the y travel for j rows, and 0.3 ms of scanner settling (CHOICE). This greedy schedule is not optimal. A planner that combines balancing and compaction, or shares passes between stack positions, would need fewer passes.

Sensitivity of the cycle time (32 x 32, 50%, one stack):

timing	cycle time
base (ramps 100 us, lane steps 400 us)	17.2 ms
low-heating (ramps 200 us, lane steps 400 us)	18.8 ms
fast lane steps (100 us)	12.5 ms
all fast (ramps 50, lanes 100, x 200 us per 40 um)	8.2 ms
scanner settling 1 ms instead of 0.3 ms	22.7 ms

Because vacuum loss grows with cycle time, the fast setting halves it (0.28 defects at 30 s). It costs heating: about 10 uK per move at 0.3 mK, or 1.2 uK at 0.5 mK (Sec. 3). Hence two regimes: slow and cold at 0.3 mK (low power), or fast at 0.5 mK.

13.3 Quantum check of the handoff

A 1D split-step propagation along y (the handoff and lane axis) starts the atom in the ground state of the static site. The y profile is the moving tweezer's, taken from the 3D grid, with gravity included. The x move is treated separately. Results are compared with classical trajectories from a Wigner sample of the same ground state:

hold / moving	ramp / lane step	energy gain, quantum	ground state kept	energy gain, classical
0.1 / 0.3 mK	100 / 100 us	9.7 uK	0.08	11.6 uK
0.1 / 0.3 mK	100 / 200 us	2.0 uK	0.21	1.4 uK
0.1 / 0.3 mK	**200 / 400 us**	**0.12 uK**	**0.98**	0.94 uK
0.1 / 0.5 mK	100 / 100 us	0.84 uK	0.84	1.53 uK
0.1 / 0.5 mK	200 / 400 us	0.13 uK	0.99	0.87 uK
0.2 / 0.6 mK	200 / 400 us	0.15 uK	0.99	1.72 uK
0.2 / 1.0 mK	100 / 100 us	1.58 uK	0.66	1.58 uK

(Trap quantum of the static site: 1.4 uK at 0.1 mK, 2.0 uK at 0.2 mK.) The x move adds 0.08 to 0.37 uK in both models (ground state kept 0.98 to 0.99). Two findings follow. (i) The lane step, not the x move, sets the heating, and it falls steeply with its duration: from 100 to 400 us, the gain at 0.1 / 0.3 mK drops 80-fold. (ii) Above about one trap quantum the classical model agrees with the quantum one to within a factor of 2, on the pessimistic side. Below it, classical Wigner sampling overstates the heating by a factor of up to 10, as expected for zero-point motion. A 3D classical run with the slow steps confirms the delivery: 3,000 of 3,000 atoms, with 0.16 uK (5 uK atoms, 0.1 / 0.3 mK) and 0.59 uK (10 uK, 0.2 / 0.6 mK) of heating, and no loss of waiting atoms.

13.4 Static-array depth uniformity

2,000 atoms per case, lane transport, three operating points (results/m18_uniformity.json):

non-uniformity	5 uK, 0.1 / 0.3 mK	5 uK, 0.1 / 0.5 mK	10 uK, 0.2 / 0.6 mK
source site deeper by 20 / 40 / 60%	1.000 / 1.000 / 0.996	1.000 / 1.000 / 1.000	1.000 / 1.000 / 1.000
target site shallower, x 0.8 / 0.6 / 0.4	1.000 / 0.999 / 0.998	1.000 / 1.000 / 1.000	1.000 / 1.000 / 0.9995
random depth errors 5 / 10 / 20% rms (two draws)	1.000 all	1.000 all	1.000 all

The handoff tolerates the static array's depth errors far beyond what a meta-optic or SLM array would show. The energy figures in `m18_uniformity.json` include the adiabatic change between unequal source and target depths and should not be read as heating.

13.5 Limits and next steps

The scheduler is greedy and uses one rearrangement cycle without repair. The loss per move is not computed: the simulations bound it below about 1e-3 (0 of 3,000 atoms), and 1e-4 is an assumption. The quantum model is 1D per axis. Photon scattering by the trapping light (estimated at a few per second at 0.3 to 0.5 mK) and imaging are not included; neither are moves of atoms whose heating would make them leave the target sites during later moves.

Next steps: (i) a repair round after imaging, with its own pass count and time; (ii) an optimised schedule (passes shared across stack positions); (iii) a 3D quantum or semiclassical check of the axial motion during the handoff.

13.6 Verification notes

value	location	type	pointer
passes, times, moves, defect estimates	Sec. 1, 2	SIM	results/m20_rearrange.json (40 loadings; review)
quantum and Wigner energy gains	Sec. 3	SIM	results/m19_quantum.json (review)
slow-lane 3D check	Sec. 3	SIM	results/m18_slowlane.json (3,000 atoms)
uniformity table	Sec. 4	SIM	results/m18_uniformity.json (2,000 atoms)
per-move loss 1e-3 / 3e-4 / 1e-4, vacuum lifetimes 30 / 300 s, scanner settling 0.3 ms	Sec. 1, 2	CHOICE	assumptions

Chapter 14

Repair rounds, scheduler variants and the axial handoff

> Follow-up (`docs/megascale.md`): scaling to 1e4, 1e5 and 1e6 sites. A 32 x 4 engine block assembles 1e6 sites in 1.5 s and holds them at the loss-limited floor rho ~ 1.5-1.8 (eps_img + T_img/tau); static trapping power, the objective's field and the 3 um lane distance are the walls.

Follow-up to `docs/rearrangement.md`. Code: `picsim/rearrange.py` (`repair`, `cycle_with_repair`, selection modes), `scripts/m21_repair.py`, `scripts/m19_quantum.py axial`; results `results/m21_*.json`, `results/m19_axial.json`. Values are SIM and valid in the regime stated. They should be reviewed before they are quoted.

14.1 Result

After the rearrangement cycle (32 x 32 block from a 50 x 45 array at 50% loading, 17 ms), one round of imaging and direct repair takes 10 to 13 ms: a 10 ms image and 0.1 to 1.3 repair passes. Each repair moves a single reservoir atom through the lanes into a hole. The repair works, but it cannot get below a floor set by the image itself. Atoms lost because of the image are not seen by it, and every atom is also exposed to vacuum loss for the round's duration. The floor is

```
defects ~ N (eps_img + t_round / tau) + p_move x (repairs),   N = 1024,
```

about 0.14 defects for eps_img = 1e-4 and tau = 300 s. A second round does not lower it. Hence the probability of a defect-free 32 x 32 block is set by the imaging loss and the vacuum lifetime, not by the transport, once the per-move loss is about 1e-4 or less:

per-move loss	vacuum lifetime	imaging loss	P(defect-free): after cycle / after 1 repair / after 2	defects after 1 repair
1e-3	300 s	1e-4	0.27 / **0.87** / 0.87	0.14
1e-4	300 s	1e-4	**0.84** / 0.87 / 0.87	0.14
1e-3	300 s	1e-3	0.27 / 0.35 / 0.33	1.04
1e-4	300 s	1e-3	**0.84 / 0.35** / 0.33	1.03
1e-3	30 s	1e-4	0.14 / 0.59 / 0.62	0.53
1e-4	30 s	1e-4	0.48 / 0.62 / 0.63	0.48

(40 loadings x 25 loss draws per row.) With good transport (1e-4) and an imaging loss of 1e-3, a repair round makes things worse: 0.84 becomes 0.35, because the image costs about one atom of the block. Imaging should then be skipped, or used only to post-select. For a 99% defect-free block one needs roughly 1e-5 for each term: per-move loss, imaging loss per image, and t_round / tau. These are properties of the platform, not of the photonic chip. The chip's contribution is 1.1 moves per atom and 17 ms per cycle.

14.2 Scheduler variants

Balancing-pass selection on 40 loadings (32 x 32, 50%, one stack):

mode	passes (all)	cycle time, mean / max
greedy (most atoms per pass)	8.05	17.8 / 23.1 ms
rate (most atoms per unit pass time)	8.05	17.6 / 22.8 ms
lookahead (best two-pass sequence among the 8 best first passes)	7.98	17.6 / 22.9 ms

The three modes agree to within 1%. The balancing stage has a floor of about four passes, set by the geometry. In one pass the 16-row window can hold at most 9 non-target rows above or below the 32-row target band, so it can feed at most 9 target rows. Feeding all 32 therefore takes at least four passes, plus two compaction passes. The greedy schedule (5.8 + 2) is within about 30% of that floor. Further gains would come from shorter passes, not fewer: ramps, lane steps and scanner settling make up 1.3 of the 2.2 ms per pass.

14.3 Axial motion during the handoff

1D split-step propagation along the optical axis from the ground state of the static site (axial trap quantum 0.40 uK at 0.1 mK). The static site's depth follows the lateral separation of the moving-trap centre, which carries the atom 2 um into the lane and back, and the moving trap's axial profile comes from the 3D grid. The parameter is the axial offset dz between the 1075 nm and 1064 nm foci (results/m19_axial.json):

hold / moving	ramp / lane step	dz = 0	0.5 um	1 um	2 um
0.1 / 0.3 mK	100 / 100 us	0.06 uK	4.1 uK	16.6 uK	40.7 uK
0.1 / 0.3 mK	200 / 400 us	< 0.01 uK	**0.19 uK**	2.0 uK	8.3 uK
0.1 / 0.5 mK	100 / 100 us	0.02 uK	6.4 uK	31.0 uK	28.3 uK
0.1 / 0.5 mK	200 / 400 us	< 0.01 uK	**0.22 uK**	1.8 uK	4.4 uK

(Quantum energy gain; the classical Wigner values agree to within 20% above 1 uK and are pessimistic below.) With the foci coincident, the axial motion is not excited: the change of axial trap frequency during the handoff is adiabatic. A focus offset displaces the axial equilibrium twice per move, and that is what heats. With the slow handoff, keeping the heating per move below the lateral 0.12 to 0.15 uK of `docs/rearrangement.md` needs dz <= 0.5 um. A 1 um offset costs about 2 uK per move, and 2 um costs 4 to 8 uK. This tightens the +/-1 um delivery tolerance of `docs/transport3d.md` into a heating tolerance of +/-0.5 um. Since a diffractive objective puts the two foci about 170 um apart, the 1064 nm path needs a focus adjustment that holds 0.5 um (0.3% of the chromatic offset), for example a slow stage on the relay's last y and x lenses, calibrated on the atoms.

14.4 Limits and next steps

The repair model moves one reservoir atom per hole, through the lanes; repaired atoms come from the nearest rows under the pass rules. Detection is assumed perfect apart from the imaging-induced loss, with no false positives or negatives. The imaging time (10 ms) and losses are CHOICE values, not measured. The axial check is 1D, with the atom assumed to sit at the moving-trap centre laterally.

Next steps: (i) image-free or low-loss detection (the floor of Sec. 1); (ii) shorter passes (Sec. 2); (iii) an active focus lock between the 1064 and 1075 nm paths to 0.5 um (Sec. 3).

14.5 Verification notes

value	location	type	pointer
repair table	Sec. 1	SIM	results/m21_repair.json (review)
scheduler modes	Sec. 2	SIM	results/m21_modes.json
axial quantum and Wigner results	Sec. 3	SIM	results/m19_axial.json (review)
eps_img 1e-3 / 1e-4, t_img 10 ms, tau 30 / 300 s, p_move 1e-3 / 1e-4	Sec. 1	CHOICE	assumptions

Chapter 15

Rearranging 10^4, 10^5 and 10^6-atom arrays at 3-4 um pitch

> Follow-up (Sec. 9): a 3 x 4 um lattice (3 um along the rows, rows 4 um apart) transports as cleanly as 4 x 4 um and has 33% more sites. This supersedes the corridor conclusion of Sec. 5. A free-space lattice needs more power than tweezers; only a cavity lattice (buildup ~100) needs less. At 10^6 sites, fast handoff steps are needed in steady state as well, at a cost of at most 10 uK per refilled atom.
>
> Follow-up (Sec. 10): engines tiled along x can sit at 3.0 mm pitch (ghost below 2e-7 of a tweezer peak), so six fit on an 18 mm chip edge. The 3 x 4 um floor of 8e-4 in Sec. 9 was an edge artefact; it is 2.4e-4 at image, as on 4 x 4 um. The defect fractions quoted in Sec. 1-9 are sampled just after an image and count imaging losses twice; the time average is about 25% lower (1.8e-4 against 2.4e-4 at the nominal point). Intermediate steps (2 uK per move) reach a time average of 2.2e-4 at 10^6 sites with the 128-engine block if the imaging period is raised from 0.1 to 0.3 s.
>
> Follow-up (Sec. 11): stacks combine in space as well as by polarisation. A slotted 45 deg comb mirror in the 184 um gaps between chip beams at the collimator plane interleaves two 16-chip stacks built at 1.0 mm pitch, at a worst-row cost of 0.45% in power and, in 2D, a full-row crosstalk of 2.3e-4 at 4 um (1.8e-4 without the comb). Two such pairs and a PBS form a 64-row block, so the 64 x 4 block (256 engines on 12 mm chip edges) is buildable.
>
> Follow-up (Sec. 12): with the real (glued, traced) collimator the comb results hold. But a cylindrical collimator at NA 0.9 defocuses the light of tweezers away from the row centre (equivalent facet focus error -99 s_x^2 um, s_x = x_a / 288 um). With the best common focus offset a row is usable over |x_a| <= 45 um, half of the 180 um segment. Earlier power and crosstalk figures hold near the row centre only; a collimator that is insensitive to the x angle (for example all-reflective) is needed for full segments.

Follow-up to `docs/scaling_beyond.md`, `docs/transport3d.md`, `docs/rearrangement.md` and `docs/repair.md`. Code: `scripts/m22_megascale.py` (budget, tiling), `picsim/megasort.py` and `scripts/m23_steady.py` (assembly and steady state), `scripts/m18_transport3d.py pitch3` (3 um pitch); `picsim/abcd.py`; results `results/m22_*.json`, `results/m23_steady.json`, `results/m18_pitch3.json`; tests `tests/test_megascale.py`. Values are SIM, DERIVED or CHOICE as marked, and valid in the regime stated. They should be reviewed before they are quoted.

15.1 Result

At 4 um pitch the transport side scales to 10^6 sites. One engine block of 32 rows x 4 engines (two 16-chip stacks, 128 engines) assembles 10^6 sites from 85% loading in 1.1 s to 1e-2 defects and 1.5 s to 1e-3, with fast handoff steps. It then holds the array at 2.3e-4 defects per site in steady state. Smaller arrays reach the same floor faster: 10^4 sites in 0.1 to 0.3 s, 10^5 sites in 0.2 to 0.6 s. The steady-state floor is not set by the transport. It follows the loss rates and the imaging cadence,

```
rho ~ (1.5 to 1.8) x (eps_img + T_img / tau),
```

where eps_img is the loss per image, T_img the imaging period and tau the vacuum lifetime. With eps_img = 1e-4, T_img = 0.1 s and tau = 3000 s this gives 2.3e-4, at every array size.

Three walls come before the transport does:

(i) **Static trapping power.** 4.5 mW per site per mK at the atoms (w0 0.85 um, 1075 nm). 10^6 sites plus reservoir need 0.5 to 0.9 kW at the atoms at a 0.1 mK hold, and 5 to 9 kW at a 1 mK loading or imaging depth (18 to 30 kW of laser at 30% delivery).

(ii) **The objective's field.** A single objective needs a corrected field of radius 0.2 to 0.3 mm for 10^4 sites, 0.7 to 0.9 mm for 10^5, and 2.1 to 2.8 mm for 10^6 (about 1.4e7 resolution cells). Zones of +/-250 um would need 2, 8 to 13, or 72 to 127 separate objectives.

(iii) **Pitch.** At 3 um pitch the lane between rows is 1.5 um from the waiting atoms. The moving tweezer then pulls 55 to 99% of them out, so 3 um needs empty corridor rows, and these cost most of the density gain.

	10^4 sites	10^5 sites	10^6 sites
side at 4 / 3 um	0.40 / 0.30 mm	1.26 / 0.95 mm	4.0 / 3.0 mm
single objective: field radius	0.28 mm	0.89 mm	2.8 mm
zones of +/-250 um (objectives)	2	13	127
static power at the atoms, 0.1 mK hold / 1 mK (85% loading, incl. reservoir)	5.3 / 53 W	53 / 530 W	0.53 / 5.3 kW
engine block (32 x 4): field positions	10	28	252
assembly to 1e-3 (85% loading, 32 x 4)	1.1 s (reload-limited)	0.12 to 0.35 s	1.5 s (fast steps)
steady-state defects, tau 3000 / 300 s	2.3e-4 / 6.0e-4	2.3e-4 / 6.0e-4	2.3e-4 (fast) / 6.0e-3 (slow, 300 s)
refills needed per second (tau 3000 s, eps 1e-4 at 10 Hz)	13	134	1,335

15.2 Optics: one large field or many zones

A square array of side L needs a corrected field of radius L/sqrt(2). In units of the resolution cell (1.06 um at NA 0.5) that is 372, 1,187 and 3,754 cells per side at 4 um pitch. The proposal's +/-100 um field covers none of these. 10^4 sites need +/-0.28 mm; 10^5 sites need +/-0.9 mm, which is within the range of refractive NA 0.5 objectives (RECALLED, verify); 10^6 sites need a lithography-class field of +/-2.8 mm at NA 0.5.

Zones (independent arrays, each with its own objective, static array and engine block) avoid the large field. They behave like the smaller arrays: a +/-250 um zone holds 7,700 sites at 4 um and assembles like the 10^4 case. But each needs its own objective, about 25 to 30 mm across at a 15 mm working distance, so 127 zones for 10^6 are impractical. Zones suit 10^4 (2 zones) to 10^5 (13). For 10^6, a single large-field objective, or a few large zones, is unavoidable.

15.3 Static trapping power

From `trap_depth_K` with alpha(1075 nm) = 1120 au and a 0.85 um waist: 4.5 mW per site per mK at the atoms. The reservoir sites must be counted too. The budget uses the minimum reservoir fraction, 1 - f: 50% of all sites at 50% loading and 15% at 85%. The simulated 3:1 layout at 85% uses 25%, which adds 13% to these powers. At the 0.1 mK hold of `docs/transport3d.md`, 10^6 sites take 0.53 kW at the atoms with 85% loading; at a 1 mK loading or imaging depth, 5.3 kW. With 30% delivery (CHOICE) the laser must supply 1.8 to 3 kW at the hold and 18 to 30 kW at 1 mK. This is the first wall at 10^6. Three levers bear on it, all outside this study: (i) loading and imaging at lower depth (power scales with depth); (ii) a smaller static waist; (iii) an optical lattice in an enhancement cavity for the static potential (as in the continuously operated lattice array of Gyger et al., arXiv:2402.04994, search summary only).

15.4 Transport hardware: engines, tiling and blocks

One engine (192 ports, 383 fast actuators) serves a 180 um row segment: 45 sites at 4 um, 60 at 3 um. Engines can be tiled in two directions.

Along y, by stacking chips (`docs/stacked_relay.md`). One chip per row, with a collimator per chip.

Along x, several engines on one chip edge. Each gets its own x lenslet; a common Fourier lens overlaps all of them in the pupil and maps their offsets to adjacent atom segments. The ABCD check (`m22_tiling`) gives magnification -60.1 for every engine and all chief rays at the pupil centre. The grating lobes are the catch: with 1.5 um emitter pitch, each engine's first lobe lands 204 um from its spot at the

atoms, inside the neighbouring segment. A shared slit plane must therefore have opaque gaps at least as wide as the slit zones to absorb the lobes. That sets an engine pitch of at least 4.4 mm on the chip (four engines per 18 mm edge) and lets one block position address every other segment (Sec. 10 supersedes this pitch: the wave-optics check allows 3.0 mm, six engines per edge). Two scanner positions cover a row; baffles between the lenslet channels are needed against stray lobe light.

Block. 32 rows (two polarisation-combined 16-chip stacks, or comb-interleaved, Sec. 11) x 4 engines per row = 128 engines, 49,024 fast actuators. It holds 5,888 simultaneous tweezers, which need 85 W at 0.3 mK if all are lit (at most during assembly). In steady state only the few refilling tweezers are lit. A 16 x 1 block (one stack) is too small for 10^6 sites: it never finishes assembling (Sec. 6).

15.5 3 um pitch needs corridors

The 3D transport model at 3 um pitch, with a 1.5 um lane, 200 us ramps and 400 us lane steps (`results/m18_pitch3.json`, 3,000 atoms):

hold / moving	transported atom	waiting atom, same row, lane 1.5 um away	waiting atom, next row, beside the drop
0.1 / 0.3 mK	1.000 delivered	0.335 stay, 0.584 lost	0.149 stay, 0.851 lost
0.1 / 0.5 mK	1.000	0.001 stay	0.006 stay
0.2 / 0.6 mK	1.000	0.225 stay	0.117 stay
0.2 / 1.0 mK	1.000	0.014 stay	0.012 stay

The moving tweezer (1.37 um FWHM across the row, with strong out-of-focus light) captures atoms 1.5 um away. At 4 um pitch the lane is 2 um away, and there waiting atoms are safe at a hold of about 20 kT. Transport at 3 um pitch must therefore run in empty corridor rows. Consider a repeating layout of computation row, corridor, reservoir row, at 3 um. Every transport path is then 3 um from occupied sites, but the computation-site density, (1/3)/9 um^-2 = 0.037 um^-2, is only about 20% above that at 4 um with a 1:1 reservoir layout (0.031 um^-2). Moves across columns would need corridor columns too. Hence 3 um pitch gains little with this spot size. The `m23` runs at 3 um pitch use the 4 um lane model and are geometry-only for that reason.

15.6 Assembly and steady state

The simulation (`picsim/megasort.py`, `m23_steady`) works as follows:

- **Layout:** computation rows interleaved with reservoir rows, 1:2 at 50% loading and 3:1 at 85%.
- **Imaging:** every 0.1 s (10 ms each); 1e-4 loss per image and 1e-4 per move; the reservoir is refreshed every 1 s.
- **Visits:** the engine block goes round-robin to tiles with known holes. Each visit runs up to four repair passes (common row shift, order-preserving x, one move per refill).

At 4 um pitch:

sites	loading	block	timing	to 1e-2	to 1e-3	steady state, tau 3000 / 300 s
10^4	50%	16 x 1 / 32 x 4	slow	0.34 / 0.12 s	same	3.2e-4 / 5.8e-4 (16 x 1)
10^4	85%	32 x 4	slow	0.12 s	1.11 s	2.3e-4 / 6.0e-4
10^5	50%	16 x 1 / 32 x 4	slow	2.0-2.7 / 0.45-0.57 s	2.0-3.0 / 0.45-0.57 s	2.3e-4 / 6.0e-4
10^5	85%	32 x 4	slow / fast	0.23 / 0.12 s	0.34 / 0.12 s	2.3e-4 / 6.0e-4
10^6	50%	32 x 4	slow	4.5 s (tau 3000)	not in 20 s	1.1e-3 / 5.5e-3
10^6	85%	32 x 4	slow	3.6 s	not in 20 s	1.2e-3 / 6.0e-3
10^6	85%	32 x 4	**fast**	**1.1 s**	**1.5 s**	**2.3e-4** (tau 3000)
10^6	50% / 85%	16 x 1	slow	never	never	3-13%

(Slow: 200 us ramps, 400 us lane steps, 0.2 to 0.3 uK per move. Fast: 50 us ramps, 100 us lane steps, 200 us per 40 um, about 1 to 10 uK per move from `docs/rearrangement.md`.)

Three findings:

(i) **Up to 10^5 sites, the floor is set by losses.** It is the same for every block and timing: 2.3e-4 (tau 3000 s) and 6e-4 (tau 300 s).

(ii) **At 10^6, latency matters.** With slow steps, a sweep of 252 block positions takes longer than the imaging period. New holes then wait several sweeps, and the floor rises five-fold. Fast steps restore the loss-limited floor; they cost heating (about 1 to 10 uK per move, but each atom moves once).

(iii) **Each refill is one move**, from a nearby reservoir row into the hole, so heating does not accumulate.

The floor. A loss becomes visible only at the next image, and an image itself loses eps_img per atom. Hence rho ~ k (eps_img + T_img / tau), with k = 1.5 to 1.8 from the simulations. Imaging more often shortens the latency but costs more imaging loss, so the eps_img term is a floor of its own:

- 1e-3 defects: eps_img + T_img/tau <= 6e-4, met by eps_img = 1e-4 and tau >= 200 s at 10 Hz imaging;
- 1e-4 defects: eps_img <= 3e-5 and tau >= 3,300 s at 10 Hz.

For comparison, Manetsch et al. (arXiv:2403.12021, abstract read 2026-11-06) report a 23 min lifetime in a room-temperature apparatus and 99.98952% imaging survival (eps_img = 1.05e-4) for a 6,100-atom Cs tweezer array. That places current technology at the 1e-3 to 3e-4 level.

Supply. In steady state at 10^6 sites the reservoir is consumed at about 1,300 atoms/s (tau 3000 s). Chiu et al. (arXiv:2506.20660, abstract read 2026-11-06) report 30,000 initialized qubits per second from lattice conveyor belts, maintaining more than 3,000 atoms for over 2 hours. That rate would cover the steady-state demand with a margin of 20.

15.7 Walls in order, and what removes each

scale	first wall	second	third	transport (this design)
10^4	field +/-0.28 mm (or 2 zones)	static power 50 W at 1 mK	losses: 2-6e-4 floor	1 stack is enough; < 0.4 s assembly
10^5	field +/-0.9 mm or 13 zones	static power 0.5 kW at 1 mK	losses	32 x 4 block: 0.1-0.6 s assembly
10^6	static power 5 kW at 1 mK	field +/-2.8 mm (SBP 1.4e7)	vacuum and imaging loss (rho >= 2e-4 now); supply 1.3k atoms/s	32 x 4 block with fast steps; 1.5 s assembly

Recommendations.

- **10^4:** one large-field objective or two zones, one 16-chip stack, 4 um pitch.
- **10^5:** a 32 x 4 engine block, 4 um pitch, a field of +/-0.9 mm or about 13 zones, and about 85% loading (it cuts the reservoir from two thirds of the sites to a quarter, and the static power by 40%).
- **10^6:** feasible on the transport side with the 32 x 4 block and fast steps. The static potential needs a different source (lattice or much lower depths) and a lithography-class objective, and the loss rates must improve to reach 1e-4 defects.

15.8 Limits and next steps

The steady-state model assumes perfect detection apart from imaging loss, an external reservoir supply, a single objective whose field reaches every site, and one move per refill. It does not schedule the reservoir refresh physically (conveyor timing) or include photon scattering. Heating per move for the fast steps is taken from the 1D quantum and 3D classical results (1 to 10 uK) and is not tracked per atom here. The x-tiling optics is first order (no baffle or lenslet aberration design). The 3 um corridor layouts are argued from the 4 um transport results, not simulated.

Next steps: (i) a corridor layout at 3 um pitch in the 3D and steady-state models; (ii) a lattice-based static potential and its power budget at 10^6; (iii) an optical design of the large-field objective or of the zone layout; (iv) per-atom heating in the steady-state model, with fast and slow steps mixed (slow for refills, fast for assembly).

15.9 Follow-up: denser lattice, static potential and step timing

3 um along the rows, 4 um across (`results/m18_aniso.json`, 3,000 atoms per case, 200 us ramps and 400 us lane steps). Only the closest approach to waiting atoms matters, and that is set by the lane (2 um from the rows), not by the pitch along the row. Hence a 3 x 4 um lattice transports as cleanly as 4 x 4 um:

layout	hold / moving	transported	heating	waiting atoms lost: in-row neighbour at 3 um, passed by the lane, next row beside the drop
3 x 4 um, lane 2 um	0.1 / 0.3 mK (5 uK)	3000/3000	0.16 uK	0 / 0 / 0
3 x 4 um, lane 2 um	0.1 / 0.5 mK (5 uK)	3000/3000	0.05 uK	0 / 0 / 3 of 3000
3 x 4 um, lane 2 um	0.2 / 0.6 mK (10 uK)	3000/3000	0.59 uK	0 / 0 / 0
3 um, corridor rows 6 um apart, lane 3 um	all three	3000/3000	0.3-1.5 uK	0 / 0 / 0

The 3 x 4 um lattice has 33% more sites per area than 4 x 4 um (0.063 against 0.047 computation sites per um^2 with a 3:1 reservoir layout), at no transport cost. The corridor layout is safe but less dense. This replaces the conclusion of Sec. 5: 3 um pitch works along the rows. Only the row spacing must stay at 4 um.

Static potential: tweezers or a lattice (`results/m22_lattice.json`, DERIVED). The figures are laser power at 30% delivery, for N sites plus a 25% reservoir at 4 x 4 um, at depths of 0.1 / 1 mK. The lattice uses two orthogonal retro-reflected beams (incoherent sum, peak 8 I per beam; coherent four-beam: 16), held within -20% of the peak over the array.

static potential	10^4 sites	10^5 sites	10^6 sites
tweezer array (w0 0.85 um)	20 / 200 W	0.2 / 2.0 kW	2.0 / 20 kW
free-space lattice, Gaussian beams	0.5 / 5.0 kW	5 / 50 kW	50 / 500 kW
free-space lattice, flat-top beams (incoherent / coherent)	85 / 42 W ; 0.85 / 0.42 kW	0.85 / 0.42 ; 8.5 / 4.2 kW	8.5 / 4.2 ; 85 / 42 kW
cavity lattice, Gaussian, buildup 10 / 100	50 / 5 W ; 0.5 / 0.05 kW	0.5 / 0.05 ; 5 / 0.5 kW	5 / 0.5 ; 50 / 5 kW

(Each pair is 10 / 100 buildup or incoherent / coherent; then ";" separates 0.1 mK and 1 mK.) A tweezer puts its power into about pi w0^2/2 = 1.1 um^2 per site. A Gaussian lattice beam that keeps the corners within 20% spreads it over about 15 a^2 / G = 31 um^2. Hence a free-space lattice loses by about 25 with Gaussian beams and by 2 to 4 with flat-top beams. Only a cavity with a buildup near 100 wins, by a factor of about 4. That cavity would need a waist near 10 mm for 10^6 sites, which is demanding. Two levers do more than the choice of potential: depth (a factor of 10 between 1 and 0.1 mK), and a higher delivery efficiency for the simpler lattice optics (up to about 2.7 if 0.8 instead of 0.3). For 10^6 sites, the static potential needs about 2 kW at the hold depth with tweezers, or about 0.5 kW with a buildup-100 cavity lattice. Loading and imaging must happen at a lower depth, or in a smaller zone.

Step timing in steady state, with heating per atom (`results/m23_mixed.json`, 32 x 4 block, 85% loading, tau 3000 s). Heating per move: 0.3 uK with slow steps and 10 uK with fast steps (Sec. 6).

sites, pitch	timing	to 1e-3	steady-state defects	atoms moved	heating: mean / max
10^5, 4 x 4	slow	0.35 s	2.5e-4	16%	0.05 / 0.3 uK
10^5, 3 x 4	slow	0.23 s	2.5e-4	16%	0.05 / 0.3 uK
10^6, 4 x 4	slow	not in 20 s	1.2e-3	17%	0.05 / 0.3 uK
10^6, 4 x 4	fast	1.45 s	2.5e-4	17%	1.7 / 10 uK
10^6, 4 x 4	adaptive (fast until 1e-2, then slow)	2.47 s	1.2e-3	17%	1.4 / 10 uK
10^6, 3 x 4	fast	1.45 s	8.1e-4	17%	1.7 / 10 uK

Every atom moves at most once (refills come straight from a reservoir row), so no atom gains more than one move's heating. Up to 10^5 sites, the slow, cold steps reach the loss-limited floor. At 10^6 sites the

refill load is too high for slow steps: switching to them after assembly brings back the latency-limited floor of 1.2e-3. Fast steps are therefore needed in steady state too, at a cost of at most 10 uK per refilled atom. The 3 x 4 um lattice at 10^6 sites carries 33% more refills per block position and rises to 8e-4. A larger block (more engines) or intermediate steps (about 2 uK per move at 200 us lane steps, from the quantum check) would restore 2.5e-4. (Correction, Sec. 10: the 8.1e-4 on 3 x 4 um was an edge artefact of the block placement, not latency.)

15.10 Follow-up: tiling wave optics, remedies at 10^6 sites, imaging cadence

Engines tiled along x: wave optics (`results/m24_tiling_wave.json`). Sec. 4 required opaque gaps as wide as the slit zones, from a first-order lobe argument, and hence an engine pitch Pe >= 4.4 mm. We now propagate the emitter near field of one engine (D1, LinearSynth tweezer at x_a = 0, 60 and 90 um, i.e. up to the segment edge) over f1 = 3.12 mm to its lenslet (clear aperture = fill x Pe), apply an ideal lens phase, propagate over f1 to the shared slit plane Sigma and measure the intensity in the neighbouring engine's slit zone. Sigma is conjugate to the atoms, so this is the ghost a neighbour segment sees. We keep only directions inside the pupil acceptance A/(2 f1), since steeper light misses the objective; an exact ray trace through the neighbouring lenslets (all angles that miss the engine's own lenslet) checks the same question geometrically.

engine pitch Pe	worst ghost peak / tweezer peak (accepted; raw)	ghost power / own slit power	accepted rays through neighbouring lenslets
8.0 mm	4.5e-13 ; 1.7e-9	2.6e-13	none
4.43 mm	3.6e-10 ; 2.5e-7	1.4e-10	none
3.5 mm	1.7e-8 ; 3.1e-6	1.0e-8	none
3.0 mm	1.7e-7 ; 7.7e-6	6.3e-8	none
2.6 mm	1.6e-6 ; 1.3e-5	7.5e-7	none
2.4 mm	6.7e-5 ; 5.6e-5	7.2e-5	none

(Worst case over x_a = 0, 60, 90 um at fill 1; it is the segment-edge tweezer, x_a = 90 um, except at 3.5 mm, where x_a = 60 um is marginally worse. A lenslet fill of 0.8 at 4.43 mm raises the accepted ghost to 2e-9.) The first-order lobes are not the limit: the lenslet aperture and the pupil acceptance remove them, and no accepted ray reaches another slit zone at any pitch scanned. What sets the pitch is the lenslet aperture needed to pass the engine's own slit zone, 2 f1 tan(asin s_cut) = 2.37 mm. Below 2.6 mm the lenslet clips the own beam and its diffraction tail enters the neighbour zone. At Pe = 3.0 mm the accepted ghost is below 2e-7 of a tweezer peak, under the coherent criterion of 1e-6 (`docs/row_coherence.md`). Hence six engines fit on an 18 mm chip edge instead of four. Two scanner positions per row are still needed: the segment pitch at the atoms, Pe f_obj / f_c = 0.092 Pe = 277 um at Pe = 3.0 mm (m22 combiner, f_c 188 mm, f_obj 17.3 mm), exceeds the 180 um segment, so one block position addresses every other segment. The model is 1D (x only), uses an ideal thin lens and no surface scatter; baffles between lenslet channels remain the defence against scattered light, which this calculation does not include.

Intermediate steps on 3 x 4 um (`results/m18_midlane.json`, 3,000 atoms per case). Ramps 100 us, lane steps 200 us, 300 us moves along the lane, transport over 9 um:

hold / moving depth (atom temperature)	transported	heating per move	bystanders at 3 um, 6 um, next row (lost / displaced)
0.1 / 0.3 mK (5 uK)	3000/3000	3.0 uK	0 / 0 / 0
0.2 / 0.6 mK (10 uK)	3000/3000	0.8 uK	0 / 0 / 0

Hence intermediate steps are clean on the dense lattice, at 1 to 3 uK per move. We use 2 uK per move below (the quantum check gave a similar figure).

Three defect fractions. Until now we quoted the defect fraction sampled just after each image. That value counts imaging losses twice: the image removes a fraction eps_img of the atoms after detecting them, so those holes stay unseen for a full period, and at the next image they are still there when the new losses are added. The simulator now also records the value just before each image and the time average (`Sim.mean_defect`, a trapezoid sum over visit, image and idle intervals; vacuum losses are uniform in time, so the rule is exact for them in expectation, while each visit's fills are booked at its start, which biases the

average low by at most one visit time, 1 to 2 ms against T_img = 100 ms). The time average is what a computation started at a random moment sees. With fast steps at 10^6 sites the three are 2.4e-4 (at image), 1.4e-4 (before image) and 1.8e-4 (average): the at-image value is close to 2 eps_img + (T_img + t_img)/tau = 2.4e-4, and the average is close to eps_img + (T_img + t_img)/(2 tau) = 1.2e-4 plus the repair latency.

Remedies at 10^6 sites (`results/m23_remedy.json`, 85% loading, tau 3000 s, eps_img 1e-4, T_img 100 ms, rows 4 um apart). Blocks of 128 to 256 engines, three step timings, heating per move 10 / 2 / 0.3 uK (fast / intermediate / slow). Defect fraction at image / time average over the last 3 s of a 20 s run, and the time from the first image to 1e-3 (at image):

block (engines)	pitch along rows	fast	intermediate	slow
32 x 4 (128)	4 um	2.5e-4 / 2.0e-4, 1.67 s	5.3e-4 / 4.8e-4, 3.14 s	1.2e-3 / 1.1e-3, not in 20 s
32 x 8 (256)	4 um	2.4e-4 / 1.8e-4, 1.01 s	3.9e-4 / 3.4e-4, 1.58 s	8.5e-4 / 8.0e-4, 4.18 s
48 x 4 (192)	4 um	2.3e-4 / 1.8e-4, 1.01 s	4.1e-4 / 3.6e-4, 2.25 s	8.7e-4 / 8.2e-4, 3.06 s
64 x 4 (256)	4 um	2.4e-4 / 1.8e-4, 0.90 s	3.4e-4 / 2.9e-4, 1.47 s	6.6e-4 / 6.1e-4, 2.72 s
32 x 4 (128)	3 um	2.4e-4 / 1.9e-4, 1.34 s	4.3e-4 / 3.8e-4, 2.35 s	1.1e-3 / 1.0e-3, 6.77 s
32 x 8 (256)	3 um	2.4e-4 / 1.8e-4, 0.79 s	3.0e-4 / 2.5e-4, 1.46 s	6.4e-4 / 5.9e-4, 4.40 s
48 x 4 (192)	3 um	2.4e-4 / 1.8e-4, 0.79 s	3.7e-4 / 3.2e-4, 1.91 s	7.5e-4 / 7.0e-4, 4.29 s
64 x 4 (256)	3 um	2.4e-4 / 1.8e-4, 0.68 s	3.2e-4 / 2.7e-4, 1.46 s	5.8e-4 / 5.3e-4, 2.39 s

Four results follow. First, the 8.1e-4 reported in Sec. 9 for 3 x 4 um was an artefact. The last row of block positions ended short of the array edge, and the holes in rows below it had no reservoir row in any block that reached them (563 of 716 remaining holes sat there). `Block.positions` now places the last block flush with the edge, and 3 x 4 um holds 2.4e-4 with the 32 x 4 block and fast steps, as 4 x 4 um does. Second, fast steps reach the loss floor with any block from 128 engines; more engines only shorten assembly (1.67 to 0.90 s at 4 um). Third, intermediate steps (2 uK per move) come within a factor of 1.4 to 1.9 of that floor with 256 engines (time average 2.5e-4 to 3.4e-4) but not with 128. Fourth, slow steps (0.3 uK) do not reach the floor with any block tried; their best is 5.3e-4. The 3 um rows do slightly better than 4 um rows with the slower timings. We attribute this to fewer block positions per sweep (a 180 um segment holds 61 sites at 3 um against 46 at 4 um, so the 32 x 4 block has 222 positions against 252); we did not isolate this cause.

The block shape is a hardware choice. A 32-row block is two 16-chip stacks combined by polarisation (Sec. 4); 48 or 64 rows need three or four stacks and a combining method beyond polarisation (designed in Sec. 11: a slotted comb mirror, combined with the PBS for 64 rows). 32 x 8 needs eight engines per chip edge: at the 3.0 mm pitch allowed above that is a 24 mm edge. At 2.6 mm (21 mm edge) the accepted ghost rises to 1.6e-6, just above the 1e-6 criterion. Hence our preferred block for 10^6 sites is 32 x 8 on a 24 mm chip edge (98,048 fast actuators), with fast steps if the 10 uK per refilled atom is acceptable and intermediate steps (time average 2.5e-4 on 3 x 4 um) if it is not.

Imaging cadence and loss (`results/m25_imaging.json`, `results/m25_imaging_fit.json`). We scan the imaging period T_img (0.03, 0.1, 0.3, 1 s), the loss per image eps_img (1e-5, 1e-4, 1e-3) and the vacuum lifetime tau (300, 3000, 30000 s) at 10^5 sites (4 x 4 um, 85% loading, 32 x 4 block, intermediate steps, t_img 10 ms, 8 s runs, averages over the second half), add t_img = 3 and 20 ms, and take three points at 10^6 sites per timing. Selected time-averaged defect fractions at 10^5 sites:

eps_img, tau	T_img 0.03 s	0.1 s	0.3 s	1 s
1e-5, 3000 s	2.0e-5	3.1e-5	6.7e-5	1.8e-4
1e-4, 300 s	2.9e-4	4.1e-4	7.8e-4	1.9e-3
1e-4, 3000 s	1.6e-4	1.3e-4	1.8e-4	3.1e-4
1e-4, 30000 s	1.4e-4	1.2e-4	1.2e-4	1.5e-4
1e-3, 3000 s	3.1e-3	1.4e-3	1.2e-3	1.3e-3

The values at image fit rho_image = 2.42 eps_img + 0.96 (T_img + t_img)/tau (median error 12%, worst 33%), close to the 2 eps_img + (T_img + t_img)/tau expected from the double counting. This replaces the rule rho ~ 1.5-1.8 (eps_img + T_img/tau) of Sec. 1 and 6, which was fitted to at-image values over a narrower range. The time average follows a hole-lifetime model. An imaging loss is unseen for one period T = T_img + t_img and is then repaired after a mean delay d; a vacuum loss waits T/2 on average for the next image, and then d. Hence

rho_mean = eps_img (T + d)/T + (T/2 + d)/tau.

One delay, d = 25 ms, fits all 40 points at 10^5 sites with a median error of 10% (d = 0 underestimates by a median 16%). The worst point (61%) is eps_img = 1e-3 at T_img = 30 ms, where the repair saturates (below). Minimising rho_mean over T gives the optimum imaging period, T* = (2 eps_img d tau)^(1/2) when d « T. For d = 25 ms and tau = 3000 s this is 40 ms, 0.12 s and 0.39 s for eps_img = 1e-5, 1e-4 and 1e-3, in agreement with the minima of the scan (0.03, 0.1 and 0.3 s). At the optimum, rho_mean ~ eps_img + (2 eps_img d / tau)^(1/2) + d/tau: for eps_img = 1e-4 and tau = 3000 s that is 1.5e-4 (simulated 1.3e-4 at T_img = 0.1 s). Hence the loss per image dominates the floor once tau exceeds a few thousand seconds. Lowering eps_img, not imaging more often, is the lever. The imaging time matters only through the dead time at short periods: at T_img = 30 ms, t_img = 3 / 10 / 20 ms gives 1.35 / 1.55 / 1.70e-4.

At 10^6 sites the delay is not a constant, because the block needs a sweep over all its positions to reach a hole (252 positions for a 32 x 4 block on 4 x 4 um; a one-pass visit takes 0.9 to 1.0 ms with fast steps and 1.35 to 1.5 ms with intermediate steps, from `Timing.pass_time`). Time averages for eps_img = 1e-4 and tau = 3000 s:

timing (32 x 4 block)	T_img 0.03 s	0.1 s	0.3 s
fast	1.2e-3	1.8e-4	2.0e-4
intermediate	2.3e-3	4.9e-4	2.2e-4

At T_img = 30 ms both timings saturate: the loss rate, eps_img / T + 1/tau = 2.8e-3 per site per second, needs 2,800 fills/s. At rho = 2e-4 a block position holds about 0.8 holes (2e-4 x 10^6 / 252), so a sweep of 252 visits (0.25 to 0.4 s) refills only about 200 atoms, and the imaging dead time takes 25% of the time. The holes therefore pile up until each visit has enough work. By Little's law, the mean hole lifetime is rho_mean / (eps_img / T + 1/tau): 0.15 s for fast steps at T_img = 0.1 s, and 0.40 and 0.34 s for intermediate steps at 0.1 and 0.3 s. Intermediate steps at T_img = 0.3 s reach 2.2e-4, within 20% of fast steps at 0.1 s, with the 128-engine block and at one fifth of the heating per refilled atom (2 against 10 uK). Hence, at 10^6 sites, a slower imaging cadence is an alternative to a larger block. The imaging period should match the sweep time of the block rather than be as short as possible.

15.11 Follow-up: combining stacks into blocks of 32 to 64 rows

Why stacks can be combined in space. The plane S, formed by the back focal planes of the per-chip collimators, is conjugate to the atoms with magnification 1/D (D = 125 at 4 um rows; `docs/stacked_relay.md`, Sec. 2). At S, each chip's beam is a stripe of +/- f_FAC NA_FAC = +/-158 um on the 0.5 mm chip pitch, so neighbouring beams leave 184 um gaps. A slotted mirror at 45 deg (a "comb") placed in these gaps interleaves two stacks built at twice the chip pitch. Stack A passes through the slots; stack B, turned by 90 deg, is reflected by the teeth. Each beam then sees a window of half-width a <= 250 um. The comb uses no polarisation, and the downstream relay is the same as for a single stack of all the rows. In addition, it doubles the chip pitch to 1.0 mm, which leaves room for the fibre and electrical access that Sec. 7 of `docs/stacked_relay.md` left open (383 fast actuators per engine, 1,532 lines per four-engine chip). Our choice of a 45 deg comb is a compromise between two considerations. First, it keeps the optical path of every row of stack B equal to that of stack A: with B's facets in the plane y = -z_c, the path to any plane z = Z is Z for all rows. However, the comb then spans the combined stack height along z, so rows at the far end meet it farther from S.

The penalty is set by the light the objective accepts (`results/m26_edge.json`). We propagate the field of one chip in y from the collimator exit (aperture f_FAC before S, sine-condition lens, beta 0.7 um, NA_FAC 0.9) to the comb at a distance z from S, apply the window, propagate back to S, and continue to the atoms through the pupil (`picsim/opa/stack.RelayY`). Without a comb this reproduces the m13 values (transmission 0.973, crosstalk at 4 um 1.09e-4 against 1.06e-4). Single tweezer, hard collimator edge, slot window of half-width a:

z from S	a = 250 um: loss / peak / crosstalk at 4 um / at 3 um	a = 225 um: loss / peak / crosstalk at 4 um
no comb	0 / 1 / 1.1e-4 / 2.9e-4	0 / 1 / 1.1e-4
10 mm	< 1e-3 / 0.998 / 1.0e-4 / 2.8e-4	6e-4 / 0.992 / 1.1e-4
20 mm	2.5e-3 / 0.992 / 1.5e-4 / 4.0e-4	7.2e-3 / 0.985 / 1.8e-4
30 mm	8.5e-3 / 0.987 / 2.7e-4 / 7.1e-4	1.6e-2 / 0.981 / 3.1e-4
45 mm	1.9e-2 / 0.973 / 6.0e-4 / 1.7e-3	3.4e-2 / 0.956 / 7.9e-4
60 mm	3.6e-2 / 0.953 / 1.3e-3 / 2.9e-3	5.3e-2 / 0.931 / 1.6e-3

The hard edge of the collimator is not the problem. Its Fresnel tail spreads into the gap, but it consists of directions steeper than the pupil accepts, and clipping it costs nothing. A graded collimator edge (raised cosine 0.8-0.9) therefore does not help: it raises the baseline crosstalk to 1.5e-4 and gains no distance. (A Fresnel-tail estimate that ignores the pupil predicts 0.7% loss at 10 mm; the simulated loss there is below 0.1%.) What matters is the light inside the pupil acceptance, which leaves S at half-angle NA/D = 4 mrad. It reaches the window edge at

z_max = (a - f_FAC NA_FAC) D / NA,

which is 23 mm for a = 250 um and 17 mm for a = 225 um. This agrees with the scan: at z = 20 mm and a = 250 um the worst row loses 0.25% and the crosstalk at 4 um rises from 1.1e-4 to 1.5e-4, while at 30 mm it reaches 2.7e-4. A single edge (block combining, rows 0-15 from one stack and 16-31 from the other) behaves similarly. It also shifts the boundary row by 1.7 nm at z = 20 mm (a = 250 um) and by 12 nm at a = 200 um, a static offset to calibrate. The rows are mutually incoherent (`docs/row_coherence.md`), so this crosstalk adds in intensity.

2D check (`results/m26_twod.json`). The 1D model leaves out the circular pupil, which couples x and y, and the full rows. We therefore pass the comb-windowed y pupil of both chips through the 2D model of `docs/stacked_relay.md` (D1 x response, circular NA 0.5 pupil). Chip m forms a full row of 41 tweezers at 3 um, and chip m+1 forms 11 tweezers offset by 1.5 um; one tweezer of chip m then moves by 0.25 um. Without a comb this reproduces the m13 values. Rows incoherent, neighbour row at 4 um (at 3 um in brackets):

comb (both chips windowed)	single tweezer: peak / crosstalk at 4 um	full row: crosstalk at 4 um, max over x	neighbour, static: depth / position	neighbour, moving tweezer: depth / position
none	1 / 1.1e-4	1.8e-4	1.7e-4 / 0.002 nm (2.0e-4)	2.4e-5 / 0.04 nm (6.2e-5)
z 15 mm, a 250 um	0.995 / 1.1e-4	1.8e-4	1.8e-4 / 0.02 nm (2.2e-4)	2.9e-5 / 0.03 nm (7.4e-5)
z 23 mm, a 250 um (worst row, bold layouts)	0.990 / 1.5e-4	2.3e-4	2.1e-4 / 0.08 nm (2.6e-4)	3.9e-5 / 0.03 nm (9.6e-5)
z 23 mm, a 225 um	0.982 / 1.6e-4	2.4e-4	2.2e-4 / 0.12 nm (2.2e-4)	3.9e-5 / 0.03 nm (8.0e-5)
z 30 mm, a 250 um (beyond z_max)	0.984 / 2.1e-4	2.9e-4	2.5e-4 / 0.14 nm (3.1e-4)	5.3e-5 / 0.03 nm (1.2e-4)
single edge, z 23 mm, a 250 um (block boundary)	0.995 / 1.3e-4	2.0e-4	1.9e-4 / 0.07 nm (2.2e-4)	3.4e-5 / 0.03 nm (8.4e-5)

Hence 2D does not change the conclusions. At the worst row the full-row crosstalk rises by a factor 1.3, and the effect of a moving tweezer on the next row by a factor 1.55 to 1.65 (to 3.9e-5 in depth at 4 um, 9.6e-5 at 3 um). This is still more than an order of magnitude below the in-row figures of D1 (0.11%, 3.5 nm). Tweezers at x = 0 and at the row end (60 um) give the same values to 0.1%, because the comb acts on y alone and the circular pupil couples the two only weakly. The 1D single-tweezer figure (1.8e-4 at z = 23 mm) slightly overstated the 2D one (1.5e-4). The block-combining edge moves the boundary row by 7 nm in 2D (peak position), a static offset to calibrate. With coherent rows the comb would raise the static depth change at 4 um from 2.6% to 2.9%; rows must stay incoherent, as they must without a comb.

Comb roll and edge straightness. A comb edge that is not parallel to x makes the window depend on x. Beyond the slit plane the x relay is telecentric (chief rays leave the slit plane parallel to the axis), so each tweezer meets the comb at its slit-plane position X = f1 s. Four engines at 3.0 mm pitch span X = +/-5.6 mm. One tweezer's footprint at the comb is at most +/-0.78 mm (17 mm beyond the slit plane, half-angle (A/2)/f1 = 0.046). Hence, for a roll of 2 mrad, the edge moves by only 3 um across one tweezer, against the tens of um over which the result changes. We therefore treat each tweezer with its local window a(X), without a non-separable propagation. In 2D, at z = 23 mm, the peak falls from 0.989 to 0.972 as a goes from 250 to 205 um (about 0.04% per um), while the crosstalk at 4 um stays at 1.4 to 1.6e-4. A roll therefore

mainly makes the depth vary along the row: about +/-0.4% for a +/-10 um change of a, which a 1.8 mrad roll gives over +/-5.6 mm. The engines set each tweezer's power in any case, so this is a calibration, not a limit. The same holds for edge straightness of +/-10 um over 12 mm.

Layouts (`results/m26_layout.json`; first comb point z_c = 7 mm, just beyond the x slit plane at 6.24 mm, CHOICE). The worst row is the one at the far end of the comb:

rows	layout	chips per stack, pitch	comb z range	worst row (a = 250 um): loss / peak / crosstalk at 4 um	polarisation-insensitive objective
32	one stack	32, 0.5 mm		(no comb)	no
32	2 stacks, PBS	16, 0.5 mm		(no comb)	yes
32	**2 stacks, comb**	**16, 1.0 mm**	**7-23 mm**	**0.45% / 0.991 / 1.8e-4**	**no**
48	4 stacks, comb + PBS	12, 1.0 mm	7-19 mm	0.21% / 0.992 / 1.4e-4	yes
48	2 stacks, comb	24, 1.0 mm	7-31 mm	0.93% / 0.987 / 2.9e-4	no
64	**4 stacks, comb + PBS**	**16, 1.0 mm**	**7-23 mm**	**0.45% / 0.991 / 1.8e-4**	**yes**
64	2 stacks, comb	32, 1.0 mm	7-39 mm	1.3% / 0.984 / 3.9e-4	no
64	2 stacks, PBS	32, 0.5 mm		(no comb)	yes

With a = 225 um (about what a 50 um tooth plate, t / (2 sqrt 2) = 18 um per edge, and +/-10 um edge placement leave) the worst rows of the two bold layouts lose 0.86% and reach 2.0e-4. A second comb level (stacks at 2.0 mm, combined pairwise and then again) would put the second comb beyond z_max, so we do not use it. The single taller stack (32 or 64 chips at 0.5 mm) needs no combiner. In first-order optics it only enlarges the y-relay field: +/-8 or +/-16 mm at the first telescope lens, and 3.7 or 7.4 mrad at the pupil. Its costs are mechanical: 16 to 32 mm of facets flush to about 100 um, and all the I/O at 0.5 mm pitch.

Tolerances (DERIVED from the m13 conjugations). A comb tilt theta about x deflects stack B by 2 theta at S. This shifts its facet image in the pupil by 2 theta F_y (F_y = 2.17 m). Matching the shift to the m13 decentre results gives 1% peak loss at theta = 0.16 mrad and 3% at 0.28 mrad. The same tilt moves B's rows by 2 theta z / D: 0.32 nm per urad at z = 20 mm, and 50 nm at 0.16 mrad. This static offset alternates between rows (A and B rows interleave), so it must be calibrated once against the static array. Keeping the alternating pitch error below 3 nm needs a tilt stability of about 10 urad (drift, not alignment). Moving the comb by dz along z changes B's path by dz, which tolerates 100 um (m13 chip piston, peak >= 0.992). It also shifts the windows by dz in y, which should stay within about +/-10 um (the a scan). Light that misses a tooth leaves at 90 deg to a dump. Light from stack A that hits the back of a tooth is reflected towards stack B and needs an absorbing back face.

Hence the blocks of Sec. 10 can be built as follows. (i) 32 rows: two 16-chip stacks at 1.0 mm pitch and one comb. This needs no polarisation optics, and it gives alternate rows from different stacks, so the passive orthogonal-polarisation decorrelation of `docs/row_coherence.md` comes for free if the objective accepts it. (ii) 64 rows: two such comb pairs, combined by a PBS. Four 16-chip stacks with four engines per chip at 3.0 mm pitch (12 mm edge) then form the 64 x 4 block: 256 engines, 98,048 fast actuators. With intermediate steps at 10^6 sites it reaches a time average of 2.9e-4 on 4 x 4 um and 2.7e-4 on 3 x 4 um (Sec. 10), without the 24 mm chip edge that 32 x 8 needs. The worst rows lose 0.9 to 1.5% of their peak depth to the comb (a = 250 / 225 um), which is recovered by the per-tweezer power the engines set in any case. The edge scan is 1D in y and the 2D check uses the same windowed pupil; both assume an ideal thin collimator and the window of each slot at one distance z; the 45 deg tilt places the two edges of a 0.5 mm slot 0.25 mm before and after that distance, which changes z by at most 4% at z_c = 7 mm. Tooth-edge roughness, scatter from the comb, and a lens design of the larger y-relay field are not included.

15.12 Follow-up: the real collimator

Sec. 11 used an ideal thin collimator in air. The adopted collimator (`docs/collimator.md`) is a fused-silica Weierstrass block glued to the facet, followed by a 10 um gap and a 250 um aplanat of index 1.8. We repeat the comb study with this traced lens (`scripts/m26_stack_combine.py lens`, `results/m26_lens.json`). We then ask how the cylindrical collimator treats light that leaves the chip at an angle along x (`skew`,

`skew_offset`; `results/m26_skew.json`, `results/m26_skew_offset.json`). The first changes little. The second limits the whole stacked relay.

The comb with the real collimator. The exit field is built from the traced lens map (exit height, wavefront and amplitude per ray) at a plane 60 um beyond the last vertex. S lies at the traced back focal plane, 0.44 mm from the facet. Without a comb this reproduces the m14 wave model: crosstalk 8.0e-5 at 4 um against 8.0e-5, and 3.8e-4 against 4.0e-4 at 3 um. The real geometry also moves the comb. The traced x lens of m14 puts the x slit plane 7.37 mm from the facet, the collimator glass adds 0.15 mm in x (t (1 - 1/n) per section), and we keep 0.5 mm of clearance. The comb therefore starts 7.6 mm from S, and the far row of a 32-row comb sits at 23.6 mm. Slot window a = 250 um (225 um in brackets):

z from S	real collimator: loss / peak / crosstalk at 4 um	thin collimator (Sec. 11)
15 mm	5e-4 / 0.994 / 7.6e-5 (3.4e-3 / 0.986 / 7.5e-5)	5e-4 / 0.994 / 1.2e-4
20 mm	2.6e-3 / 0.991 / 9.2e-5 (7.6e-3 / 0.985 / 9.7e-5)	2.5e-3 / 0.992 / 1.5e-4
23.6 mm (far row)	5.2e-3 / 0.991 / 1.1e-4 (9.6e-3 / 0.985 / 1.2e-4)	4.7e-3 / 0.991 / 1.8e-4
30 mm	9.0e-3 / 0.988 / 1.9e-4 (1.7e-2 / 0.983 / 2.0e-4)	8.5e-3 / 0.987 / 2.7e-4

The immersed facet puts more light near the edge of the beam at S, but loss and peak follow the thin-lens values within 12%, and the crosstalk is lower. The rule z_max = (a - f_FAC NA_FAC) D / NA therefore still holds. In 2D at the far row (z = 23.6 mm, a = 250 um), the full-row crosstalk at 4 um is 1.6e-4 (1.75e-4 without the comb). A moving tweezer changes the next row by 2.8e-5 in depth at 4 um (1.2e-5 without the comb) and by 7.0e-5 at 3 um (3.6e-5). Hence the comb conclusions of Sec. 11 hold for the real collimator, at the slightly larger distances of the real geometry.

Light that leaves the chip at an angle along x. A tweezer at x_a along the row is formed by light that leaves the emitters with x direction sine s_x = x_a / x_scale (x_scale = 288 um for D1). That is up to 0.31 at the end of a 180 um segment and 0.355 at the slit edge. The collimator is cylindrical. Its surface normals have no x component, so k_x is conserved, and the y-z problem is exactly the 2D trace with every index n replaced by sqrt(n^2 - s_x^2). A unit test checks this against 3D vector refraction. All the earlier y models (m13, m14, and Sec. 11 above) used s_x = 0. For s_x > 0 the effective index ratios of the lens rise: 1.45 -> 1.50 for the block and 1.80 -> 1.89 for the aplanat at s_x = 0.355. The lens is then no longer aplanatic for the facet. Peak relative to an ideal collimator at the same s_x (1D, 2D in brackets), equivalent facet focus error, and crosstalk at 4 um:

x_a (s_x)	peak	equivalent facet defocus	residual after refocusing the facet	crosstalk at 4 um
0	1.000 (1.000)	0	0	8.4e-5
15 um (0.052)	0.999 (0.997)	-0.27 um	0.001 waves	2.2e-4
30 um (0.104)	0.971 (0.942)	-1.10 um	0.005 waves	3.0e-3
45 um (0.156)	0.794 (0.725)	-2.46 um	0.014 waves	8.2e-3
60 um (0.208)	0.481 (0.440)	-4.36 um	0.029 waves	6.9e-3
90 um (0.312)	0.231	-9.67 um	0.096 waves	9.0e-3

The error is almost purely a focus shift of the facet, dz = -99 s_x^2 um. A thin glass lens in air would give -f (n + 1) s_x^2 / (2 n) = -148 s_x^2 um, the same order. Refocusing the facet for one tweezer restores its peak fully; the residual after refocusing stays below 0.1 waves up to 90 um. But all tweezers of a row share the facet and the collimator. The phase sits at S, which is conjugate to the atoms, so refocusing the atom plane does not help either. In 2D the best atom-plane focus recovers the peak up to x_a = 30 um (1.02, with a 1.5 um y-focus shift, outside the +/-0.5 um axial registration of `docs/transport3d.md`), and not beyond. Beyond about 60 um the 2D spot breaks up (peak 0.11 at 75 um), because the y pupil is then filled to its rim and the circular pupil cuts it differently for each x pupil column.

What a common focus offset recovers. A single facet offset for the whole row (the length of the glued block, set at fabrication) trades the centre against the ends. With dz = -1.25 um the peak stays at or above 0.96 out to |x_a| = 45 um, and the static crosstalk at 4 um stays at or below 3.6e-3. That is within the criteria of `docs/row_coherence.md` (others' depth change below 3e-3 when a tweezer moves), since for incoherent rows the moving-tweezer effect was about 0.13 of the static crosstalk in m13. We did not recompute it for these cases. Beyond 50 um no offset helps: 0.90 at 50 um and 0.66 at 60 um. No offset brings the worst peak over |x_a| <= 90 um above 0.43.

Hence a refractive cylindrical collimator at NA 0.9 serves about half of the 180 um segment (|x_a| <= 45 um, about 23 sites at 4 um), not all of it. Every power and crosstalk figure of m13, m14 and Sec. 11

holds near the row centre and is optimistic for the outer half. For the large arrays this doubles the number of engines per row at fixed coverage: 90 um segments instead of 180 um. Alternatively, the collimator must become insensitive to s_x. Serving |x_a| <= 90 um within the +/-0.5 um focus tolerance, with the best common offset, needs |dz/s_xˆ2| <= 10 um: ten times less than now. We see four routes, none of them designed here.

(i) An all-reflective collimator in one medium. The law of reflection and a uniform medium are unchanged by the effective-index map, so W = 0 for every s_x; a flat exit face normal to the collimated beam keeps it so (DERIVED). It does not obey the sine condition and needs a folded geometry within the 0.5 mm chip pitch.

(ii) A cemented refractive design. Glass-glass interfaces are less sensitive than glass-air ones (relative power change 0.41 s_xˆ2 for 1.45 | 2.0 against 0.84 s_xˆ2 for 1.45 | air, DERIVED), but a factor of two is not enough on its own.

(iii) A corrector at the x slit plane, where x encodes s_x. This fails at large s_x. At x_a = 90 um the error acts at S like a y lens of 4 mm focal length (2.9 waves at 158 um), so the beam has passed a focus before the slit plane, 7 mm later.

(iv) A larger vertical facet mode (lower facet NA). This helps by less than a factor of two, because the optimal pupil fill, and with it f_FAC, grows with the mode.

The downstream cylindrical lenses of the y relay see x angles below about 0.05 and are not modelled here.

15.13 Verification notes

value	location	type	pointer
geometry, SBP, zones, static power, blocks, loss rates	Sec. 1-4	DERIVED	results/m22_budget.json
x-tiling lobe geometry and ABCD	Sec. 4	DERIVED	results/m22_tiling.json
3 um pitch transport	Sec. 5	SIM	results/m18_pitch3.json (3,000 atoms; review)
assembly and steady state	Sec. 1, 6	SIM	results/m23_steady.json (review)
rho ~ 1.5-1.8 (eps + T_img/tau)	Sec. 1, 6	SIM/DERIVED	fit to m23 floors (at-image values; superseded by Sec. 10)
23 min lifetime, 99.98952% imaging survival, 6,100 atoms	Sec. 6	SOURCE	Manetsch et al., arXiv:2403.12021, abstract read 2026-11-06
30,000 qubits/s, > 3,000 atoms for > 2 h	Sec. 6	SOURCE	Chiu et al., arXiv:2506.20660, abstract read 2026-11-06
2,088 sites at 6 K, 828-atom rearrangement	context	SOURCE	Pichard et al., arXiv:2405.19503, abstract read 2026-11-06
continuous lattice array > 1,000 atoms	Sec. 3	SEARCH SUMMARY	Gyger et al., arXiv:2402.04994 (abstract not read)
eta 0.3, hold 0.1 mK, load 1 mK, w0 0.85 um, eps_img, p_move, T_img, t_reload	Sec. 3, 6	CHOICE	stated assumptions
refractive NA 0.5 objective fields	Sec. 2	RECALLED	verify
anisotropic and corridor transport	Sec. 9	SIM	results/m18_aniso.json (3,000 atoms; review)
tweezer and lattice power	Sec. 9	DERIVED	results/m22_lattice.json (interference gain 8 / 16, -20% uniformity, eta 0.3, cavity buildup CHOICE)
mixed timing and heating	Sec. 9	SIM	results/m23_mixed.json (review)
ghost in the neighbour slit zone vs engine pitch	Sec. 10	SIM	results/m24_tiling_wave.json (1D, ideal thin lenslet, no surface scatter; review)
segment pitch at the atoms 0.092 Pe	Sec. 10	DERIVED	f_obj / f_c from results/m22_tiling.json
intermediate-step transport on 3 x 4 um	Sec. 10	SIM	results/m18_midlane.json (3,000 atoms; review)
remedies at 10ˆ6 sites, edge-artefact correction	Sec. 10	SIM	results/m23_remedy.json (one seed per case; review)
imaging map, rho_image fit, latency d = 25 ms	Sec. 10	SIM	results/m25_imaging.json, results/m25_imaging_fit.json (one seed per case; review)
T* = (2 eps d tau)ˆ(1/2), Little's-law hole lifetimes, holes per position	Sec. 10	DERIVED	from the hole-lifetime model and the SIM values
coherent ghost criterion 1e-6	Sec. 10	DERIVED	docs/row_coherence.md
24 mm chip edge for 32 x 8	Sec. 10	open	not designed
comb edge penalty vs distance and window	Sec. 11	SIM	results/m26_edge.json (1D, thin collimator, both slot edges at one z; review)

value	location	type	pointer
z_max = (a - f_FAC NA_FAC) D / NA	Sec. 11	DERIVED	in-band half-angle NA/D at S
layouts, worst-row penalty, y-relay field	Sec. 11	SIM/DERIVED	results/m26_layout.json; scripts/m13_stack.design
comb tilt tolerance, row shift per urad	Sec. 11	DERIVED	m13 decentre tolerance (results/m13_tolerance.json) and conjugations
z_c = 7 mm, tooth plate 50 um, +/-10 um edge placement	Sec. 11	CHOICE	stated assumptions
2D comb check: full row, neighbour static and moving	Sec. 11	SIM	results/m26_twod.json (m13 Focus2D; review)
roll and straightness by a local window a(X)	Sec. 11	DERIVED	telecentric slit space; footprint (A/2)/f1
comb fabrication, edge quality, scatter; mount stability ~10 urad	Sec. 11	open	[REF?]
comb with the traced collimator, real slit-plane distance	Sec. 12	SIM	results/m26_lens.json (1D and 2D; review)
skew-ray detuning of the collimator vs x_a	Sec. 12	SIM	results/m26_skew.json (effective-index trace, exact for cylindrical surfaces; downstream relay ideal; review)
common facet offset	Sec. 12	SIM	results/m26_skew_offset.json (first-order focus phase, 1D; review)
thin-lens skew estimate, glass-glass sensitivity, corrector focal length	Sec. 12	DERIVED	formulas in the text
moving-tweezer / static crosstalk ratio 0.13	Sec. 12	SIM	m13 (results/m13_crosstalk.json); not recomputed for skewed tweezers
skew-insensitive collimator (reflective)	Sec. 12	open	not designed